

Supplementary record

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Paper under double-blind review

This record accompanies the submission. It is the companion note of an earlier, longer draft of the paper, preceded by this cover note and by the protocol records of that draft (Table R1). It lets a reader check every result the paper summarises. The paper is self-contained and does not depend on it.

What to find where. The full inventory of tests, with their seeds, families, thresholds, statistics and statuses, is in Tables S24 to S28 (§S22). The Holm audit, the minimum detectable effects and the GPU audit cited by the paper are in §S25, which also gives the full tables of the campaigns added for this version. Table R1 dates every protocol of the earlier campaigns. The protocols of the campaigns added for this version are listed in the paper itself.

Reading this record against the paper. References to “the main paper” in this record point to the earlier draft, so their section and table numbers differ from those of the submitted paper. The record keeps the earlier vocabulary. “Pre-registered” there means prespecified in internal version-control records, not registered externally. “Budget-matched” corresponds to the paper’s comparable-budget reference. The configuration names of the record (`top`, `bottom`, `dual_top`, `B@dom`, `A@min` and so on) map to the paper’s B frozen and A frozen on the top or bottom band. Where the record and the paper differ, the paper prevails.

Values. Two values of the earlier draft are corrected in this record. The first Holm step requires $|t| \geq 11.34$ for a two-sided test with four degrees of freedom, and 82.5 % of the runs with a recorded GPU model ran on RTX 4000 Ada. This record reports random spectral indices at rank 4 as 70.23 %, at the rate that its rank-4 measure fixed in advance. The paper reports 70.81 %, at the rate its selection rule gives, and says so.

campaign	committed (2026)	used in
Qwen position sweep, fifteen seeds	19/09, 12:46	§S3.4
Llama extension, seeds 9 to 16	21/09, 14:31	§§S3.3 and S3.5
rate effect at rank 4 and on Qwen	21/09, 20:42	§S23.4
random_ortho on Qwen, seeds 7 to 16	21/09, 21:13	§S3.5
top_sigma, rate sweep	22/09, 14:09	§S9
common-rate inversion on Qwen2.5-1.5B	22/09, 16:04	§S23.4
ordering at rank 4, top_sigma counted, seeds 2 to 8	23/09, 09:32	§S11.4
top_scalar at rank 2	23/09, 09:32	§S9.3
rank 16 on OpenBookQA, Qwen2.5-1.5B	23/09, 09:46	§S23.5
replication on ARC-Easy and ARC-Challenge	24/09, 00:03	§S23.5
format at ranks 8 and 16; ARC addendum	24/09	the earlier draft
rate shift at ranks 8 and 16, a prediction	24/09	§S16.2
bottom_sigma; top_sigma vs top_scalar, 15 seeds; OpenBookQA at rank 2	24/09	the earlier draft, §S9.4
second budget; control at equal norm	25/09	§§S11.1 and S3.6
top_unfrozen: isolating the cost of freezing	22/09	the earlier draft
its addendum: rank 2, bottom, one night, angles, intervals	25–26/09	the earlier draft
rank 8 on fresh seeds, addendum	24/09	the earlier draft
top_sigma vs top_scalar, 40 fresh seeds (C2)	25/09	§S9.3
OpenBookQA tie, addendum: after the selection sweeps, before the reported runs	25/09	§S23.4
norm control, deviation	25/09	§§S3.6 and S12
ARC-Challenge at rank 32	25/09	the earlier draft
ARC-Challenge at rank 2	26/09	the earlier draft
freezing A instead of B (the version run)	26/09	the earlier draft
its note: dual_random_ortho, a code defect fixed	26/09	§S15
OpenBookQA: freezing B , isolated	26/09	the earlier draft
its addendum: GPU model, after launch, before reading	26/09	the earlier draft
its follow-up: one GPU model, after reading	26/09	the earlier draft
freezing A on OpenBookQA	26/09	the earlier draft
the scalar as a per-module rate (top_lrscaler)	26/09	§S9.4
the format task on Qwen2.5-1.5B, ranks 2 and 16	26/09	the earlier draft
the rate shift on five seeds	26/09	§S16.2
its reading rules: after launch, before reading	26/09	§S16.2
freezing B isolated at rank 16, with its planned step and oracle	26/09	the earlier draft
bracket		
the central result on a larger model (probe, rule fixed in advance, selection then report)	26/09	the earlier draft
addendum: SVD check recalibrated for large matrices, before the probe was rerun	27/09	the earlier draft
second addendum: eleven selection runs rerun after machine failures	27/09	the earlier draft
rates selected on seeds 0 and 1, report grids committed before their runs	27/09	the earlier draft
freezing B isolated on the format task with Qwen2.5-1.5B	27/09	the earlier draft
the trajectories of the per-module rate and the scalar (clipping, ϵ , precision)	27/09	§S9.4
the rank-2 comparison of the introduction on fresh seeds	27/09	the earlier draft
the central result on Mistral-7B (probe, rule fixed in advance, no fallback)	27/09	the earlier draft
its selection grids, committed before their runs	27/09	the earlier draft
the rank-2 isolation with four times more training (rates selected again)	27/09	the earlier draft
freezing A against freezing B at equal budget, on square modules	27/09	§S15.2
addendum: equal budgets, collapse rule fixed in advance	27/09	the earlier draft
the isolation at rank 16 on OpenBookQA (references and G committed)	27/09	the earlier draft
the A-frozen arms at rank 16 on ARC-Challenge and OpenBookQA	27/09	the earlier draft
addendum: budget, reading fixed in advance, comparison with free LoRA	27/09	§S19
B trained at rank 1 on the two 7-billion models (the budget-matched comparison)	27/09	the earlier draft
its report grids, committed before their runs	27/09	the earlier draft
addendum: three selection runs killed by the time limit, rerun once	27/09	the earlier draft
extension: the scripts and definitions of the post hoc analyses and of the analysis of where free LoRA writes, before execution	28/09	the earlier draft
addendum: its reading rule, before the OpenBookQA analysis	28/09	the earlier draft
extension: the budget, B frozen at $r=3$ and 4 against A frozen at $r=2$ (OBQA)	28/09	the earlier draft
extension: the rank trend, ranks 4 and 8 (OBQA, Qwen2.5-1.5B)	28/09	the earlier draft
extension: rank 16 on Qwen2.5-7B (OBQA)	28/09	the earlier draft
addendum: a second relaunch for two selection runs	28/09	the earlier draft
extension: phase 2, B frozen on the learned direction; launch declared	28/09	the earlier draft
extension: the two replications of phase 2 (direction seed 7 on OBQA; format, Llama-3.2-1B), before their first runs	28/09	the earlier draft
extension: B frozen on the initial gradient's directions (grad), committed at 23:29	28/09	the earlier draft
declared fixes before any basis or run: SVD in double precision (00:00), and the wording of the verdict (00:35), values and rule unchanged	29/09	the earlier draft

Table R1: Protocols of the earlier campaigns, with the date of the commit that registered each and where it is reported: a section of this record, or the earlier draft. Each protocol fixes the seeds, the comparison, the threshold and what is concluded in each case. The protocols of the campaigns added for this version are listed in the paper itself.

Companion note

Frozen factor or spectral band? What comparisons of frozen-subspace LoRA variants measure at low rank

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Paper under double-blind review

Abstract

This companion note holds, complete and with every number and status tag unchanged, the results, tables, figures and status tags that the main paper “Frozen factor or spectral band? What comparisons of frozen-subspace LoRA variants measure at low rank” moves out of its body and appendix. Each section names what it holds, and its references to the main paper follow that paper’s current numbering.

S1 Overview

This note holds the material moved out of the main paper (Anonymous, 2026), complete: its text, tables, figures and status tags, with every number and status tag unchanged and the notation of the main paper. References to the main paper’s sections, tables and figures are live cross-references, updated with its numbering. The main paper points to the relevant section wherever it mentions one of these results.

Status tags. Every claim about the arms carries a two-part tag, [provenance · outcome], defined as in the main paper. The provenance says when the comparison was fixed.

- [Pre-registered]: the comparison, its seeds and its threshold were fixed in a protocol committed before the first run of its campaign returned a result.
- [Planned]: fixed before its reported seeds were seen, as our records confirm, but without a committed protocol.
- [Post hoc]: decided after the reported seeds were seen, or of a timing our records do not confirm. It carries no confirmatory weight.

The outcome says what the evidence shows.

- [Established]: the result clears its stated threshold.
- [Not established]: a test was run and the result falls short of its threshold.
- [Descriptive]: no test was run, or the paper declares the result descriptive or rests no claim on it.

A pass is *marginal*, marked ^m, when its statistic lies within 15 % of its threshold.

The main claim tagged here. Gap minus spread, a descriptive summary: format $r=2$ to 8 and Qwen2.5-1.5B $r=2$ / format $r=16$ / ARC-C $r=16$, 32: PH · E ($r=2$), PR · E / PR · NE / PR · E (reversed). This row of the main paper’s table of claims now lives here (§S11.1).

The full records follow.

S1.1 Secondary findings

Four further results are reported in full in the sections below, each with its status tag. On Llama at rank 2, the minor band trails every other main arm at its selected rate, established against each of the three ($t = 4.39$ to 7.42), while on Qwen2.5-0.5B no deficit is detected (§S4). The direction of forgetting follows the learning rate on every seed, not past a corrected threshold (§S23.1). The knowledge task does not separate the arms at ranks 1 to 16, and at rank 64 the free arm, at $2.15\times$ the budget, leads every frozen arm by 20.8 to 26.6 points (§S5). A spectral criterion, $\sigma_{\min}/\sigma_{\max}$ below 10^{-4} , separates the degenerate modules of three models where a fixed list fails (§S6).

S2 Notation

S3 Position along the spectrum: an interior band at rank 2

At rank 2 the best band is interior ($\beta = 0.75$) and position is of the same order as the gap to a budget-matched free LoRA; at rank 4 no position separates from the best, and the position effect is smaller than the other two. §S3.1 describes the sweep, and §S3.2 sizes the position effect. §§S3.3 and S3.4 place the optimum on Llama and on Qwen. §S3.5 compares it with a random basis, and §S3.6 asks whether the axis is position or update norm.

S3.1 The position sweep

On Llama, the selected rate is flat across the four upper positions and drops only at the minor band: a cliff, not a gradient. PiSSA, PiCa, MiLoRA and MiCA each contrast two extremes of the spectrum and argue for one of them. Quercia et al. sweep the position on initialisations, at a common rate. We sweep it on frozen subspaces instead, over the five β -positions of Appendix C.2 of the main paper, each at its own selected rate. $\beta = 0$ reproduces **top**, and $\beta = 1$ reproduces **bottom**.

On Llama at rank 2, the selected rate is 10^{-2} at every position except $\beta = 1$ (Table S2). At $\beta = 1$, $5 \cdot 10^{-3}$ wins in validation by 5.3 points over its runner-up, $2 \cdot 10^{-3}$, and by 8.3 over 10^{-2} . At $\beta = 0.25$, the selection is a tie inside the noise: 10^{-2} leads $5 \cdot 10^{-3}$ by 0.006 on two seeds. The swept grid runs from 10^{-3} to $2 \cdot 10^{-2}$, so every selected rate is interior. The reported seeds, 2 to 8, are disjoint from the selection seeds, 0 and 1.

We had expected a gradient instead: an optimal rate rising continuously as the band moves up. That expectation is not borne out, though not by non-monotonicity: the selected rate stays at 10^{-2} from $\beta = 0$ through 0.75, and moves only at the extreme band.

What per-position rates change here. On Llama, the four upper positions select the same rate, so per-position selection departs from a common rate only at $\beta = 1$. The survival of the interior optimum is therefore tested against that one departure on Llama, and against the positions' own selected rates on Qwen.

Status: [Post hoc · Descriptive].

Caveats. The $\beta = 0.25$ selection is itself a tie, so the cliff may begin earlier than it appears.

S3.2 How large the position effect is

At rank 2, position matters more than the main table suggests; at rank 4, the positions converge. At rank 2, the spread of the four upper positions is 4.0 points, four times the 0.91 points that separate **top**, **random** and **random_ortho**. This 4.0 leaves out $\beta = 1$ deliberately: **top**, **random**, **random_ortho** and the four upper positions all run at 10^{-2} , while $\beta = 1$ runs at $5 \cdot 10^{-3}$, so the four upper positions are the like-for-like comparison.

Those three arms are not comparable positions: two of them are drawn at random and do not sit on the spectrum at all. The position effect spans all five positions: 8.4 points on seeds 2 to 8 (Table S2) and 9.1 on seeds 2 to 6 (Table S12). Paired with the free reference on seeds 2 to 6, this 9.1 stands against a gap to a

term	meaning
<i>arms</i>	
dominant band, minor band	The r leading singular directions of W_0 ; the r trailing ones, all with nonzero singular value.
frozen arm, B-frozen arm	An arm whose B is fixed; only A is trained. The main text says B-frozen wherever an A-frozen arm could be confused with it.
top , bottom	Frozen arms with B on the dominant band and on the minor band.
random	Frozen arm with B on r columns of U drawn without replacement.
random_ortho	Frozen arm with B drawn uniformly among orthonormal matrices, independently of W_0 .
main arms	top , bottom , random , and random_ortho where it was run (rank 2).
β -position	Frozen band a fraction $\beta \in \{0, 0.25, 0.5, 0.75, 1\}$ of the way down the spectrum; $\beta = 0$ is top and $\beta = 1$ is bottom (Appendix C.2 of the main paper).
top_sigma	top with Σ restored, $B = U_r \sqrt{\Sigma_r}$, at the same budget (§S9).
top_scalar	top with one scalar per module in place of $\sqrt{\Sigma_r}$, $B = \sqrt{\sigma} U_r$, norm-matched to top_sigma (§S9.3).
free	Standard LoRA, both factors trained; the reference for the gap to a budget-matched free LoRA.
init_top , ...	Both factors initialised on a band and trained, as in PiSSA and MiLoRA (§S13).
<i>budgets and populations</i>	
budget-matched	free at half the frozen arms' rank. It holds 7.5% more parameters than a frozen arm over all seven modules, 72% of its budget on five modules (gate and up dropped), and exactly as many on q and o alone (§S11).
2.15× the budget population	free at the frozen arms' own rank. Such rows are shown, not relied on. The arms compared on one rank, one task, one choice of modules and one seed set (Fig. S6).
<i>effects</i>	
gap to a budget-matched free LoRA	free minus the best frozen arm of the same population; where the budgets differ (§S11.3, §S11.5), <i>gap to free LoRA</i> .
recovery at matched budget, R	The arm that trains B from its band's basis, at the free reference's rank, minus the frozen arm at its own rank; the recovery at matched budget, R , a robustness check in §3 of the main paper and detailed in its Appendix B. The protocol's criterion is the whole 95% interval of R above $0.7G$. When the interval straddles $0.7G$, the verdict is <i>inconclusive</i> , which the protocols called "shared".
cost of freezing B at equal rank	B trained minus B frozen, at the same rank and initialisation; the arm that trains B holds about twice the parameters.
spread among frozen arms	The best minus the worst frozen arm of the same population. It grows with the number of arms compared.
position effect	The spread among frozen arms, computed over the β -positions only.
swing	bottom 's lead over the best other frozen arm at the low common rate, plus its deficit to the best other frozen arm at the high one. Leads and deficits are differences of means unless marked paired.
<i>learning rates</i>	
common rate	One learning rate for every arm compared. The low and high common rates are $2 \cdot 10^{-3}$ and 10^{-2} at ranks 1 to 4, and 10^{-3} and $2 \cdot 10^{-3}$ at rank 16 (§S23.5).
selected rate	The learning rate an arm is reported at (Appendix C.4 of the main paper).
rate shift	At rank 2, the other arms' selected rate (10^{-2}) divided by bottom 's ($5 \cdot 10^{-3}$): two, on both models. On Llama, the five-seed selection places its size between two and five (§S16.2).
<i>training failures</i>	
diverge	The update norm explodes during training. Norms are logged only at the end of a run, so we read it from a training loss that rises far above its starting value. The criterion is the trajectory, not an absolute value: a large final norm reached without an explosion is not a divergence.
collapse	The score drops sharply while the training loss does not rise, whatever the final update norm.
<i>tests</i>	
marginal pass	A pass whose statistic lies within 15% of its threshold (§S22); marked ^m .

Table S1: Notation and terminology. Every later section uses these terms in this sense only.

budget-matched free LoRA of 9.0 (Table S12): at rank 2, position and freezing are of the same order. Along the spectrum itself, position therefore matters more than the main table suggests.

β	0	0.25	0.5	0.75	1
score	0.6547	0.6368	0.6671	0.6768	0.5928
σ	0.011	0.020	0.019	0.017	0.019

Table S2: Valid-format accuracy of the five positions on Llama at rank 2, 614 items, seeds 2 to 8, one run per cell, each position at its selected rate. $\beta = 0.75$ scores highest. The selected rates are given in the text.

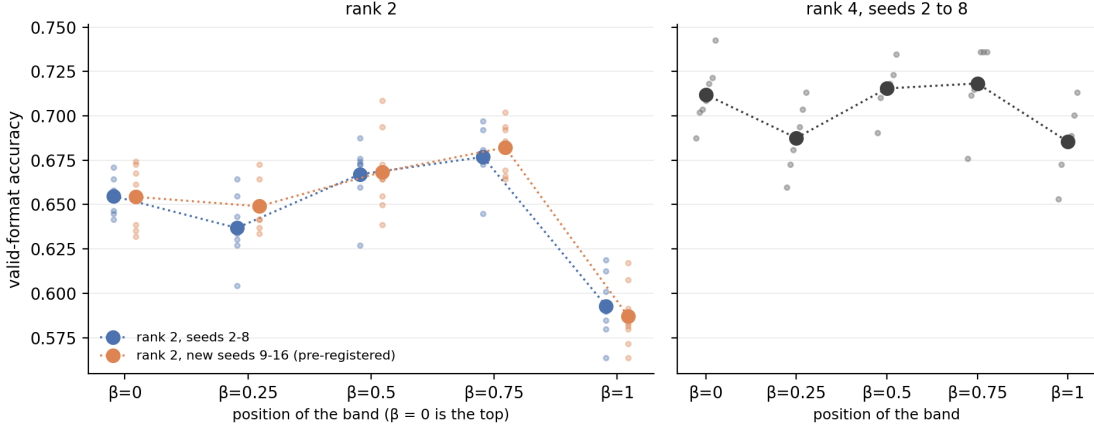


Figure S1: **Valid-format accuracy against band position β on Llama.** Left, rank 2: the first seven seeds and the eight fresh seeds of the pre-registered replication, at the same rates. Right, rank 4: seven seeds. Grey dots are seeds; points are joined by dotted lines because the five positions are discrete. $\beta = 0.75$ is best in both panels, but at rank 4 it separates from neither the top nor its neighbour.

At rank 4, the same sweep on seeds 2 to 8, each position at a rate selected on seeds 0 and 1, gives 0.7120, 0.6875, 0.7154, 0.7182 and 0.6854 from $\beta = 0$ to $\beta = 1$ (Fig. S1, right). $\beta = 0.75$ is again the best of five, but it separates from neither the top (+0.63 points, $[-1.9, +3.2]$, $t = 0.60$) nor its neighbour (+0.28, $[-1.8, +2.4]$, $t = 0.33$; 95 % intervals). Only $\beta = 0.25$ and the bottom sit lower, by about three points. The position effect falls from 8.4 points at rank 2 to 3.3 at rank 4, while the gap to a budget-matched free LoRA stays at 8.4 to 10.1 (Table S12). No β sweep was run at rank 16; there, on OpenBookQA, the spread of the three frozen arms (4.24 points) exceeds the gap (2.08) in point estimate, and the two do not separate (§S23.5). Paired with the free reference on seeds 2 to 6, the position effect at rank 4 is 3.6 points against a gap to a budget-matched free LoRA of 8.4, the best frozen arm there being **top** at its own selected rate (Fig. S13).

At both ranks, the two values of the position effect are the same quantity on two seed sets: seeds 2 to 8, and seeds 2 to 6, where the free reference exists and the comparison is paired. The interior optimum is thus a rank-2 result. What holds at rank 4 is the full ordering of the three effects, confirmed by a pre-registered test with the Σ arm counted (§S11.4), not the location of the best band.

Status: [Post hoc · Descriptive].

Caveats. The rank-4 sweep is descriptive: no test on it was fixed in advance.

S3.3 Llama: $\beta = 0.75$ beats both extremes

On Llama at rank 2, $\beta = 0.75$ beats both extremes of the spectrum, and the pre-registered fresh seeds establish it. On the first seven seeds, $\beta = 0.75$ wins on all seven paired seeds against both the top and the bottom (Table S2, Fig. S1, left). The seven seeds are shared across positions, so the comparison is paired. Against $\beta = 0$, 0.25, 0.5 and 1, the gaps are +2.21, +4.00, +0.98 and +8.40 points, with exact p of 0.024, 0.011, 0.36 and 0.0003 against a Bonferroni threshold of 0.0125 (Table S3). Two of the four pass, against $\beta = 0.25$, a marginal pass, and against the bottom, while the margins against the top and against

$\beta = 0.5$ do not. Without assuming normality, the sign test and the exact Wilcoxon test give $p = 0.016$ against $\beta = 0, 0.25$ and 1 , and nothing against $\beta = 0.5$ ($p = 1.0$ and 0.47). At seven seeds, 0.016 is the smallest p either test can reach, and it lies above the corrected threshold. No non-parametric test can pass at this sample size, which is why the replication below adds eight seeds and tests them on their own. Llama is the model the hypothesis was found on, so nothing pre-specified a direction there: these seven-seed tests are two-sided and Post hoc (Table S3).

A pre-registered replication. Seven seeds leave the margin against the top below the corrected threshold. We therefore ran eight fresh seeds, 9 to 16, under a protocol written and committed before launch (Appendix F of the main paper). Seed 9 had in fact already been run at the selected rates of $\beta = 0$ to 0.75 : the protocol declares it and states that those scores had not been looked at, a declaration the history cannot verify; seed 9 at $\beta = 1$, which existed only at 10^{-2} , was rerun at its selected rate. The protocol keeps the rates of Table S2 unchanged and uses one-sided paired tests in the direction the first seven seeds fixed. It applies Bonferroni over five comparisons, the four other positions and `random_ortho`, and reports all five whichever way they fall. The decision to extend was taken after seeing seeds 2 to 8, which is why the primary test uses the eight fresh seeds alone. On them, $\beta = 0.75$ beats the top ($+2.79$ points, $t = 3.08$ against 2.998 , a marginal pass, $7/8$ seeds), $\beta = 0.25$ ($t = 3.76$) and the bottom ($t = 11.59$, $8/8$). Against $\beta = 0.5$ it does not ($+1.38$ points, $t = 1.09$, adjusted one-sided lower bound -2.4), and against `random_ortho` the margin is not established ($t = 2.75$, §S3.5). The margin against the top is narrow. The exact Wilcoxon test agrees with it ($p = 0.008$, one-sided), but the sign test does not pass the corrected threshold ($p = 0.035$ against 0.01). No seed was added after these. A one-sided test in the direction the first seeds showed is what a confirmatory replication is. The fresh seeds alone carry it, so that its error rate does not depend on the decision to extend.

PiSSA and PiCa argue for the dominant components, MiLoRA and MiCA for the minor ones, and neither is where the optimum sits. Quercia et al. reached the same conclusion on initialisations (§7 of the main paper); here it holds on frozen subspaces, with each position at its own selected rate.

Status: [Pre-registered · Established], on the fresh seeds. Table S3 gives the status of each comparison, and Table S24 that of the non-parametric tests.

Caveats. The optimum’s position is resolved to no better than a quarter of the spectrum. The maximum is interior on the point estimate, and $\beta = 0.75$ beats both extremes on every one of the first seven seeds. Against the top, the first seven seeds do not pass ($t = 3.00$) and the fresh seeds then do. Against the neighbouring position, $\beta = 0.5$, neither the first seven seeds ($4/7$ seeds, $t = 0.99$) nor the fresh seeds ($t = 1.09$) separate it.

S3.4 Qwen: $\beta = 0.75$ is again the best band

On Qwen at rank 2, $\beta = 0.75$ is again the best band, and three of its four comparisons pass. We ran the same sweep on Qwen2.5-0.5B to test the direction Llama had already fixed. On the first five seeds, no comparison reached significance ($t = 1.08$ to 1.79). The decision to extend was taken after seeing those five, so we first wrote down and committed a protocol and then launched ten added seeds. The protocol fixes fifteen paired seeds, each position at its own selected rate, a one-sided paired t per comparison and Bonferroni over the four, and reports all four whichever way they fall. On the fifteen seeds, $\beta = 0.75$ beats $\beta = 0, 0.25$ and 0.5 ($t = 4.22, 5.50$ and 6.20 , against 2.51), but not the bottom ($+2.02$ points, $t = 2.30$, adjusted one-sided lower bound -0.2 ; Table S3).

The ten added seeds alone are seeds the decision to extend could not have influenced. The protocol did not specify a test on them, and we ran this check afterwards. On them, three of the four comparisons pass at $t > 2.685$ ($4.65, 6.35$ and 8.04 , with 9 or 10 seeds of 10), and the bottom resists again ($+2.28$ points, $t = 1.97$, adjusted one-sided lower bound -0.8). Every run in the sweep shares one evaluation size, one validation size, one module set, one clipping threshold, one checker version and one dataset, and no cell holds a replicate. Leaving out any single seed moves the worst t to $3.82, 5.04$ and 5.69 on the three that pass, so no verdict rests on one run.

Where the two models agree. Both models put the maximum inside the spectrum, and neither resolves it to better than a quarter. On Llama the comparison that fails, on the first seven seeds and on the fresh

$\beta = 0.75$ vs	$\beta = 0$	$\beta = 0.25$	$\beta = 0.5$	$\beta = 1$
<i>Llama-3.2-1B, first seven seeds (post hoc)</i>				
gap	+2.21	+4.00	+0.98	+8.40
<i>t</i>	3.00	3.66^m	0.99	7.59
<i>Llama, eight fresh seeds (pre-registered)</i>				
gap	+2.79	+3.32	+1.38	+9.53
<i>t</i>	3.08^m	3.76	1.09	11.59
<i>Qwen2.5-0.5B, fifteen seeds (pre-registered, descriptive)</i>				
gap	+4.03	+2.63	+2.66	+2.02
<i>t</i>	4.22	5.50	6.20	2.30
seeds won	13/15	14/15	14/15	11/15
<i>Qwen, the ten added seeds alone (post hoc)</i>				
gap	+5.37	+2.96	+3.14	+2.28
<i>t</i>	4.65	6.35	8.04	1.97

Table S3: Paired t for $\beta = 0.75$ against each other position: two-sided on the first seven Llama seeds, one-sided on the fresh Llama seeds and on Qwen. What resists is the neighbouring position on Llama and the low extreme on Qwen. Bold passes Bonferroni: $t > 3.52$ at 6 d.f. two-sided over four; $t > 2.51$ at 14 one-sided over four; for the fresh Llama seeds, $t > 2.998$ at 7 d.f. over five (the four positions and **random_ortho**); for the ten added Qwen seeds, $t > 2.685$ at 9 d.f. over four. Block labels give each block’s provenance; bold cells are Established and the others Not established, except in the fifteen-seed Qwen block, which is Descriptive throughout. ^m: a marginal pass (§S22).

ones, is against $\beta = 0.5$; on Qwen it is against $\beta = 1$. The two models agree on the best band but not on the worst: on Qwen the top is lowest (0.470, fifteen seeds), on Llama the bottom (0.593, seeds 2 to 8).

Status: [Pre-registered · Descriptive], on the fifteen seeds. The fifteen-seed population is not confirmatory, because five of its seeds had been seen before the protocol was committed. Table S3 gives the status of each comparison, including the post hoc check on the ten added seeds.

S3.5 Against a random basis: positive, not established

$\beta = 0.75$ leads the random orthonormal basis on both models, but the margin is established on neither under a common correction. **random_ortho**, an orthonormal basis drawn with no reference to W_0 , edges out the other main arms at rank 2 on both models (Tables S14 and S5).

On Llama’s first seven seeds, at the same rate, $\beta = 0.75$ leads it by 2.42 points on six seeds of seven ($t = 2.94$, adjusted interval $[-0.6, +5.5]$; exact Wilcoxon $p = 0.047$, sign test $p = 0.125$). That clears the threshold of a test taken alone, 2.45. But this comparison was added after the four above, and counted in their family it would need 3.71. The interior optimum is therefore not shown to beat a random frozen basis. On the eight fresh seeds of the pre-registered replication, the margin is 3.36 points on seven seeds of eight, with $t = 2.75$ against 2.998 and an adjusted one-sided lower bound of -0.3 : still not established. On all fifteen seeds pooled, it clears the pooled threshold ($t = 3.94$ against 2.624), but the protocol ranks that test second, and we rest no claim on it.

On Qwen the comparison had only seeds 2 to 6, so we ran **random_ortho** on the added seeds, 7 to 16, at its selected rate, under a protocol fixed before launch (Appendix F of the main paper). The protocol fixes one comparison: $\beta = 0.75$ against **random_ortho** on the ten added seeds alone, with a one-sided paired t and a threshold of 1.833. The scores of $\beta = 0.75$ on those seeds were already known; those of **random_ortho** were not. The test passes, marginally: +1.24 points, $t = 2.08$, exact Wilcoxon $p = 0.040$. It passes on little. $\beta = 0.75$ wins six seeds of ten, and the sign test gives $p = 0.38$. The gap is smaller than what a single run moves at a fixed seed (Appendix C.3 of the main paper). On all fifteen seeds, the protocol’s secondary test, the margin falls to +0.48 points ($t = 0.75$, against a one-sided threshold of 2.51 at 14 d.f., Bonferroni over four; adjusted one-sided lower bound -1.1).

The two models were not tested under the same convention. On Llama, `random_ortho` was one of five comparisons (threshold 2.998); on Qwen, it was the only one (1.833). Under Llama’s convention the Qwen test would not pass (2.08 against about 2.82), and under Qwen’s the Llama test would (2.75 against about 1.895). The margin is positive on both models and established on neither under a common correction.

PiCa (Hwang et al., 2026) report the opposite ordering at the dominant end in their own ablation, where the frozen factor is A rather than B . On Gemma-2B, projecting onto the dominant column space beats a random projection by 4.42 points on commonsense reasoning (their Table 5, rank 256, with cross-layer sharing). Here `random_ortho` scores above `top` on Llama at rank 2, in the means (Table S5). The two settings differ in model, task, rank and sharing, and we do not know which of these decides it. At rank 16 on OpenBookQA, a task of the family PiCa reports on, `top` is ahead of `random_ortho` at every rate we ran: by 1.6 points at 10^{-3} , 0.7 at $2 \cdot 10^{-3}$ and 3.5 at their selected rates, in the direction PiCa reports for a frozen A ; these gaps are not tested.

Status: [Post hoc · Not established], under a common correction. Table S24 gives the status of each test in this subsection.

S3.6 Position, or update norm?

The ordering along β is not read off the update norm, although we have not matched norm by construction. Across arms, the score and the update norm covary (Fig. S2, left). The curves cross between 2.2 and 2.9: below that norm `bottom` leads, and above it `bottom` trails. This raises an objection: the axis the score is ordered on could be the norm rather than the position. The position sweep answers it (Fig. S2, right).

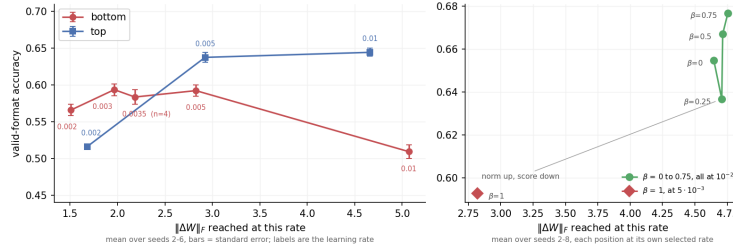


Figure S2: **Valid-format accuracy against update norm.** Left: across arms, means over seeds 2 to 6. Right: the position sweep on its own axes, seeds 2 to 8. Across arms the curves cross between 2.2 and 2.9; along the sweep, the norm rises from 4.646 to 4.755 over the four positions at 10^{-2} while the score does not follow. $\beta = 1$ is drawn apart because it runs at $5 \cdot 10^{-3}$.

Among the four positions that run at the same rate, the update norm rises with β : 4.6463, 4.7074, 4.7145 and 4.7546. The seeds resolve the rise from $\beta = 0$ to 0.25 and from 0.5 to 0.75, but not between the two middle positions. The score is not monotone in the norm: from $\beta = 0$ to $\beta = 0.25$ the norm goes *up* and the score goes *down*, from 0.6547 to 0.6368. That does not rule the norm out, because if there is an optimal update norm, the score is non-monotone in it by construction. It does mean that the ordering along β is not read off the norm. The two middle positions differ by 0.0071 in norm, six times less than the within-position spread of 0.04, while their scores differ by 3.0 points. $\beta = 1$ sits at 2.82 rather than 4.7, and it is also the one position that runs at $5 \cdot 10^{-3}$: its norm gap follows the rate. We therefore do not attribute the interior maximum to the update norm.

Status: [Post hoc · Descriptive].

Caveats. We have not matched norm by construction, and the two covary on both models (Appendix B.4 of the main paper). Rescaling every update to a common norm did not give a usable control: neither of the two norms we tried lets all four arms learn, and Fig. S2 is the descriptive alternative (§S12).

Position. The interior optimum is a rank-2 result: at rank 4 no position is separated from the best (§S3.2). Spectral position and update norm covary on both models, and rescaling every arm to a common norm leaves no norm at which all four learn (above, and §S12).

S4 Secondary: at rank 2 the minor band trails on Llama; no deficit is detected on Qwen2.5-0.5B

On Llama at rank 2, the minor band trails every other main arm; on Qwen2.5-0.5B at rank 2, no such deficit is detected. At rank 16 on OpenBookQA, with Qwen2.5-1.5B, a deficit is established at both common rates (§S23.5), so the contrast between the two models at rank 2 does not extend to every rank, model size and task. This section and §§S23.1, S5 and S6 report the secondary findings summarised in §S1.1. §S4.1 measures the deficit on Llama, §S4.2 looks for it on Qwen, and §S4.3 asks why the gradient points at this band.

S4.1 On Llama, bottom trails every other main arm

On Llama at rank 2, bottom trails each of the other three main arms by at least 5.21 points at its selected rate. At bottom’s former rate, $2 \cdot 10^{-3}$, the smallest gap was 7.85 points (Table S22). Against the best arm, bottom is 6.12 points behind. At that earlier rate the deficit survived removing the two degenerate modules, `q_proj` and `o_proj` (§S6): it reached 12.08 points there, with a t interval of [9.7, 14.5], and that survival is what this control is for (Table S4).

	all 7	5 non-degenerate k,v,gate,up,down
gap, $5 \cdot 10^{-3}$ (selected)	5.21	<i>not computed</i>
gap, $2 \cdot 10^{-3}$ (former)	7.85	12.08
t CI, $2 \cdot 10^{-3}$	[5.2, 10.7]	[9.7, 14.5]

Table S4: Gap in points from `bottom` to the nearest other main arm on Llama at rank 2, with its t confidence interval, on all seven modules and without the degenerate `q_proj` and `o_proj`, at bottom’s selected rate and at its former rate.

Status: [Post hoc · Established]. At bottom’s selected rate, $5 \cdot 10^{-3}$, it trails `top`, `random` and `random_ortho` by 5.21, 5.99 and 6.12 points ($t = 4.39, 5.97$ and 7.42 , each above this family’s 3.96, the first marginally, five seeds of five); the best arm leads it by 6.12 points, [+4.69, +7.88]. On five modules it trails `top` by 5.85 ($t = 5.45$) and `random` by 7.69 ($t = 8.80$).

Caveats. The control does not show that degeneracy was masking the deficit. Dropping `gate_proj` and `up_proj` instead, which are *not* degenerate, gives 11.59 points to the nearest arm (Table S13). Removing any two modules therefore moves the gap. With two sets of modules we cannot separate *degenerate* from *fewer modules*, and the intervals overlap.

S4.2 On Qwen2.5-0.5B, no deficit is detected

On Qwen2.5-0.5B at rank 2, no deficit of the minor band is detected at its selected rate. On Llama, bottom is 6.12 points behind the best arm (§S4.1). On Qwen2.5-0.5B, the frozen arms fall within 2.8 points of one another (Table S5). At its selected rate on Qwen, $5 \cdot 10^{-3}$, bottom passes `top` and `random`. It trails `random_ortho`, the one arm drawn outside the spectrum, by 1.50 points, and this gap is not detected ($t = 0.96$). At $2 \cdot 10^{-3}$, bottom passes `top` by 0.42 points and trails `random_ortho` by 2.41 on every seed ($t = 3.77$). That gap clears the uncorrected threshold (2.78) but not the corrected one for three comparisons (3.96). Against the other arms built from U , the deficit does not reproduce: on Qwen2.5-0.5B, bottom is ahead of `top` and `random` in the means, and neither gap is detected.

This is not a difficulty ceiling. **free** reaches 0.6306 ± 0.0069 , 14.66 points above **bottom** at $2 \cdot 10^{-3}$ and 13.74 above it at its selected $5 \cdot 10^{-3}$. The task is learnable, and the absence of a detected gap between the frozen arms is a result about position.

	Llama	Qwen
free (ref.)	0.7684	0.6306
random_ortho	0.6534	0.5081
random	0.6521	0.4919
top	0.6443	0.4798
bottom , $2 \cdot 10^{-3}$ (common rate)	0.5658	0.4840
bottom , $5 \cdot 10^{-3}$ (selected)	0.5922	0.4932

Table S5: Valid-format accuracy at rank 2 on each model: the free reference and the main arms, with **bottom** at both of its rates. The frozen arms fall within 2.8 points on Qwen.

Status: [Post hoc · Not established], a non-detection. Table S24 gives the status of each comparison.

Caveats. We do not know which property of the architecture separates Qwen from Llama here.

S4.3 The gradient points where the worst arm on Llama writes

Before any training, the gradient puts its energy at the bottom of the spectrum, where the worst arm on Llama writes, and this is not a contradiction. We measure the gradient probe before any training, over every layer, module and decile of the spectrum, for both tasks (Fig. S3). Pooled over the two tasks, the probe places its energy at the bottom of the spectrum, which is where **bottom** operates. Between the tasks we detect no difference: paired by (layer, module), knowledge leads format in 47 of 80 cases, with a sign test $p = 0.15$. That is a non-detection on 80 pairs, not evidence that the profiles agree. The gradient signal-to-noise ratio inside the band does not separate the arms (18.05 against 19.27). It is one of the four ruled-out mechanisms (Table S32).

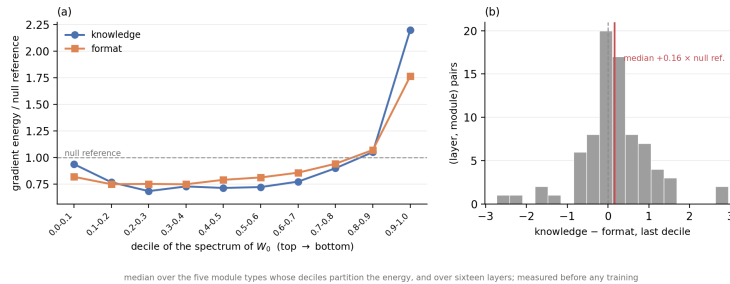


Figure S3: Gradient energy by decile of the spectrum of W_0 , before any training, relative to the null reference. Pooled over the two tasks, the energy concentrates at the bottom of the spectrum; between the tasks, no difference is detected.

A reader is entitled to ask why the band the gradient favours is the band that trains worst. Our own measurements describe the same directions from four other angles. They are ten times stiffer at initialisation (§S23.2.2), and they take half the learning rate (§S16.1). They perturb W_0 proportionally twice as much (Table S10). At a common rate, training there raises the base model’s perplexity by three more units of perplexity than training at the top, though at each arm’s selected rate it forgets less than the top (§S23.1). A large gradient in the minor band, which carries little of W_0 ’s energy, says the loss is sensitive to those directions. Every other measurement says that this sensitivity is expensive to exploit. This is a consistent reading of five measurements, not a mechanism.

Status: [Post hoc · Descriptive]. Table S24 gives the status of the sign test between tasks.

What the format metric measures. A valid format requires every sub-rule at once. At rank 2 on Llama, **bottom**’s deficit sits in the sub-rules that must be computed: against the mean of the other frozen arms, it is 6.9 points on the signature of initials and 3.7 on the number of entities, and less than half a point on the keys and on the order (Table S6); on the valid format as a whole it is 5.5.

arm	n	valid	signature	keys	count	order
top	5	0.644	0.747	0.992	0.952	0.836
bottom	5	0.592	0.676	0.990	0.896	0.843
random	10	0.645	0.736	0.992	0.930	0.853
random_ortho	5	0.653	0.751	0.990	0.916	0.849
free	5	0.768	0.830	0.996	0.967	0.906

Table S6: Valid format and its sub-rules at rank 2 (Llama-3.2-1B, seeds 2 to 6, each arm at its selected rate). **random** counts ten runs: two measurements of the same five seeds.

S5 Secondary: the knowledge task carries signal only at rank 64

The knowledge task separates the arms **only at rank 64**, and there what separates them is **freezing at $2.15\times$ the budget, not position**. §S5.1 covers ranks 1 to 16, and §S5.2 covers rank 64.

S5.1 At ranks 1 to 16, the knowledge task does not separate the arms

At ranks 1, 2, 4 and 16, **no arm exceeds 0.36 on the multiple-choice task, against a chance level of 0.25**. This is the synthetic knowledge task; OpenBookQA at rank 16 is §S23.5. The arms sit above chance and within reach of one another. In this regime the task does not separate the arms, whatever signal it carries. Whether frozen bands differ in how much knowledge they acquire cannot be tested in this regime.

Status: [Post hoc · Descriptive].

S5.2 At rank 64

At rank 64 the task comes alive: the free arm, at $2.15\times$ the budget, leads every frozen arm by 20.8 to 26.6 points at their selected rates, while the frozen arms, above chance, lie within six points of one another (Table S7). Paired tests of **free** give $t = 4.90$ against **top**, 7.08 against **random** and 8.74 against **bottom**, each at its selected rate; at a common 10^{-3} , $t = 4.90$, 6.28 and 4.92 against **top**, **bottom** and **random**.

What this row shows, at $2.15\times$ the budget and with the frozen arms too close to one another to be separated, is that the free arm dominates at rank 64. It does not measure a gap to a budget-matched free LoRA. Whether the three frozen arms differ from one another is not settled: they sit within 5.8 points of each other at their selected rates, and within 2.7 at the common rate.

	rate	mean	gap	paired t
free	10^{-3}	0.5480	—	—
top	10^{-3}	0.3400	−20.8	4.90
random	$2 \cdot 10^{-3}$	0.3065	−24.2	7.08
bottom	$5 \cdot 10^{-4}$	0.2825	−26.6	8.74
<i>chance</i>		0.2500	—	—

Table S7: Four-way multiple-choice accuracy at rank 64, seeds 2 to 6, 400 items, each arm at the rate it selects in validation on seeds 0 and 1, with the gap to **free** in points and the paired t . At a common rate of 10^{-3} the gaps are 18.9 (**random**), 20.8 (**top**) and 21.5 (**bottom**), with $t = 4.92$, 4.90 and 6.28. The free arm leads every frozen arm either way.

Per-arm rates, and a common rate as a secondary line. Each arm runs at the rate it selects in validation on seeds 0 and 1: 10^{-3} for **free** and **top**, $2 \cdot 10^{-3}$ for **random**, and $5 \cdot 10^{-4}$ for **bottom** (Table S7). **bottom**’s selection first sat at the lower edge of its grid, since $2.5 \cdot 10^{-4}$ had been run for **free** only. As Appendix C.4 of the main paper requires, the grid was extended by one step: at $2.5 \cdot 10^{-4}$ on seeds 0 and 1, **bottom** scores 0.3225 in validation, below 0.3550 at $5 \cdot 10^{-4}$, so its selection is an interior maximum.

The first run of this comparison used one common rate, 10^{-3} , for every arm; we keep it as a secondary line. It rested on lower rates run after the first selection: $5 \cdot 10^{-4}$ on seven seeds for **bottom** and **free** and on two for **top** and **random**, and $2.5 \cdot 10^{-4}$ on four seeds for **free** only. Across those seeds, **bottom**’s validation accuracy peaks at 10^{-3} (0.366, against 0.335 at $5 \cdot 10^{-4}$ and 0.316 at $2 \cdot 10^{-3}$); 10^{-3} is also the best rate for **top** and **free**, and ties with $2 \cdot 10^{-3}$ for **random** (0.399 at both). At that common rate the gaps to **free** are 18.9 to 21.5 points ($t = 4.90, 6.28$ and 4.92 against **top**, **bottom** and **random**): the free arm leads every frozen arm under either choice of rates.

Status: [Post hoc · Established]: the three tests form a family, with Bonferroni threshold $t > 3.96$ at 4 d.f., two-sided, and all three pass, at the selected rates as at the common rate.

Caveats. The per-arm rates are selected on seeds 0 and 1; the common rate of the secondary line was chosen on a validation split of runs that include the reported seeds (Appendix C.4 of the main paper). The frozen arms may sit near a floor. They score 0.28 to 0.34 at their selected rates, and 0.33 to 0.36 at the common rate, against a chance level of 0.25; at lower ranks the task does not separate the arms (§S5.1). That the task does not separate them here is therefore also compatible with all three being near a floor. What that reading does not explain is **free** at 0.5480 on the same task and the same seeds. The task is learnable, and the distance between the free arm and all three frozen arms is the finding of this row.

S6 Secondary: a spectral criterion for excluding modules

A fixed list fails on OLMo; a spectral threshold separates the same two groups on all three models, although its window was read off those models. Adapting a module whose band carries almost no energy gives a frozen arm directions the pretrained weight barely uses; whether that harms adaptation is not something this criterion establishes. Which modules those are is usually decided by a hard-coded list.

The criterion. Compute $\sigma_{\min}/\sigma_{\max}$ per module from the same reduced SVD the method already needs, and exclude anything below 10^{-4} . We call the modules below that threshold *degenerate*. The criterion costs nothing beyond a decomposition that has to be done anyway. It is the difference between measuring a subspace and measuring directions that carry almost none of the weight’s energy.

$\sigma_{\min}/\sigma_{\max}$	Llama	Qwen	OLMo
q_proj	6.0e-6	3.4e-5	7.7e-7
o_proj	2.4e-5	7.7e-5	2.3e-5
k_proj	1.4e-2	4.5e-2	1.6e-5
v_proj	1.3e-1	2.9e-1	5.2e-5
MLP (3 modules)	2–6e-2	7–11e-2	2–5e-2

Table S8: $\sigma_{\min}/\sigma_{\max}$ per module, median over layers; bold marks values below 10^{-4} . Only attention projections fall below the threshold, and which ones differs by model.

Which modules fall below the threshold. They are always attention projections and never MLP ones, but which ones changes from model to model (Table S8). There are two on Llama and Qwen, and all four on OLMo (OLMo Team et al., 2025). A paper that excludes **q_proj** and **o_proj** by name is right on two of our three models and wrong on the third.

The threshold window. 10^{-4} sits two orders of magnitude above the fp32 noise floor of an SVD on matrices of this size, and two to three orders below the smallest well-conditioned module we measure. The

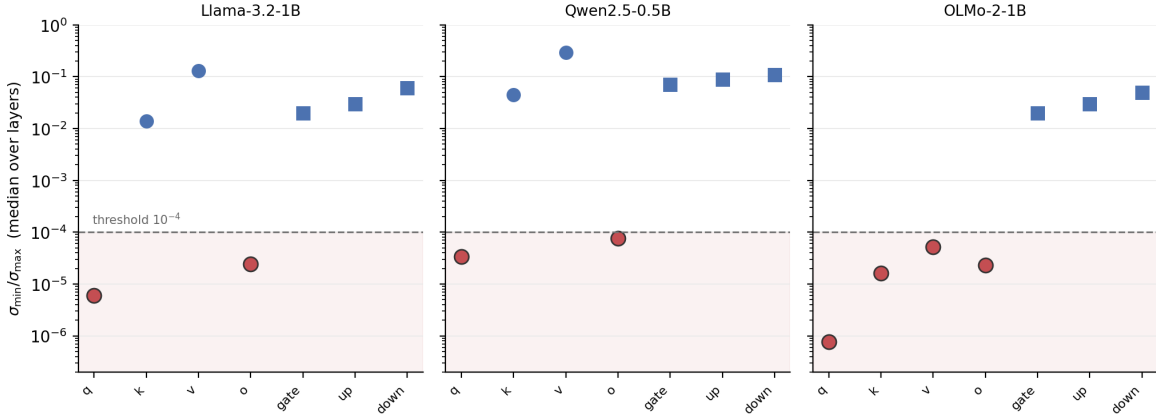


Figure S4: $\sigma_{\min}/\sigma_{\max}$ per module on the three models examined, median over layers. Circles are attention projections and squares MLP ones; red marks what falls below 10^{-4} . On all three models, the gap between the two groups spans more than two and a half orders of magnitude, with nothing in between, and the modules it catches are not the same from one model to the next: which modules to exclude is read from the spectrum of each model, not from a fixed list.

median for q on OLMo-2-1B, about $8 \cdot 10^{-7}$, lies at that floor: its exact value is not resolved, only that it falls far below the threshold. Any cut in that window separates the same two groups. No value of the threshold between 10^{-4} and 10^{-2} would change the classification on any of the three models (Fig. S4).

Status: [Post hoc · Descriptive].

Caveats. The window the threshold sits in was read off these same three models. The values compared are medians over the layers of a module, so a layer-by-layer classification is not what we check. ‘Degenerate’ has no independent definition here. What transports is the *separation* into two groups and which modules fall in the lower one, not a validated label.

Our own experiments include the degenerate modules. The main arms adapt all seven projections, q and o included, which this criterion flags on Llama. The five-module row of Table S12 is *not* that check, because it drops gate and up (Table S13). Removing q and o instead leaves the deficit of the minor band intact and larger (5.85 points behind top at $5 \cdot 10^{-3}$, $t = 5.45$, against 5.21 on seven modules; Table S4), though dropping any two modules moves it (§S4.1). We have no budget-matched reference on that module set, so the comparison of the gap with the spread is not available there. The rows on seven modules are kept because they are the setting the compared methods use. On OLMo, where four of the seven modules are degenerate, the rate sweep on the three remaining modules found no interior optimum, and the model does not enter the comparisons (§S23.6).

S7 What the constraint actually constrains

A band energy measured on the side of a frozen factor equals 1 by construction, so wherever a factor is frozen it is vacuous as evidence.

Remark S1. If $\Delta W = BA \neq 0$ for any fixed B , the fraction of $\|\Delta W\|_F^2$ carried by the column space of B equals 1 identically. At initialisation, where $\Delta W = 0$ in every arm, the fraction is undefined.

Proof. Every column of ΔW is a linear combination of the columns of B , so $\text{col}(\Delta W) \subseteq \text{col}(B)$. The orthogonal projector P onto $\text{col}(B)$ therefore satisfies $P\Delta W = \Delta W$, and $\|P\Delta W\|_F^2 = \|\Delta W\|_F^2$. The ratio is 1 with no assumption on B beyond being fixed, none on A beyond $BA \neq 0$, and none at all on the data, the optimiser or the rank. $\square$

We write E_{left} for the fraction of $\|\Delta W\|_F^2$ carried by the band’s left singular directions u_i , and E_{right} for the fraction carried by its right singular directions v_i . $U_{\mathcal{I}}$ denotes the columns of U in the band $\mathcal{I}$. In a frozen arm, E_{left} is the fraction of Remark S1.

Scope of the remark. The remark does not require B to be orthonormal, so it also covers frozen factors that carry Σ , such as LoRA-XS’s, whose fixed factors include Σ (Appendix C.2 of the main paper), and our `top_sigma` ($B = U_r \sqrt{\Sigma_r}$). A method that freezes *both* factors, $\Delta W = U_r C V_r^\top$ with a trained $r \times r$ core C , has both fractions equal to 1, by the same argument applied to $B = U_r$ and to $B^\top = V_r^\top$. LoRA-XS is such a method, with $\Delta W = U_r \Sigma_r C V_r^\top$, so its fractions are 1 identically, vacuous on both sides (Table S9). PiSSA and MiLoRA initialise rather than freeze, so neither of their fractions is fixed.

	freezes	E_{left}	E_{right}
LoRA-XS	both	= 1	= 1
MiCA	$B = U_{\mathcal{I}}$	= 1	free
PiCa	$A = V_r^\top$	free	= 1
this work	$B = U_{\mathcal{I}}$	= 1	free
PiSSA, MiLoRA	neither	free	free

Table S9: Which band energies each method fixes: = 1 means identically one, by construction, because the corresponding factor is frozen; ‘free’ means the fraction is not fixed. Wherever a factor is frozen, the band energy on that side cannot serve as evidence.

Who reports this quantity. We do not claim any published paper reports this number. We checked the three methods the proposition or its mirror image covers. MiCA freezes B and reports no such measurement. PiCa (Hwang et al., 2026) freezes the other factor: in PyTorch’s convention its fixed projector is $A = V_r^\top$, the leading right singular vectors, which its paper, writing weights input by output, calls the column space; the mirror image of Remark S1 fixes its E_{right} instead. Its validation (Theorem 1 and Appendix C.2 of that paper) measures the alignment between U_r and the singular vectors of an independent, *unconstrained* full fine-tuning update. That is not the energy of PiCa’s own constrained update: it is a different and valid quantity. LoRA-XS freezes both factors and does project ΔW onto spectral bands. But it projects the ΔW of *full*, unconstrained fine-tuning, and it reads accuracy off the projection rather than energy: a valid measurement, and not the one the proposition rules out. OPLoRA (Xiong and Xie, 2026) reports the complementary quantity, the share of update energy that falls in the dominant subspace. For its own projected update, that share is zero by construction, up to its low-rank SVD approximation. Its comparison with unconstrained baselines stays informative, but its own value could not have been otherwise. The quantity is available and it looks like evidence, but wherever the corresponding factor is fixed it cannot be anything but 1, or 0 on the complement. The same instrument does separate the bands on a planted target (§S12).

What the construction fixes. The construction pins the column space of the update, and nothing else. It does not adapt the components $\sigma_i u_i v_i^\top$. It writes along the directions u_i of the band, while reading along directions that nothing constrains.

On the right-hand side the fraction is free. We measure $E_{\text{right}} = 0.053$, measured over every adapted module, against a null reference $r/d_{\text{in}} = 0.00098$: fifty-four times the null. The reference is computed per module; 0.00098 is its value on the six modules whose input dimension is 2048, and it is four times smaller on `down_proj`, whose input dimension is 8192.

The factor A reads preferentially along the directions v_i of the same band B writes into. Part of this is inherited, not learned. A starts at zero and its gradient is $(\alpha/r) U_{\mathcal{I}}^\top \nabla_W L$, so its rows are built from what the loss gradient sends through the band. That this lands on the v_i of the *same* band is not implied by the construction: it is a property of the data and of the gradient, not an identity.

Status: Remark S1 is proved, so it carries no tag; the measurement of E_{right} is [Post hoc · Descriptive].

Caveats. We do not measure how much of the concentration of E_{right} is already present at the first step.

S8 The relative perturbation

Relative to the norm of the weight it modifies, bottom’s update perturbs far more than top’s.

Writing into the minor band, which carries little of W_0 ’s energy, moves the weight proportionally more (Table S10).

arm	ratio	95 % CI
bottom	5.13	[3.83, 7.22]
top	2.64	[1.85, 3.71]
free	0.85	[0.46, 1.24]

Table S10: Relative perturbation per arm, with its 95 % confidence interval. **bottom**’s ratio is about twice **top**’s.

Status: [Post hoc · Descriptive].

Caveats. The intervals are descriptive, not a test. The denominator is the energy W_0 holds in the adapted band, which is small for **bottom** by construction, so disjoint intervals are close to automatic here.

S9 Restoring Σ on the dominant band

Our arms differ from PiSSA in two ways: they freeze a factor, and they drop Σ . The arm **top_sigma** keeps the freezing and restores Σ : $B = U_r \sqrt{\Sigma_r}$ on the dominant band, frozen, with A trained from zero at the same budget as **top**. Section S9.1 reports its selected rate, §S9.2 its score, and §S9.3 a single scalar per module that takes the place of Σ .

S9.1 The selected rate

Restoring Σ moves the dominant band’s selected rate from 10^{-2} to $5 \cdot 10^{-3}$, the minor band’s selected rate. Under a protocol written before launch (Appendix F of the main paper), the rate of **top_sigma** was swept on seeds 0 and 1 over ten values from $2 \cdot 10^{-5}$ to $2 \cdot 10^{-2}$. The conclusion was fixed in advance. A selected rate at 10^{-3} or below would attribute to Σ the reversal of the rate shift relative to the direction reported for PiSSA (Lee et al., 2026), one at 10^{-2} or above would exclude Σ , and anything between would be reported as partial, without attribution.

The sweep selects $5 \cdot 10^{-3}$, an interior maximum (validation 0.7010, against 0.6280 at $2 \cdot 10^{-3}$ and 0.5950 at 10^{-2}). This is the third case, reported without attribution.

Status: [Pre-registered · Descriptive]; the protocol fixed a decision rule, not a test.

Caveats. Two limits of that protocol are worth naming. Its attribution threshold (10^{-3} or below) lies below the minor band’s selected rate, $5 \cdot 10^{-3}$, by design: only a dominant band that needs a lower rate than the minor band would reproduce the direction reported for PiSSA.

top_sigma lands exactly on **bottom**’s selected rate, $5 \cdot 10^{-3}$, and the protocol’s verdict stays partial. Part of the move is expected, since $\sqrt{\Sigma}$ enlarges every step at a common rate by the factor measured in §S13.

S9.2 The score

On the reported seeds, restoring Σ also moves the score: top_sigma beats top and random_ortho. At its selected rate, **top_sigma** scores 0.6804 on seeds 2 to 6, against 0.6443 for **top** at its selected rate. That is +3.61 points, with paired $t = 4.22$ on five seeds against a two-sided threshold of 2.776, ahead on every

seed. It also passes **random_ortho** (+2.70 points, $t = 12.79$, 5/5). On Llama, the dominant band with Σ restored is therefore the one arm whose margin over **random_ortho** is established, in a post hoc test; the same band without Σ does not beat **random_ortho**. It does not separate from $\beta = 0.75$ (+0.23 points, $t = 0.27$). Counting this arm, the gap to a budget-matched free LoRA at rank 2 falls from 11.5 to 8.80 points, which is the gap between **free** at $r = 1$ and **top_sigma**. What a frozen arm gives up therefore depends on how the frozen factor is scaled, not only on where its band sits. At rank 4, under the pre-registered protocol of §S11.4, **top_sigma** gains 2.82 points over **top** without overtaking **random**; that its advantage shrinks with rank is descriptive only.

Status: [Post hoc · Established]. The protocol fixed the rate decision and nothing else, and we ran these comparisons afterwards on seeds it had already named. Table S24 gives the status of each comparison.

S9.3 A per-module scalar

A single scalar per module recovers most of Σ 's gain over **top (2.31 of 3.61 points, 64%, $t = 3.32$, five seeds of five); the remainder (1.30 points, $t = 2.21$ against 2.776) is neither established nor excluded at five seeds; two replications on fresh seeds settle it, the second establishing an equivalence within one point on 40 seeds (§S9.4).** The arm **top_scalar** keeps **top**'s frozen basis U_r and replaces $\sqrt{\Sigma_r}$ by one scalar per module, $\sqrt{\bar{\sigma}}$, where $\bar{\sigma}$ is the mean singular value of the band: $B = \sqrt{\bar{\sigma}} U_r$. This matches the Frobenius norm of $U_r \sqrt{\Sigma_r}$, so the two arms differ only in how the scale is spread across the band's directions. It ran at rank 2 on Llama, under a protocol written before launch (Appendix F of the main paper); its selected rate is $5 \cdot 10^{-3}$, as for **top_sigma**, and on seeds 2 to 6 it scores 0.6674. The protocol's primary test is the scalar's gain over **top**, g . Its secondary reading compares g with 2.4 points, and at 2.31 it reads *shared*; the rule we registered would have read the non-significant remainder as an attribution to scale, and we report both readings. Under Adam, a fixed scalar on the frozen factor rescales every step on ΔW by that scalar, as a per-module learning rate would. The result is consistent with the scale of the frozen factor acting largely as a per-module rate on this arm, at rank 2 on Llama, and does not establish it.

Status: the gain g is [Pre-registered · Established]; the remainder is [Pre-registered · Not established].

The replications on fresh seeds. **top_scalar** keeps **top**'s basis and replaces $\sqrt{\Sigma_r}$ by a single scalar per module of the same Frobenius norm, so it differs from **top_sigma** only in how the scale is spread over the band's directions. That spread is sizeable: within the dominant band at rank 2, the median of σ_1/σ_2 ranges from 1.17 to 2.12 across module types, v alone being close to one (1.03; Table S11). On five seeds **top_scalar** recovered most of Σ 's gain over **top**, 64% in point estimate, with a remainder neither established nor excluded (above). Two replications on fresh seeds settle it. On 15 seeds, **top_sigma** minus **top_scalar** is -0.27 points ($t = -0.54$), short of equivalence. On 40 seeds, sized by a power calculation written before the runs, it is $+0.02$ points, with a 90% interval of $[-0.57, +0.60]$ inside the pre-registered margin of one point, and **equivalence is established**. Σ thus acts on a frozen factor as a per-module scale, approximately a per-module learning rate.

A per-module learning rate comes within about a point. Moving the scalar from B into a per-module learning rate, under a pre-registered protocol on the same 40 seeds, reproduces **top_scalar** within about one point. The difference, -0.83 ($[-1.64, -0.02]$), passes the protocol's threshold only marginally ($t = -2.08$ against 2.023); equivalence within one point is not established, and the gap keeps its size on the 13 seeds that share a GPU model. A diagnostic run on one seed locates it, in a post hoc reading (§S9.4): with no clipping, $\epsilon = 10^{-20}$ and fp32, the two arms follow the same trajectory to fp32 round-off over the first steps, while global clipping, Adam's usual ϵ and bf16 each separate them within one or two steps. The two parametrisations are thus the same optimisation problem, and each of these three factors breaks the identity. Chaotic amplification of small differences alone would move the two arms apart in either direction from seed to seed and average out; the gap of -0.83 points over 40 seeds is systematic, so it more likely reflects a systematic effect of clipping or ϵ on the effective step size. This one-seed diagnostic does not separate the three factors. The pre-registered rule, which compared the trajectories at the end of training, could not conclude, because two identical runs themselves diverge by then. The magnitude reading of Zhang et al. (2025b) is consistent with these results, for frozen factors, to within about a point.

Caveats. The equivalence between the scalar and a per-module learning rate under Adam is approximate, since gradients are clipped at 1.0. The result holds for this arm, at rank 2, on Llama, and we do not generalise it.

S9.4 Further records on Σ and the per-module scale

With Σ , the direction reverses. Under Adam, multiplying the frozen factor by a constant multiplies every step on ΔW by it, up to ϵ and clipping, so the scale of the frozen factor acts as a per-module learning rate, and $B = U_r D$ gives each direction the rate ηd_i (§S10). Zhang et al. (2026a) prove a related bound for initialisations that are then trained (their Theorem 4.1). The prediction written before the runs was that, with Σ restored, the minor band would select a strictly higher rate. With $B = U_r \sqrt{\Sigma_r}$ frozen, the dominant band (**top_sigma**) selects $5 \cdot 10^{-3}$ and the minor band (**bottom_sigma**) $2 \cdot 10^{-2}$, with all seven modules and without the two degenerate ones (Fig. S8). The reversal agrees with LoRA-XS (Balazy et al., 2024), which carries Σ in its frozen factors, with the lower rate PiSSA needs in Lee et al. (2026), and with the higher curvature of principal directions reported under PiSSA in reinforcement learning with verifiable rewards (Zhu et al., 2025).

One scale per module accounts for Σ . A single scalar per module, of the same norm as $\sqrt{\Sigma_r}$, reproduces **top_sigma** within one point on 40 seeds (equivalence established), and moving that scalar into a per-module learning rate reproduces it within about a point (§S9.3).

A per-module scale recovers part of the gap with B frozen. On Llama at rank 2, one scalar per module on the frozen dominant band (**top_scalar**) adds 2.31 points to **top**, against a gap to budget-matched free LoRA of 12.41 points (§S9.3).

Σ and the per-module scalar. In selection, **top_sigma** peaks at $5 \cdot 10^{-3}$ (0.701 in validation) and **bottom_sigma** at $2 \cdot 10^{-2}$ (0.555), collapsing from $5 \cdot 10^{-2}$; with the five non-degenerate modules, $5 \cdot 10^{-3}$ (0.643) and $2 \cdot 10^{-2}$ (0.533). The two replications of **top_sigma** against **top_scalar** use fresh seeds and are analysed separately, never pooled. On seeds 50 to 64, the scores are 0.6631 and 0.6658: the difference, -0.27 points ($t = -0.54$), is neither established nor equivalent within one point (its 90 % interval, $[-1.16, +0.61]$, crosses -1). On seeds 65 to 104, sized so that a true difference of zero would be declared equivalent nine times in ten, the scores are 0.6660 and 0.6658, the difference $+0.02$ points ($t = 0.05$), and the 90 % interval, $[-0.57, +0.60]$, lies inside ± 1 point: equivalence is established.

A per-module learning rate. **top_lrscale** keeps $B = U_r$ and multiplies A 's learning rate, module by module, by the scalar **top_scalar** puts in B ; a test checks that, with no clipping and $\epsilon \rightarrow 0$, the two follow the same trajectory on a small model. On the 40 seeds of §S9.3, at **top_scalar**'s rate, pinned to one GPU model: 0.6575 against 0.6658, a difference of -0.83 points (95 % interval $[-1.64, -0.02]$, 90 % interval $[-1.50, -0.16]$, $t = -2.08$ against a threshold of 2.023). The difference is established, marginally; equivalence within one point is not. The runs of **top_scalar** ran on four GPU models; on the 13 seeds where both arms share one, the difference is -1.00 ($t = -1.84$, 12 d.f.), the same size. Clipping fires on 5 % of **top_lrscale**'s steps (mean pre-clip norm 0.32, maximum 24.8); for **top_scalar** only the mean pre-clip norm was recorded (0.54), so the two cannot be compared step by step. Status: [Pre-registered · Established] (a difference).

The trajectories of the per-module rate and the scalar. The protocol (PROTOCOL_lrscale_trajectory.md) trains **top_scalar** and **top_lrscale** on one fresh seed at $5 \cdot 10^{-3}$, saving both factors at every step, under five conditions: no clipping (a threshold of 10^6), $\epsilon = 10^{-20}$ and fp32 (C0); C0 with clipping at 1.0; C0 with $\epsilon = 10^{-8}$; C0 in bf16; and the paper's setting. A control reruns **top_scalar** in C0. The measure is $D = \|\Delta W_{\text{lrscale}} - \Delta W_{\text{scalar}}\|_F / \|\Delta W_{\text{scalar}}\|_F$ over all modules, computed exactly from $r \times r$ products. The two control runs coincide exactly over the first eight steps, then diverge ($8 \cdot 10^{-7}$ at step 16, 0.26 at the end). In C0, D stays at fp32 round-off over the first sixteen steps ($4 \cdot 10^{-8}$ to $3 \cdot 10^{-6}$); with clipping it reaches 10^{-3} at step 2, with the usual $\epsilon 3 \cdot 10^{-3}$ at step 1, and in bf16 $9 \cdot 10^{-3}$ at step 2. The pre-registered rule compared final distances with ten times the control's, 0.26; no final distance can pass it, so it is not conclusive. The reading of the first steps is post hoc.

Singular values inside the bands. Table S11 gives, per module type, the spread of the singular values inside the frozen bands and the mean scale that Σ would give them.

module	σ_1/σ_2	σ_{m-1}/σ_m	σ_1/σ_{16}	$\sqrt{\sigma}$, top	$\sqrt{\sigma}$, bottom
q	2.12	5.24	2.77	2.815	0.0137
k	1.82	1.16	2.50	2.472	0.3659
v	1.03	1.06	1.17	1.029	0.4024
o	1.22	3.76	1.51	1.593	0.0120
gate	2.09	1.06	2.93	2.997	0.5744
up	1.17	1.11	1.45	1.929	0.5278
down	1.22	1.08	1.72	2.123	0.4628

Table S11: Singular values of Llama-3.2-1B inside the frozen bands, median over the 16 layers: the spread in the dominant band at rank 2 (σ_1/σ_2), in the minor band at rank 2 (σ_{m-1}/σ_m), in the dominant band at rank 16 (σ_1/σ_{16}), and the mean of $\sqrt{\sigma}$ over each rank-2 band. On q and o, the minor band’s scale is about 0.01: with Σ , these modules barely move.

S10 Why the scale of a frozen factor acts as a learning rate

Under Adam, a frozen factor $B = U_r D$ gives the step along its i -th direction a learning rate proportional to d_i . Take one module, $\Delta W = \frac{\alpha}{r} B A$, with $B \in \mathbb{R}^{d_{\text{out}} \times r}$ frozen and A trained by Adam from $A_0 = 0$, and write $B = U_r D$ with U_r orthonormal and $D = \text{diag}(d_1, \dots, d_r) \succ 0$. With $G = \nabla_{\Delta W} \mathcal{L}$, the gradient with respect to A is $\nabla_A \mathcal{L} = \frac{\alpha}{r} D U_r^\top G$: its i -th row is d_i times the i -th row of $\frac{\alpha}{r} U_r^\top G$. Adam’s step on a coordinate, $-\eta \hat{m}/\sqrt{\hat{v}}$, is unchanged when that coordinate’s gradients are all multiplied by the same positive constant, since $\hat{m}$ scales by the constant and $\sqrt{\hat{v}}$ by its absolute value. At a given state, the step δA_i on the i -th row therefore does not depend on d_i , and the step on the update is

$$\delta(\Delta W) = \frac{\alpha}{r} \sum_{i=1}^r d_i u_i \delta A_i,$$

so that direction u_i moves with an effective learning rate ηd_i .

Three consequences. Multiplying B by a constant c multiplies the module’s learning rate by c , which is what `top_scalar` does with $c = \sqrt{\sigma}$; that its selected rate and its score match `top_sigma`’s within the equivalence margin (§S16.3) says that, in our setting, the spread of Σ across a band’s directions matters less than its mean scale. With $B = U_r \sqrt{\Sigma_r}$, direction i moves at $\eta \sqrt{\sigma_i}$: faster on the dominant band than on the minor one. If $\sqrt{\sigma_i}$ exceeds one on the dominant band and falls below one on the minor band, then, holding the rest fixed, `top_sigma` should select a lower η than `top`, and `bottom_sigma` a higher one than `bottom`—the direction observed ($5 \cdot 10^{-3}$ against 10^{-2} , and $2 \cdot 10^{-2}$ against $5 \cdot 10^{-3}$, Fig. S8). On Llama-3.2-1B the condition holds: $\sqrt{\sigma}$ exceeds one on the dominant band at rank 2 in 110 of the 112 modules, and falls below one on the minor band in all 112 (Table S11). A related bound, for factors that are then trained and on the first step, is Theorem 4.1 of Zhang et al. (2026a); that tuning α under Adam is roughly tuning the learning rate was noted by Hu et al. (2022).

Where it fails. The equivalence is exact only if $\epsilon \ll \sqrt{\hat{v}}$ and no clipping acts. It breaks first where the scaled gradients are tiny: on a numerically degenerate minor band, σ_i is near zero, and ϵ dominates $\sqrt{\hat{v}}$ (§S6). Clipping on the global norm couples every module, so a large scale on one module shortens the steps of all the others; and a decoupled weight decay on A is not rescaled with B . The argument predicts a direction, not a size: the ratio of the selected rates need not equal the ratio of the scales, since the grid is coarse and clipping acts.

What it does not explain. Without Σ , every direction of every frozen band has $d_i = 1$, and the argument is silent: that the minor band then selects a lower rate is a matter of the curvature along its directions, which we measure (§S23.3) but do not derive.

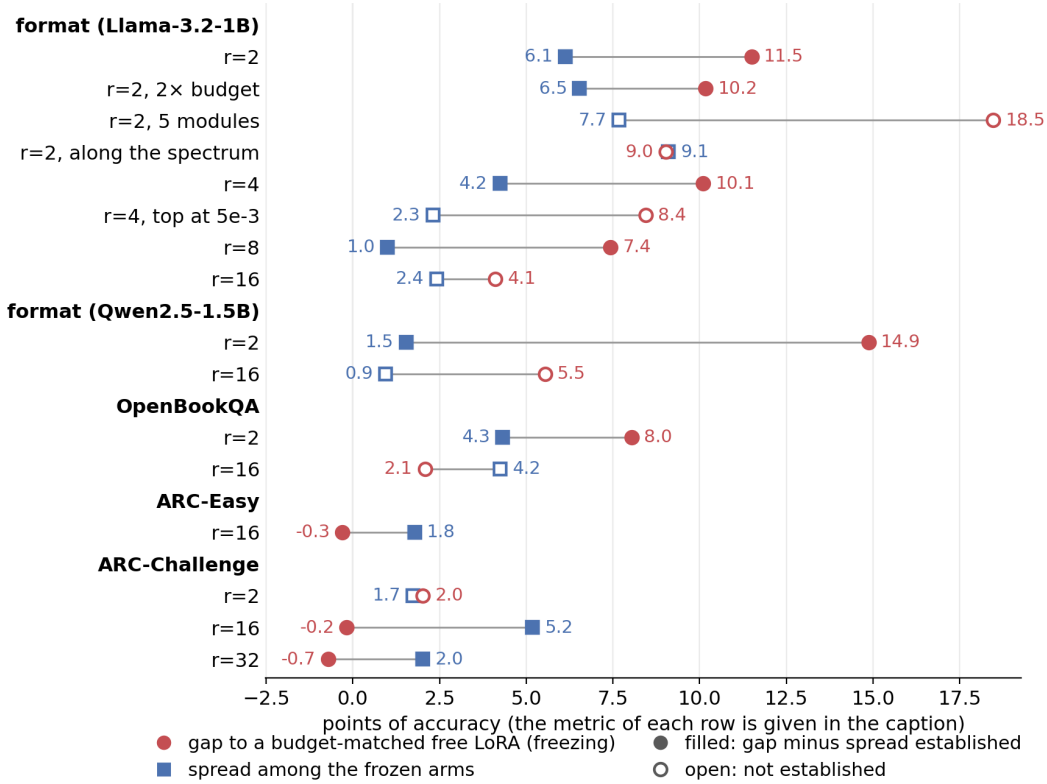


Figure S5: **Gap to free LoRA and spread among frozen arms, population by population.** Rows are grouped by task (the format task on Llama-3.2-1B, the format task on Qwen2.5-1.5B, OpenBookQA, ARC-Easy, ARC-Challenge), then by rank, with each arm at its selected rate. Red disc: gap to a budget-matched free LoRA; blue square: spread among the frozen arms. Filled markers: gap minus spread is established, in either direction; open markers: it is not, or it was not tested. The free reference holds 7.5 % more trainable parameters than the frozen arms on Llama-3.2-1B at every rank; on Qwen2.5-1.5B it holds 13.4 % more at rank 2 and 0.8 % fewer at ranks 16 and 32. Metrics: valid-format accuracy on the format task, multiple-choice accuracy on OpenBookQA and ARC. Rows that carry no claim are drawn in the full version, Fig. S6. Its rank-4 rows are on seeds 2 to 6. The pre-registered rank-4 measure, on seeds 2 to 8, gives a gap of 10.00 and a spread of 4.56 (Fig. 9 of the main paper).

S11 Gap to free LoRA and spread among frozen arms

In the three populations at ranks 2 and 4 where the budgets can be matched, the gap to free LoRA is 1.9 to 2.4 times the spread among the main frozen arms. Along the spectrum at rank 2 the two are of the same order, and counting the Σ arm there brings them level; at rank 4, a pre-registered test with the Σ arm counted confirms that the gap exceeds the spread. At rank 16 on OpenBookQA the two do not separate (§S23.5). §S11.2 compares the two quantities population by population, §S11.3 shows that the budget works against the free reference, §S11.4 counts the Σ arm at ranks 2 and 4, and §S11.5 gives the rows where the free reference holds 2.15× the budget. Both quantities are defined in Table S1. The spread is a maximum minus a minimum, so it grows with the number of arms: each new arm can only widen it.

S11.1 Gap minus spread across populations

The *spread among frozen arms* is the difference between the best and the worst frozen arm, each at its selected rate. Fig. S5 draws it beside the gap to free LoRA, and we summarise the two by their difference, *gap minus*

spread. The summary is descriptive. It compares a minimum, the gap of the best frozen arm, with a range that grows with the number of arms compared, and noise inflates that range (below). Counting `top_sigma` as a fifth frozen arm at rank 2, for instance, brings the gap and the spread level (8.80 against 8.82 points). With that caveat, the gap exceeds the spread on the format task at ranks 2 to 8, with both models at rank 2, on OpenBookQA at rank 2 and at twice the training budget. At rank 16 the two do not separate on the format task or on OpenBookQA, and on both ARC benchmarks at rank 16, and on ARC-Challenge at rank 32, the spread exceeds the gap (Fig. S14b). Table S24 gives every interval and test.

The range statistic. Noise inflates the spread between the best and the worst frozen arm, which biases gap minus spread towards the spread. A null simulation keeps each population’s free reference, gives every frozen arm the same true mean, and draws the noise from the observed variance between seeds (Table S15). The two reversals on ARC fall far outside what identical arms produce ($p < 10^{-4}$). At rank 8, by contrast, the spread among frozen arms is at the level of noise ($P = 0.64$), so the order there rests entirely on the cost of freezing. The simulation is slightly liberal: under a true null it gives $p < 0.05$ in 6.5 % of datasets.

S11.2 Population by population

Among the main arms, the gap to free LoRA exceeds the spread among frozen arms in each of the three populations at ranks 2 and 4 where the budgets can be matched, by a factor of 1.9 to 2.4. Rank 16 on OpenBookQA is a fourth budget-matched population, where the two do not separate (§S23.5). Table S12 gives both quantities for every population, and Fig. S6 draws them. At rank 2, `free` at $r = 1$ holds 7.5 % more parameters than the frozen arms at $r = 2$. Its gap to each frozen arm depends on the arm: 12.4 points against `top`, 11.5 against `random_ortho` and 17.6 against `bottom`.

The gap to a budget-matched free LoRA, against the best frozen arm, is the smallest of these, 11.5 points, against a spread among frozen arms of 6.1: a ratio of 1.9. At rank 4, with `free` at $r = 2$, the two quantities are 10.1 and 4.2, a ratio of 2.4; under `top`’s 614-item selection they are 8.4 and 2.3, a ratio of 3.7. On five modules, where `free` holds 72 % of the budget (§S11.3), they are 18.5 and 7.7, a ratio of 2.4.

A paired bootstrap over the five seeds, recomputing the best and the worst arm in every resample, gives gap minus spread = +5.4 points [+2.3, +7.0] at rank 2, with `bottom` at its selected rate, and +5.9 [+2.7, +7.8] at rank 4 among the main arms (95 % percentile intervals, approximate at five seeds).

The five-module population is not bootstrapped.

	gap	spread	ratio
<i>budget-matched</i>			
rank 2 (free $r=1$, +7.5 %)	11.5	6.1	1.9
rank 4 (free $r=2$, +7.5 %)	10.1	4.2	2.4
<code>top</code> , 614-item selection	8.4	2.3	3.7
5 modules (free at 72 %)	18.5	7.7	2.4
2 modules (exact)	42.5	—	
along the spectrum, rank 2	9.0	9.1	1.0
<i>free at 2.15× the budget, shown, not relied on</i>			
rank 1	24.0	7.4	3.2
rank 64, knowledge (§S5.2)	20.8	5.8	3.6

Table S12: Gap to free LoRA (*gap*) and spread among frozen arms (*spread*), over every frozen arm of each population, in points of valid-format accuracy; the free reference is budget-matched in the upper rows, except where marked, and holds 2.15× the budget in the lower ones. —: not applicable. Its rank-1 row uses the earlier selection. With the extended grid of §S25.1, that row reads a gap of 24.1 and a spread of 7.3.

Along the spectrum, the five β -positions count as frozen arms, paired with the reference on seeds 2 to 6. At rank 2, the gap to free LoRA is 9.0 points and the spread among frozen arms 9.1, a ratio of 1.0, and the bootstrap interval on their difference includes zero ($-0.1 [-4.3, +4.0]$). At rank 4, the difference is +4.9 [+0.6, +5.8], a marginal pass. The spectral row counts the best position as the best frozen arm and spans all

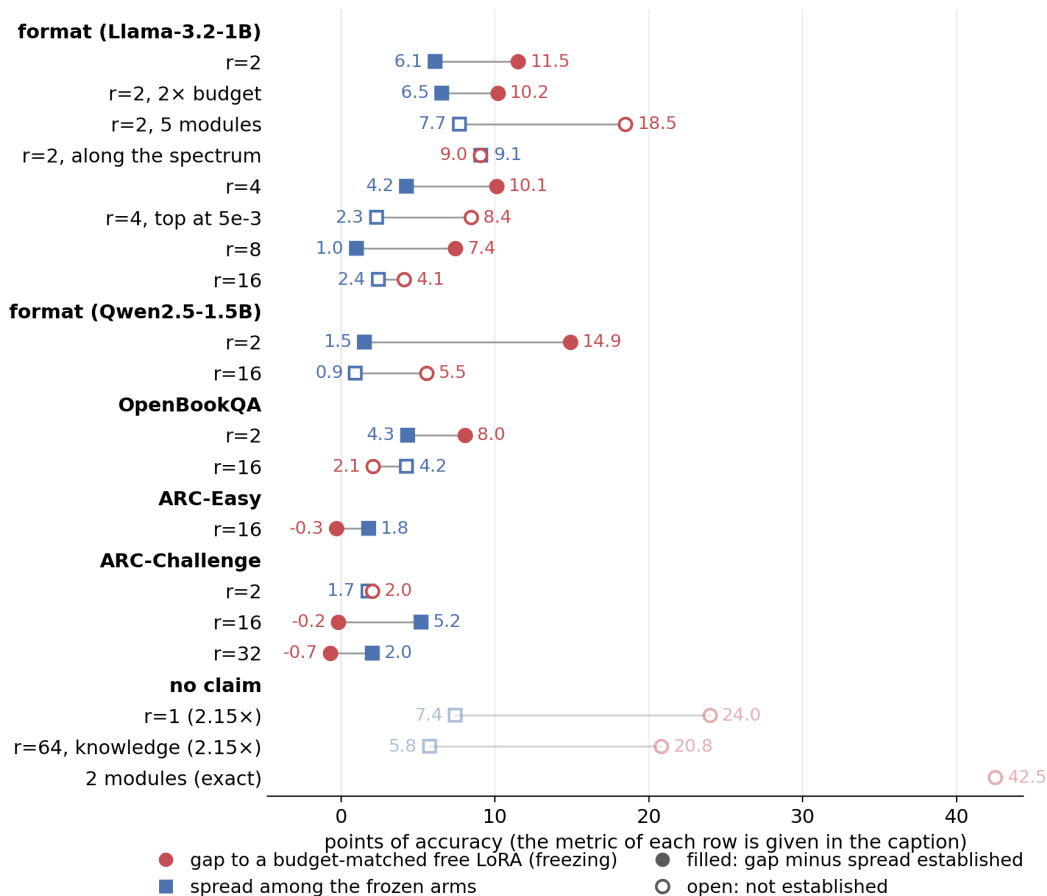


Figure S6: **Gap to free LoRA and spread among frozen arms, every population.** The full version of Fig. S5, including the rows that carry no claim. Each row is one population: a rank, a task, a training budget or a choice of modules, with each arm at its selected rate; the free references and their budgets are those stated in the caption of Fig. S5. Red disc: gap to free LoRA; blue square: spread among the frozen arms; a line joins the two values of one population, and only those two are comparable. Filled and open markers as in Fig. S5. Faded rows carry no claim: two modules at exactly equal budget (§S11.3), and the free reference at 2.15× the budget at rank 1 and on the knowledge task at rank 64 (§S11.5). The “2 modules (exact)” row has no spread among the frozen arms, since Table S12 gives none for it. Metrics: valid-format accuracy on the format task, multiple-choice accuracy on OpenBookQA, ARC and the knowledge task. Its rank-1 row uses the earlier selection. With the extended grid of §S25.1, that row reads a gap of 24.1 and a spread of 7.3.

five positions (9.1), not the four upper ones (4.0, §S3): the narrower range would define the spread among frozen arms differently from every other row. The spread grows with the number of arms, which is why, at rank 2, the five-position row (9.1) exceeds the four-arm row (6.1). At rank 2 the two quantities are therefore of the same order, and the full ordering rests on rank 4.

Status: [Post hoc · Established] at ranks 2 and 4 among the main arms, where the bootstrap intervals exclude zero; the five-module ratio is [Post hoc · Descriptive]. Table S24 gives the two spectral rows.

Caveats. The claim holds within a population, not between them. The two ranges overlap: the spread among frozen arms reaches 7.7 points on five modules, and the gap to free LoRA falls to 2.1 at rank 16. Comparing one population’s spread among frozen arms with another’s gap to free LoRA would therefore not mean anything; in Fig. S6, only the two values joined by one line are comparable. The last two rows of

Table S12 give the free reference $2.15\times$ the budget, because a free arm at rank r trains $r(d_{\text{in}}+d_{\text{out}})$ parameters against $r d_{\text{in}}$; they are shown for completeness and carry no claim (§S11.5).

S11.3 The budget works against the free reference

With the budget tilted against the free reference, the gap to free LoRA grows, from 11.5 to 18.5 points. At $r = 1$, **free**'s budget exceeds a frozen arm's at $r = 2$ by 7.5 % overall, but the split is uneven: on **gate_proj** and **up_proj** it has 2.5 times more parameters. We remove those two modules (Table S13, upper half). **free** then holds 72 % of the frozen arms' budget (376,832 parameters against 524,288) and scores 0.7941, against 0.6094 for **random**, 0.5910 for **top** and 0.5326 for **bottom** at its selected rate. The gap to free LoRA, measured against the best frozen arm as everywhere else, grows from 11.5 to 18.5 points.

	parameters	score
<i>gate and up dropped; q, k, v, o, down kept</i>		
free $r=1$	376,832	0.7941
random $r=2$	524,288	0.6094
top $r=2$	524,288	0.5910
bottom $r=2$	524,288	0.5326
<i>q, o only, exactly equal budget</i>		
free $r=1$	131,072	0.7049
top $r=2$	131,072	0.2801

Table S13: Parameter counts, read from the run logs, and valid-format accuracy. Five seeds (2 to 6), shared by the arms, so the comparison stays paired. Bold: the smaller budget.

Status: [Post hoc · Descriptive].

Caveats. The one population at exactly equal budget is reported but is not the primary control. The lower half of Table S13 restricts both arms to **q** and **o**, with 131,072 parameters each. **top** collapses there (0.2801) while **free** holds (0.7049): its training loss falls steadily from 2.3 to 0.13 on every seed and never rises, so nothing diverges, and its update norm ends at 6.8, against 1.6 for **free**.

The degeneracy of **q** and **o** (§S6) concerns the bottom of their spectrum, not the dominant band **top** writes in, so it does not explain the collapse. The collapse sits in one sub-rule: over five seeds, the keys are right in 0.968 of the outputs but the signature of initials in 0.339 only. The outputs have the structure and the entities; the loss, averaged over tokens, stays low, while the valid format, all or nothing, fails on the one field that must be computed from the others. The row carries no claim; it is faded in Fig. S6.

S11.4 Restoring Σ brings the gap and the spread level at rank 2

At rank 2, counting top_sigma as a frozen arm brings the spread among frozen arms level with the gap to a budget-matched free LoRA. **top_sigma** is the dominant band with Σ restored and B still frozen, at the same budget (Appendix C.2 of the main paper, §S9). It scores 0.6804 on seeds 2 to 6, beats **top**, and does not separate from $\beta = 0.75$ (+0.23 points, $t = 0.27$, 95 % interval $[-2.1, +2.6]$; §S9.2). Counting it as a frozen arm, the gap to free LoRA is 8.80 points and the spread among frozen arms 8.82: the ratio falls to 1.0, and the two are equal. The arm is legitimate, and in the means it is the best frozen arm we measured at this rank. Once the scale of the frozen factor is a degree of freedom, the ordering therefore does not hold at rank 2.

Status: at rank 2, the comparison of the two quantities is [Post hoc · Descriptive], since no test compares them, and the scalar's recovery is [Pre-registered · Established] (§S9.3); at rank 4, the ordering is [Pre-registered · Established].

Caveats. Part of that change is definitional, since each new arm can only widen the spread among frozen arms. At rank 4 the ordering holds with **top_sigma** counted, on seven paired seeds under a protocol written

before the runs: the gap to free LoRA is 10.00 points and the spread of the three original positions 4.56, a difference of +5.44 (95 % bootstrap interval [+2.98, +7.42]). **top_sigma** gains 2.82 points over **top** there, but does not overtake **random**.

A per-module scalar recovers most of it. Under Adam, multiplying a frozen B by a constant leaves the step of A nearly unchanged but multiplies the update, so restoring Σ acts in part as a change of rate. To separate the two, **top_scalar** keeps the dominant band and replaces $\sqrt{\Sigma_r}$ by a single norm-matched scalar per module, $\sqrt{\sigma}$, at the same budget; its selected rate is also $5 \cdot 10^{-3}$. On seeds 2 to 6 it scores 0.6674, and so recovers 2.31 of the 3.61 points **top_sigma** gains over **top** (64 %, $t = 3.32$, five seeds of five). The remaining 1.30 points (95 % interval $[-0.3, +2.9]$, $t = 2.21$, four seeds of five) are neither established nor excluded at five seeds. This is what one arm shows, at one rank, on one model: it is consistent with scaling a frozen factor acting largely as a per-module rate, and does not establish it. The rule we registered would have read the non-significant remainder as an attribution to scale; we report both readings instead. Two replications on fresh seeds settle the remainder: -0.27 points on 15 seeds ($t = -0.54$), and on 40 seeds an equivalence within one point, established (+0.02 points, 90 % interval $[-0.57, +0.60]$; §S9.4).

S11.5 At $2.15\times$ the budget, freezing weighs more at lower rank

With free at the frozen arms' rank, and so at $2.15\times$ their budget, the gap to free LoRA falls with rank: 24.1 points at rank 1, 15.1 at rank 2 and 11.6 at rank 4. Table S14 gives the scores. **random** is the best of the three arms swept at each of these ranks, and **top** goes from second to last. At rank 16, on OpenBookQA with Qwen2.5-1.5B and a budget-matched reference, the gap is 2.08 points (§S23.5): smaller still, but the task, the model and the budget all differ, so it does not extend this series. **bottom** is at its selected rate at ranks 1 and 4 (10^{-2} and $2 \cdot 10^{-3}$); at rank 1, at $5 \cdot 10^{-3}$, it scores 0.4622. At rank 2 it is at its selected rate, $5 \cdot 10^{-3}$ (Table S22). **random_ortho** was run at $r = 2$ only, where it edges **random** out (0.6534 against 0.6521). The gap to free LoRA here is measured against the best of the three arms shown, which is why its rank-2 entry reads 15.1 where Table S12, which has all four arms, reads 11.5 against a budget-matched reference. At rank 4, **top** is at its 400-item selected rate, $2 \cdot 10^{-3}$. The $\beta = 0$ sweep, the same arm, selects $5 \cdot 10^{-3}$ on 614 items and gives 0.7189 on the same seeds, which would make it the best of the three. Two honest selections of one arm thus disagree by six points, and Table S12 reports both.

	$r = 1$	$r = 2$	$r = 4$
free	0.7684	0.8033	0.8186
random	0.5277	0.6521	0.7023
top	0.4834*	0.6443	0.6599
bottom	0.4547	0.5922	0.6958
<i>gap to free LoRA</i>	24.1	15.1	11.6

Table S14: Valid-format accuracy on Llama, five seeds, each arm at its selected rate; bold marks the best frozen arm shown. Last row: gap to free LoRA against the best of the three, with **free** at the same rank ($2.15\times$ the budget). *At $2 \cdot 10^{-2}$, selected on the grid extended in §S25.1, at the edge of the tested grid. At 10^{-2} , **top** at rank 1 scores 0.5107. **random** at rank 1 is reported on its main configuration alone. The earlier value, 0.5288, included an isolated run from another experiment.

Status: [Post hoc · Descriptive].

Caveats. These rows are not budget-matched and carry no claim beyond the trend. We do not extrapolate it: the direct measurement at $r = 64$ is in Table S7.

S11.6 The gap to free LoRA across ranks, and against a null

The gap to free LoRA falls with rank. Measured against a budget-matched free LoRA, the gap of the best frozen arm falls from 11.5 points at rank 2 to 4.10 at rank 16 on the format task with Llama-3.2-1B,

and from 14.89 to 5.54 with Qwen2.5-1.5B (Table S12 and Table 13 of the main paper). On ARC-Challenge it is 2.02 points at rank 2 and is not detected at rank 16 ($-0.18, [-1.46, +0.92]$); at rank 32 the best frozen arm scores above free LoRA, and on ARC-Easy at rank 16 it lies 0.30 points above. On ARC-Challenge the fall owes part to the reference: across the three campaigns, free LoRA scores 0.504 at rank 1, 0.495 at rank 7 and 0.489 at rank 14 (Table 13 of the main paper), and at rank 7 it beats the base model (0.436) by only 5.9 points, so the task leaves little to lose; the minor band’s frozen arm at rank 16 scores 0.445.

Gap and spread. Beside the gap to free LoRA, §S11.1 reports the spread among frozen arms and a descriptive summary of the two, their difference: the gap exceeds the spread on the format task at ranks 2 to 8, while on both ARC benchmarks at rank 16, and on ARC-Challenge at rank 32, the spread exceeds the gap.

The range statistic, against a null. The null keeps each population’s free reference, gives every frozen arm the mean of the frozen means, and draws each arm’s score per seed with the pooled residual variance of the arm-by-seed table; 20,000 draws per population (`scripts/sim_freeze_choose_null.py`). Each population must first reproduce its published F and C to 0.02 point. Under a true null, the procedure gives $p < 0.05$ in 6.5 % of 400 simulated datasets: it is slightly liberal.

population	$F - C$	C	null C	$P(C_0 \geq C)$	null $F - C$, 95 %
format, $r=2$	+5.37	6.12	1.16	$< 10^{-4}$	[+9.95, +13.09]
format, $r=8$	+6.42	1.01	1.24	0.64	[+4.29, +7.64]
format, $r=16$	+1.69	2.41	1.01	0.0026	[+1.99, +4.71]
format, $r=2, 2\times$	+3.62	6.55	1.31	$< 10^{-4}$	[+8.11, +11.64]
OpenBookQA, $r=2$	+3.72	4.32	1.32	10^{-4}	[+5.29, +8.84]
ARC-Easy, $r=16$	-2.08	1.78	0.42	$< 10^{-4}$	[-1.06, +0.34]
ARC-Challenge, $r=16$	-5.36	5.18	0.87	$< 10^{-4}$	[-1.03, +1.86]
ARC-Challenge, $r=2$	+0.28	1.74	0.71	0.0098	[+0.68, +3.06]
ARC-Challenge, $r=32$	-2.74	2.02	0.61	0.0001	[-1.67, +0.37]

Table S15: Gap minus spread against a null in which the frozen arms have no true difference. C : the spread among frozen arms; null C : its mean under the null; the last column is the 95 % range of $F - C$ under the null. On ARC the reversal lies far outside it, at ranks 16 and 32; at rank 2 on ARC-Challenge the spread among frozen arms is real, though gap minus spread is not established; at rank 8 on the format task the spread is at the level of noise.

S12 Controls

This section reports two controls: whether a band-energy measurement can detect a spectral effect at all (§S12.1), and whether rescaling every update to a common norm is a usable control (§S12.2).

S12.1 Does the instrument detect what it claims?

On a planted target, a frozen arm recovers the target only when it lies in its own band, and the free arm finds the right band on its own. We plant a synthetic target by hand in one band of a toy model (Fig. S7, left and centre). The toy is a two-layer linear network, $y = xW_0^\top W_2^\top$ with W_2 fixed and random, of width $d = 64$, adapted at rank $r = 8$ on $n = 2000$ inputs, with Adam at 10^{-2} for 1500 steps, over ten seeds; the two targets are placed in the top and the bottom band. A frozen arm recovers the target only when it lies in that arm’s own band, and the free arm recovers it either way. The free arm puts 0.99

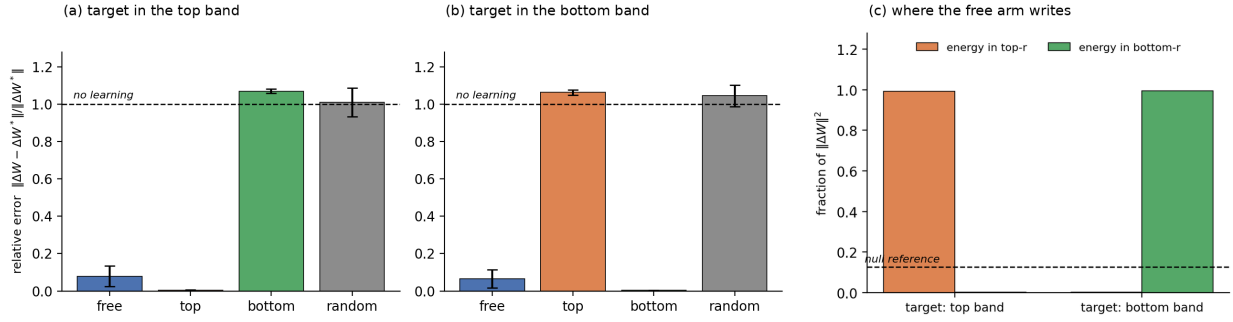


Figure S7: **Recovery of a target planted by hand in one band of a toy model.** Left and centre: recovery by each arm, with the target in each of the two bands. Right: the share of the free arm’s energy in the band where the target was planted, against a null reference of $r/d = 0.125$. A frozen arm recovers the target only when it lies in its own band, and the free arm finds the right band by itself.

of its energy in the band where the target was planted, whichever band that is, against a null reference of $r/d = 0.125$ (Fig. S7, right).

Status: [Post hoc · Descriptive].

Caveats. The control shows that the instrument can detect a spectral effect. It does not show that one of this kind exists in real weights.

S12.2 Matching the norm by rescaling

Seven common norms, under a committed protocol. Each arm ends training at its own update norm, $\|(\alpha/r)BA\|_F$: about 2.8 for **bottom**, 4.5 to 4.9 for the others. We rescaled each trained update to seven common norms, from 1.5 to 4.5, under a committed protocol (Table S16). No norm lets the four arms learn: **bottom** works between 2.0 and 3.0 and fails from 3.5, while **random** and **random_ortho** need 3.0 or more and **top** 3.5 or more, so every norm leaves one of them below the usable score of 0.3 that the protocol fixed. At a norm of 3.0, **bottom** is the best arm (0.564) while **top** stays below the usable score (0.250), and at 4.5 the order is reversed (0.000 against 0.639). The control is thus unusable in the sense its protocol defined, with all 20 pairs and with the 9 whose reloaded evaluation reproduced, and it leaves position and norm unseparated.

An earlier attempt, with two norms. Neither of the two norms we tried lets all four arms learn, so we could not make a matched-norm comparison by rescaling. We rescaled every update to a common Frobenius norm, at two targets, with each arm at its selected rate, on five seeds. At a target of 1.5, only **bottom** trains (0.573), while **top**, **random** and **random_ortho** fall to 0.000 to 0.006. At 4.5, the three train (0.639 to 0.651, within a point of their unscaled scores), and **bottom** falls to 0.000. We therefore did not separate position from norm this way; Fig. S2 is the descriptive alternative (§S3.6).

Status: [Post hoc · Descriptive].

Caveats. The comparison uses two targets, 1.5 and 4.5, and none between them, although the curves of Fig. S2 cross between 2.2 and 2.9.

Position or norm. Every run is trained once, then reloaded and evaluated at its own norm and at seven common norms; for 11 pairs, the seven norms were measured by retraining instead (below). At its own norm, **bottom** ends near 2.8 and the other arms between 4.5 and 4.9. No common norm leaves all four arms above the usable score of 0.3 (Table S16), so the control is unusable in the sense its protocol defined: position and norm stay unseparated. The reloaded evaluation reproduced the recorded score for 9 of the 20 runs; for the 11 others, it differed by 0.2 to 1.8 points. All 11 were re-evaluated on another GPU model than the one they were trained on, while the 8 re-evaluated on the same model all reproduce: the cause is the hardware

norm	1.5	2.0	2.5	3.0	3.5	4.0	4.5
top (5)	0.000	0.004	0.141	0.250	0.442	0.580	0.639
random (5)	0.000	0.041	0.145	0.315	0.485	0.583	0.627
random_ortho (5)	0.000	0.067	0.240	0.349	0.495	0.613	0.643
bottom (5)	0.062	0.347	0.564	0.564	0.245	0.000	0.000

Table S16: Valid-format accuracy of each trained update at a common norm (mean $\|\Delta W\|_F$), Llama at rank 2, seeds 105 to 109; in parentheses, the number of runs. For 9 pairs the norms are measured by reloading the trained factors and rescaling them; for the other 11, whose reloaded evaluation did not reproduce, by retraining with the rescaling applied in the same process.

arithmetic, not the reload, which restores the factors exactly. As a deviation written before any new run, the 11 pairs were measured again by retraining with the rescaling applied in the same process. With all 20 pairs, as with the 9 reloaded ones alone, the outcome is the same: the control is unusable.

S13 In PiSSA’s and MiLoRA’s own setup

In PiSSA’s and MiLoRA’s own setup, the initialisation scale changes the step size at a common rate, so comparing these variants does not isolate spectral position. The `init_top`, `init_bottom` and `init_random` arms follow PiSSA and MiLoRA exactly. The band is subtracted from W_0 , and the adapter is initialised to $B_0 = U_r \sqrt{\Sigma_r}$, $A_0 = \sqrt{\Sigma_r} V_r^\top$ (divided by the α/r scaling), so that the model is unchanged at step 0. Both factors are then trained, at rank 2, with 1,409,024 parameters, the budget of `free` at $r = 2$.

Initialising on the band instead of freezing it, `init_bottom` (0.7182) and `init_random` (0.7280) lie within 1 point of each other. `init_top` (0.8078) scores about 8 points above `init_random` and 9 above `init_bottom`. It is within half a point of `free` (0.8033), a reference at the same budget, not a costlier one. `init_top` triples the base model’s perplexity after training: 42.1, against 16.1 for `init_bottom` and 15.9 for `init_random`, and a reference of 14.1.

This perplexity is not a perturbation at initialisation. It is what relearning the dominant band, whose norm is most of the layer, does to the model.

The cause is the initialisation scale: $B_0 = U_r \sqrt{\Sigma_r}$ carries the magnitude of the singular values across. At a common rate, the variants take steps differing by a median factor of five on the five non-degenerate modules (§S6), and of two hundred on `q_proj` and `o_proj`. Their comparison therefore does not isolate spectral position.

Status: [Post hoc · Descriptive].

Caveats. The score gaps between the init arms are differences of means, with no test or interval.

S14 Extended related work

Frozen and initialised spectral subspaces. PiCa (Hwang et al., 2026) sets a rate per method, and compares its dominant projection with a random one, not with other positions along the spectrum. It freezes the input factor A on the dominant subspace, the mirror image of a frozen `top` arm, and also shares its trained factor across layers of the same functional group, a second axis this paper does not vary. In no case is a fixed `top` compared with a fixed `bottom` while one factor stays frozen. OPLoRA (Xiong and Xie, 2026) constrains the update to the orthogonal complement of the dominant singular subspace, projected on both sides, to limit forgetting. It is a fixed minor-side constraint, which compares no positions. SVFT (Lingam et al., 2024) fixes both singular bases of W_0 and learns sparse coefficients between them, so its update lives in a fixed spectral frame without choosing a band.

Parametrisation and magnitude. Rescaling a frozen factor leaves the set of reachable updates unchanged and changes only the coordinates the optimiser works in. LoRA Done RITE (Yen et al., 2025) shows that

common optimisers are not invariant to rescaling and rotating LoRA factors; Zhang et al. (2025b) argue that the benefit of spectral initialisations comes mostly from the magnitude of the update, with the rate, the scaling factor and the initialisation as its regulators; and the scaling factor and the rate do not act alike (Zhang et al., 2026b). Chen et al. (2026) relate the optimal rate to rank, initialisation and scaling factor, which bears on our cross-rank comparisons at $\alpha/r = 1$. Spectral Adapter (Zhang and Pilanci, 2024) adapts dominant and lower singular directions with both factors trained, and finds an advantage for the dominant ones.

The interior optimum on initialisations. Quercia et al. (2026) reached the interior optimum first, on *initialisations*. On their language tasks, one configuration changes only the rate, and at a high rate the accuracy curve over the starting component is an inverted U that favours intermediate positions; their priority extends to task performance, not only to the trade-off with forgetting. Sweeping the starting component of a PiSSA-style adapter, they find that forgetting is U-shaped across the spectrum, highest at both ends. They also find that intermediate components give the best performance–forgetting trade-off, on ViT-B and LLaMA-2 7B. Their positions are compared at a common rate. We reach the same place from the other side: *frozen* subspaces rather than initialisations, and each position at its own selected rate rather than at a rate shared across positions. We find an interior optimum under both changes at rank 2, on the task score (§S3.3). At rank 4 no position separates (§S3.2), and their own claim concerns a performance–forgetting trade-off rather than the score alone. Liu and Ghezelbash (2026) sweep a different continuum, the strength of curvature whitening applied to the gradient before extracting the LoRA subspace. They find that the best point is task-dependent and frequently interior to that continuum, away from the published endpoints.

Subspaces chosen by other means. EVA (Paischer et al., 2024) initialises the adapter from the principal directions of the activations rather than of the weights. SC-LoRA (Luo et al., 2025) picks its subspace from data rather than from W_0 : it initialises the adapter in the directions the fine-tuning data occupies and the preserved knowledge does not. It reports both task gain and forgetting, the trade-off §S23.1 measures for fixed spectral bands. Two further methods fix a subspace by other means. HiP-LoRA (Chen and Tan, 2026) splits the update into a channel inside the dominant singular subspace and a residual channel in its complement. LoRA-Null (Tang et al., 2026) places the adapter in the null space of the input activations rather than of W_0 , with one variant freezing the down-projection. Neither compares positions along the spectrum with a rate per position.

The factors, the scale and the evaluation. Both bear on arms that train A alone, from zero. rsLoRA (Kalajdzievski, 2023) argues that the scaling should be $\alpha/\sqrt{r}$ rather than α/r , for learning to stay stable across ranks; we hold α/r fixed. VeRA (Kopiczko et al., 2024) freezes random matrices and trains scaling vectors, close in spirit to our `random_ortho` control. DoRA (Liu et al., 2024) separates the magnitude of the update from its direction, the confound of norm and position we face in §S3.6. Shuttleworth et al. (2024) relate the spectral structure of LoRA updates to forgetting. Under Adam, the curvature that governs stability is the preconditioned one (Cohen et al., 2022); we measure the raw Hessian (§S23.3).

Tuning confounds. A large empirical study of LoRA variants (He et al., 2026) sweeps the learning rate across variants and finds that their rankings move with it, on other families than ours. Lee et al. (2026) report that PiSSA needs a learning rate about ten times lower than vanilla LoRA, and, for Qwen3-0.6B on mathematical reasoning (their Figure 1 and Section 4.2), MiLoRA about 3.2 times lower, and attribute these differences to the largest Hessian eigenvalue (§S23.2.2). The confound is the same, but its effect runs the other way. In the data of Lee et al., per-method tuning closes the gaps a common rate created; in ours, per-position tuning reveals a gap a common rate inverted (§S23.4), and it does not close the gap between frozen- B arms and a budget-matched LoRA (§3 of the main paper). The two results are not in tension once the object being varied is kept in mind: their spectral variants, PiSSA and MiLoRA, train both factors after a spectral initialisation, whereas ours freeze one factor for the entire run; RandLoRA (Albert et al., 2025), which freezes random bases and trains diagonal scalings, is the nearest case of frozen bases in that literature. Lee & Lee (Lee and Lee, 2026) identify a parallel confound in batch size, which they argue must be tuned jointly with the learning rate (§S17.2).

Position along the spectrum, on initialisations. GOAT (Fan et al., 2025) initialises the experts of a mixture of LoRA adapters from different segments of the singular value decomposition, at ranks 32 and 128, and finds that the useful segment varies with the task; the adapters are then trained, so the comparison is between initialisations, not frozen subspaces. In reinforcement learning with verifiable rewards, the orthonormality of the initialisation alone improves training, while learning in the principal subspace is not harmful but more fragile (Zhang et al., 2026a); and under PiSSA the principal directions carry a higher curvature, which makes higher rates unstable (Zhu et al., 2025). We compare frozen subspaces, each at its own rate, in supervised fine-tuning.

S14.1 Related work: frozen subspaces, factors and tuning

Frozen and initialised spectral subspaces. Table S23 lists the works that fix or initialise a spectral subspace of W_0 , and whether each tunes the learning rate per arm. None compares positions with the rate tuned per position. PiCa (Hwang et al., 2026) sets a rate per method and compares its dominant projection with a random one; it also shares its trained factor across layers of a functional group, an axis we do not vary. ROSA (Hameed et al., 2024) tunes the rate per arm but resamples the subspace during training. The nearest antecedent is LoRA-XS (Balazy et al., 2024), which freezes both singular bases and trains an $r \times r$ core. In its Appendix G, on QNLI, **top** leads the bottom basis 90.88 to 73.88 at $5 \cdot 10^{-4}$ and trails it 50.54 to 81.29 at $5 \cdot 10^{-3}$, where it collapses to the majority class. An inversion under a common rate is therefore not new. We add a single frozen factor, per-position rates selected on seeds disjoint from those reported, and a budget-matched free reference. LoRA-XS also keeps Σ in the frozen factor, and there the minor band reaches its best score at the highest rate of the grid on all four tasks, the reverse of what we find without Σ ; its Appendix H, where removing Σ moves **top**’s best rate to the top of the grid, is a direct antecedent of our **top_sigma** arm.

Factors, scale and magnitude. Zhu et al. (2024) find that freezing A and training B does better than the reverse, with random orthonormal factors and with B on the dominant left singular vectors. They also find that a frozen random factor does best when it is orthogonal. LoRA+ (Hayou et al., 2024b) shows that the two factors want different learning rates, and Hayou et al. (2024a) that the choice of factor starting at zero changes the dynamics. Zhang et al. (2025b) argue that the benefit of spectral initialisations comes mostly from the magnitude of the update, with the rate, the scaling factor and the initialisation as its regulators, and LoRA Done RITE (Yen et al., 2025) shows that common optimisers are not invariant to rescaling and rotating LoRA factors.

Tuning confounds. Unequal tuning across the variants compared has reordered published rankings in language modelling (Melis et al., 2018), in generative models (Lucic et al., 2018), and across benchmarks (Dodge et al., 2019; Bouthillier et al., 2021). Lee et al. (2026) re-evaluate nine LoRA variants under a full learning-rate search and find that the gaps between them largely close once the rate is tuned per variant; PiSSA needs a rate about ten times lower than vanilla LoRA. Their spectral variants train both factors after a spectral initialisation, whereas ours freeze one factor for the entire run. The contrast is part of our result: for variants that only initialise on a band, tuning the rate per variant closes the gap to LoRA; for variants that freeze B on a band at rank 2, it does not (§3 of the main paper).

S15 The late campaigns: full records

These are the records of the late campaigns that Table 12 of the main paper condenses, as they stood.

The cost of freezing B , isolated. The protocol (PROTOCOL_top_unfrozen.md, committed on 22/09 and first run in the night of 25 to 26/09) and its addendum fix the comparisons: R = unfrozen at rank 1 minus frozen at rank 2, against G = free at rank 1 minus frozen at rank 2 (12.41 for **top**, fixed in advance); freezing explains the gap if the whole 95 % interval of R lies above $0.7G$, and the rank-2 comparison is tested at $t > 3.495$. Every selected rate is interior to its grid: $5 \cdot 10^{-3}$ for both unfrozen arms at rank 1, $5 \cdot 10^{-3}$ for **top_unfrozen** and $2 \cdot 10^{-3}$ for **bottom_unfrozen** at rank 2.

The same isolation on OpenBookQA. The protocol of Table S17, run on OpenBookQA at rank 2 with Qwen2.5-1.5B (Table S18), passes the same tests: G reproduces the values fixed in advance, every rate is selected inside the grid, R lies above $0.7G$ at both ends, and the cost at equal rank is established ($t = 12.79$ and 19.15). Trained, B moves less than on the format task, by about 30 degrees. The reduction of the gap between bands, pre-registered here, is established (3.84 , $[+1.25, +6.43]$, $t = 4.12$), while the unfrozen bands are neither shown different nor equivalent within one point (-0.48 , 90% interval $[-1.35, +0.39]$). The GPU-model control of the addendum could be computed for one test only, because few seeds share a GPU model across the compared runs. A follow-up, beyond the addendum’s rule and written after reading, retrained the ten frozen references with their factors and re-evaluated the thirty compared checkpoints on one GPU model: every verdict holds, and the reduction becomes 3.68 ($[+2.56, +4.80]$, $t = 9.13$). The retrained references reproduce the originals within 0.3 points on average, and within one to two points per seed. Status: [Pre-registered · Established].

Freezing A instead of B . The protocol freezes A on the rows of V^T of each band and trains B from zero (`dual_top`, `dual_bottom`); the rate is selected on seeds 0 and 1 over seven values, and both arms select 10^{-2} , inside the grid. On the reported seeds, freezing A costs 12.96 and 18.96 points less than freezing B (Table S19), and both tests pass the corrected threshold, 3.495. The budget differs, and the protocol says so: an arm that freezes A trains $r \times d_{\text{out}}$ parameters, 15% more than an arm that freezes B on Llama. In a post hoc test, the gap between the bands goes from -5.21 points to $+0.78$ ($[-1.06, +2.62]$), a change of 5.99 ($[+1.74, +10.25]$, $t = 3.91$); the bands are not shown equivalent. A third arm, `dual_random_ortho`, meant only for the descriptive spread among the arms that freeze A , failed on all its runs before training, on a code defect: a check for a decomposition this arm does not use. The defect is fixed, and the arm was rerun after the primary results were read: the three arms that freeze A then reach or pass free LoRA, the best by 1.34 points, and span 1.34 points; `random_ortho` selects $2 \cdot 10^{-2}$, the upper edge of the grid. Status: [Pre-registered · Established] for the two primary tests; the comparison of the bands is post hoc.

Freezing A on OpenBookQA. The same protocol on OpenBookQA at rank 2 with Qwen2.5-1.5B, pinned to one GPU model, finds the same order (Table S20). The rates are selected inside the grid ($5 \cdot 10^{-3}$ for `dual_top`, 10^{-2} for `dual_bottom`); the tests hold on the seeds whose compared runs share a GPU model ($t = 9.66$ on five, 26.69 on three). The margin is smaller than on the format task: freezing A recovers about 80% of G , and the two spectral arms stay below free LoRA, although an arm that freezes A trains 26.8% more parameters than one that freezes B on this model, and 11.8% more than free LoRA at rank 1, because k and v are narrow in output. The comparison of the bands, post hoc in the original format campaign and pre-registered on its fresh seeds, was pre-registered here too: from -4.32 points under a frozen B to -0.60 under a frozen A , a change of 3.72 ($[+1.57, +5.87]$, $t = 4.81$). `dual_random_ortho` reaches 0.5464, above free LoRA (0.5456) in the means, without a test. Status: [Pre-registered · Established].

The format task on Qwen2.5-1.5B. Under a protocol committed before the runs, on fresh seeds (selection on 20 and 21, report on 22 to 26) and pinned to one GPU model, the format task runs on Qwen2.5-1.5B at ranks 2 and 16, with the arms of ARC (Table S21 and Table 13 of the main paper). At rank 2, the gap to free LoRA exceeds the spread among frozen arms: $+13.36$ ($[+11.30, +14.20]$). At rank 16 the order does not reverse: $+4.59$ ($[-0.52, +6.09]$), not established, where ARC-Challenge on the same model gives -5.36 ; at the oracle rate the interval is $[+1.01, +5.37]$. The reversal therefore goes with the task, not the model. `bottom` selects a lower rate than `top` and `random_ortho` at both ranks, and at rank 2 it is the best spectral arm ($+1.53$ over `random_ortho`, $t = 1.62$), so its deficit at rank 2 on Llama appears specific to that model. `random_ortho` selects $2 \cdot 10^{-2}$ at rank 2, the upper edge of the grid. Status: [Pre-registered · Established] at rank 2, [Pre-registered · Not established] at rank 16.

The rank-2 comparison on fresh seeds. The protocol (PROTOCOL_rank2_fresh.md) reruns the comparison of the introduction on fresh seeds 120 to 124, pinned to one GPU model, each arm at the rate the paper selected, with no new selection; it adds the two arms that freeze A . Scores, with their difference to seeds 2 to 6: `top` 0.6593 ($+1.50$), `bottom` 0.5954 ($+0.32$), `random` 0.6612, `random_ortho` 0.6570, free LoRA at rank 1 0.7664 (-0.20); B trained at rank 2, 0.8042 and 0.8081; at rank 1, 0.8078 and 0.7997; A frozen, 0.7772 and 0.7818. G , recomputed on these seeds, is 10.72 and 17.10 (against 12.41 and 17.62). R is 14.85 and 20.42,

above $0.7G$ at both ends. The gap between the bands shrinks by 6.78 when B trains ($t = 28.31$) and by 6.84 when A is frozen instead ($t = 3.43$, the narrowest of the four tests). Gap minus spread, reported without a test, is $+3.94$ ($+5.4$ on seeds 2 to 6). Status: [Pre-registered · Established].

Freezing B isolated on the format task with Qwen2.5-1.5B. The protocol (PROTOCOL_exp1_format_qwen.md) adds the unfrozen arms to that task’s campaign on this model, on its seeds (20 and 21 for selection, 22 to 26 for report) and its GPU model, with G fixed in advance at 16.22 and 14.89 points. All four rates are selected at $5 \cdot 10^{-3}$, inside the grid, and the oracle rate is the same. B trained scores 0.7896 and 0.7811 at rank 1, 0.7775 and 0.7857 at rank 2. The band-gap test is reported below its floor: -0.52 ($[-2.55, +1.51]$).

The central result on Qwen2.5-7B. The protocol (PROTOCOL_scale.md) was written before a probe run, with a rule fixed in advance for the model: Qwen2.5-7B if the probe fitted under 18.5 GB and finished in time, Qwen2.5-3B otherwise. The probe peaked at 17.18 GB and took 19.5 minutes. Two addenda, committed before the runs they concern, recalibrated a numerical check of the SVD that stopped a first probe on large matrices, and reran eleven selection runs that had failed for machine reasons (memory, environment, local disk) on machines that had already completed a 7B run. Rates were selected on seeds 0 and 1 among five, then seeds 2 to 6 were run at the selected rate only, all on one GPU model: **top** 0.5316 at $5 \cdot 10^{-3}$, **bottom** 0.4840 at $5 \cdot 10^{-4}$ (the lowest rate of the grid, a limit), B trained 0.6292 and 0.6312 at 10^{-3} , A frozen 0.6364 and 0.6148 at $2 \cdot 10^{-3}$, free LoRA at rank 1 0.6284 at $2 \cdot 10^{-3}$. Gap minus spread, reported without a test, is $+4.92$ ($[+3.52, +6.28]$). Status: [Pre-registered · Established].

B trained at rank 1 on the two 7-billion models. The protocol (PROTOCOL_7b_unfrozen_r1.md) trains **top_unfrozen** and **bottom_unfrozen** at rank 1 on OpenBookQA with Qwen2.5-7B and Mistral-7B-v0.3, on the base of the two 7B campaigns, five rates each, selected on seeds 0 and 1 and reported on seeds 2 to 6, on the machines where a run of the same model had completed; G is fixed in advance from those campaigns, and every reference gives back its value exactly. Six selection runs failed for lack of GPU memory and were rerun once, as the protocol provides; all succeeded. All four selected rates are $2 \cdot 10^{-3}$, interior. Scores: Qwen2.5-7B 0.6232 and 0.6264, Mistral-7B 0.5924 and 0.5924, against free LoRA at 0.6284 and 0.6012. Status: [Pre-registered · Established] for the four tests of R .

Freezing A against freezing B on square modules. The protocol (PROTOCOL_square_modules.md) adapts only **q_proj** and **o_proj** of Llama-3.2-1B on the format task at rank 2, where the arms that freeze B , those that freeze A and free LoRA at rank 1 all train 4,096 parameters per module. Rates are selected again among five on fresh seeds 130 and 131 and reported on seeds 140 to 144, pinned to one GPU model. The addendum, committed before any run, fixes a collapse threshold of 0.45 on **top**, since an earlier campaign saw **top** collapse on these modules. Selected rates: $2 \cdot 10^{-2}$ for **top** and **bottom**, 10^{-2} for **dual_top**, $5 \cdot 10^{-2}$ for **dual_bottom** (the highest of its grid), $5 \cdot 10^{-3}$ for free LoRA. The frozen- B outputs are structured (the keys are right in 97 % of them, 1 to 2 % are empty) but fail on the number of entries and the signature.

The rank-2 isolation with four times more training. The protocol (PROTOCOL_long_training.md) trains five arms on the format task with Llama-3.2-1B at rank 2 on 2.4 million tokens, each arm’s rate selected again among four on seeds 0 and 1, then run on seeds 2 to 6. Three selection runs stopped after about 115 minutes, against 25 for a normal run, and were killed by the two-hour limit; an addendum committed before the rerun ran them once more, on machines that had completed a run of this campaign. Eight report runs were left unstarted in a first pass because other jobs occupied their GPU, and ran in a second. Rates: **top** 10^{-2} and **bottom** $5 \cdot 10^{-3}$, as in the paper; **top_unfrozen** 10^{-3} (the lowest of its grid; $5 \cdot 10^{-3}$ in the paper), **bottom_unfrozen** $2 \cdot 10^{-3}$, free LoRA $2 \cdot 10^{-3}$ (the highest of its grid; 10^{-3} in the paper). Scores, with their change from 600,000 tokens: **top** 0.6837 ($+3.94$), **bottom** 0.6257 ($+3.35$), B trained 0.8036 ($+0.36$) and 0.8081 ($+0.09$), free LoRA 0.7811 ($+1.27$). Status: [Pre-registered · Established].

The central result on Mistral-7B. The protocol (PROTOCOL_scale_mistral.md) follows the one for Qwen2.5-7B, with no fallback model. The probe peaked at 15.84 GB and took 28 minutes. Of the 70 selection runs, 13 failed for machine reasons (seven out of memory, six on a full local disk) and were rerun once, as

	top	bottom
frozen, $r=2$	0.6443	0.5922
unfrozen, $r=1$	0.7928	0.7954
unfrozen, $r=2$	0.8000	0.8072
free , $r=1$	0.7684	0.7684
G	12.41	17.62
R	14.85	20.33
95 % interval	[13.14, 16.57]	[18.47, 22.18]
0.7 G	8.69	12.34
initialisation term	-2.44	-2.70
rank term	-0.72	-1.17
cost at equal rank	15.57	21.50
95 % interval	[13.32, 17.82]	[19.56, 23.43]
angle, $r=1$ / $r=2$	55.5° / 59.5°	59.4° / 41.6°

Table S17: Freezing B , isolated (Llama-3.2-1B, format task, seeds 2 to 6). Scores in valid-format accuracy; differences in points. Initialisation term: **free** minus unfrozen, at rank 1; rank term: unfrozen at rank 1 minus at rank 2; cost at equal rank: unfrozen minus frozen, at rank 2. The intervals of the initialisation and rank terms are $[-5.60, +0.72]$ and $[-1.71, +0.28]$ for **top**, $[-4.23, -1.17]$ and $[-3.35, +1.00]$ for **bottom**. The angle is the largest principal angle between the initial and the final B , averaged over modules and seeds.

	top	bottom
B frozen, $r=2$	0.6443	0.5922
A frozen, $r=2$, at 10^{-2}	0.7739	0.7818
free , $r=1$	0.7684	0.7684
A frozen minus B frozen	12.96	18.96
95 % interval	[9.76, 16.17]	[17.50, 20.42]
t (threshold 3.495)	11.24	35.99

Table S19: Freezing A instead of B (Llama-3.2-1B, format task, rank 2, seeds 2 to 6). Scores in valid-format accuracy; differences in points. B frozen at each band's selected rate, 10^{-2} for **top** and $5 \cdot 10^{-3}$ for **bottom**.

	top	bottom
frozen, $r=2$	0.4652	0.4220
unfrozen, $r=1$	0.5524	0.5496
unfrozen, $r=2$	0.5576	0.5528
free , $r=1$	0.5456	0.5456
G	8.04	12.36
R	8.72	12.76
95 % interval	[7.73, 9.71]	[10.21, 15.31]
0.7 G	5.63	8.65
initialisation term	-0.68	-0.40
rank term	-0.52	-0.32
cost at equal rank	9.24	13.08
95 % interval	[7.24, 11.24]	[11.18, 14.98]
angle, $r=1$ / $r=2$	29.8° / 32.9°	29.8° / 32.5°
one GPU model: R	9.24	12.76
cost at equal rank	9.64	13.32

Table S18: Freezing B , isolated, on OpenBookQA (Qwen2.5-1.5B, rank 2, seeds 2 to 6). Scores in multiple-choice accuracy; differences in points; terms as in Table S17. The intervals of the initialisation and rank terms are $[-2.51, +1.15]$ and $[-1.79, +0.75]$ for **top**, $[-2.70, +1.90]$ and $[-1.80, +1.16]$ for **bottom**. The last two rows re-evaluate the compared checkpoints on one GPU model, with the frozen references retrained.

	top	bottom
B frozen, $r=2$	0.4652	0.4220
A frozen, $r=2$	0.5308	0.5248
free , $r=1$	0.5456	0.5456
A frozen minus B frozen	6.56	10.28
95 % interval	[4.67, 8.45]	[7.77, 12.79]
t (threshold 3.495)	9.66	11.39

Table S20: Freezing A instead of B on OpenBookQA (Qwen2.5-1.5B, rank 2, seeds 2 to 6). Scores in multiple-choice accuracy; differences in points. A frozen at $5 \cdot 10^{-3}$ for **top** and 10^{-2} for **bottom**.

the protocol provides, on machines that had completed a Mistral run. Of the 35 report runs, six were left unstarted by the training code in a first pass because other jobs occupied their GPU, and ran in a second pass. Rates: 10^{-3} for six arms, $5 \cdot 10^{-4}$ for the dominant A-frozen arm (the lowest of its grid, a limit). Gap minus spread, reported without a test, is +12.24. Status: [Pre-registered · Established] for the cost of freezing B and for A-frozen minus B-frozen; [Pre-registered · Not established] for the band gap, whose frozen value is 0.12 points.

S15.1 Per-pair detail of the two rank-2 pairs isolated first

S15.2 The extension campaigns: full records

Every run of these campaigns ran on RTX 4000 Ada only, and every selected rate lies inside its grid. The protocols are committed on 28/09 at 00:40, before any run; the tables of the main paper's Appendix D.2 give the tested quantities.

B1. The budget: B frozen at rank 3 against A frozen at rank 2 (OpenBookQA). Protocol `PROTOCOL_budget_r3r4.md`. At rank 3, B frozen trains more parameters than A frozen at rank 2. Test:

A frozen ($r = 2$) – B frozen ($r = 3$), per band, paired over seeds 2–6, $t > 3.495$ (two bands); rank 4 reported without a test. Parameters per layer: on Qwen2.5-1.5B, A frozen at rank 2 46,080, B frozen at rank 3 54,528 (+18.3%), at rank 4 72,704 (+57.8%); on Qwen2.5-7B, 99,328, 121,344 (+22.2%) and 161,792 (+62.9%). Qwen2.5-1.5B: **top** +6.28 [+4.48, +8.08], $t = 9.71$ (B at rank 3 at $5 \cdot 10^{-3}$, 0.4680), rank 4 +6.64 [+5.50, +7.78], $t = 16.12$ ($5 \cdot 10^{-3}$, 0.4644); **bottom** +9.20 [+6.68, +11.72], $t = 10.14$ ($2 \cdot 10^{-3}$, 0.4328), rank 4 +7.24 [+4.80, +9.68], $t = 8.23$ ($2 \cdot 10^{-3}$, 0.4524). Qwen2.5-7B: **top** +9.00 [+8.26, +9.74], $t = 33.54$ ($2 \cdot 10^{-3}$, 0.5464), rank 4 +9.00 [+7.55, +10.45], $t = 17.26$ ($5 \cdot 10^{-3}$, 0.5464); **bottom** +12.24 [+10.74, +13.74], $t = 22.59$ ($5 \cdot 10^{-4}$, 0.4924), rank 4 +10.12 [+9.40, +10.84], $t = 39.27$ (10^{-3} , 0.5136). References (A frozen at rank 2, published): 0.5308 / 0.5248 (Qwen2.5-1.5B), 0.6364 / 0.6148 (Qwen2.5-7B). Fixed reading reached on both models: A frozen at rank 2 stays ahead of B frozen at rank 3 on both bands, with fewer parameters, so the parameter budget does not explain the advantage of freezing A . On Qwen2.5-7B, **top** scores 0.5464 at rank 3 ($2 \cdot 10^{-3}$) and at rank 4 ($5 \cdot 10^{-3}$), a coincidence of totals like the exact ties of the main paper.

B2. The rank trend: ranks 4 and 8 (OpenBookQA, Qwen2.5-1.5B). Protocol `PROTOCOL_rank_trend_obqa.md`. Free LoRA at the budget-closest rank: rank 2 for B frozen at rank 4, rank 4 for rank 8 (+13.4%, the same relation as rank 1 for rank 2). $G = \text{free} - B$ frozen was computed and committed before the second wave of arms it judges. Tests per rank and band: R against $0.7G$ (floor $G \geq 2$); cost at equal rank, $t > 3.495$; A frozen – B frozen, $t > 3.495$; the band gap, oriented by the sign of the frozen gap (a sign change with $|\text{gap}| \geq 2$ is a reversal, not a shrinkage). R/G and $I = G - R$ with 95% paired-bootstrap intervals. Rank 4, **top** / **bottom**: G +8.80 / +10.00; R +8.96 [+7.53, +10.39] / +9.96 [+7.56, +12.36], freezing explains most on both; R/G 1.02 [0.90, 1.17] / 1.00 [0.87, 1.15]; I -0.16 [-1.36, +0.92] / +0.04 [-1.36, +1.44]; cost at equal rank +8.60 [+8.14, +9.06], $t = 51.39$ / +8.96 [+5.76, +12.16], $t = 7.78$; A frozen – B frozen +7.00 [+5.42, +8.58], $t = 12.30$ / +8.64 [+6.90, +10.38], $t = 13.77$. Rank 8: G +5.52 / +8.88; R +5.12 [+4.13, +6.11], freezing explains most / +7.64 [+6.06, +9.22], inconclusive; R/G 0.93 [0.80, 1.07] / 0.86 [0.74, 1.02]; I +0.40 [-0.32, +1.36] / +1.24 [-0.16, +2.56]; cost at equal rank +6.44 [+5.50, +7.38], $t = 19.11$ / +9.00 [+8.07, +9.93], $t = 26.89$; A frozen – B frozen +6.36 [+4.80, +7.92], $t = 11.29$ / +6.88 [+4.18, +9.58], $t = 7.07$. All freezing-alone and A frozen – B frozen tests are established. Band gap: at rank 4 the frozen gap (-1.20) is below the two-point floor and is not read; at rank 8, with B trained, $-3.36 \rightarrow -0.80$, oriented statistic +2.56 [+1.92, +3.20], $t = 11.06$, shrinkage established (expected, B leaves its band); with A frozen, $-3.36 \rightarrow -2.84$, +0.52 [-2.55, +3.59], $t = 0.47$, not established. Pre-registered prediction (G and A frozen – B frozen decrease monotonically from rank 2 to rank 16), read descriptively: held on the minor band (G 12.36, 10.00, 8.88, 6.32; A frozen – B frozen 10.28, 8.64, 6.88, 3.80), not in the strict sense on the dominant band (G 8.04, 8.80, 5.52, 2.08; A frozen – B frozen 6.56, 7.00, 6.36, 1.36: a small rise at rank 4). The advantage of freezing A on the dominant band still holds at rank 8 and is no longer established at rank 16.

B3. Rank 16 at scale (OpenBookQA, Qwen2.5-7B). Protocol `PROTOCOL_7b_rank16.md`; memory probe passed; free LoRA at rank 7 (-2.5% budget); the same tests as B2. **top** / **bottom**: G +3.00 / +6.44; R +2.28 [+1.74, +2.82], inconclusive ($0.7G = 2.10$) / +6.00 [+4.43, +7.57], inconclusive ($0.7G = 4.51$); R/G 0.76 [0.56, 1.16] / 0.93 [0.85, 1.03]; I +0.72 [-0.32, +1.56] / +0.44 [-0.20, +1.12]; cost at equal rank +3.12 [+1.43, +4.81], $t = 5.13$, established / +5.24 [+2.60, +7.88], $t = 5.52$, established; A frozen – B frozen +2.20 [+0.69, +3.71], $t = 4.04$, established / +5.80 [+3.35, +8.25], $t = 6.58$, established. Band gap: with B trained, $-3.44 \rightarrow -1.32$, +2.12 [-1.00, +5.24], $t = 1.88$, not established; with A frozen, $-3.44 \rightarrow +0.16$, +3.60 [+0.30, +6.90], $t = 3.03$, shrinkage established, a marginal pass. Prediction (ranks 2 to 16): G 9.68 $\rightarrow$ 3.00 and 14.44 $\rightarrow$ 6.44; A frozen – B frozen 10.48 $\rightarrow$ 2.20 and 13.08 $\rightarrow$ 5.80: decreases on both bands. On Qwen2.5-7B the advantage of freezing A is still established at rank 16 on both bands, whereas on Qwen2.5-1.5B it is not on the dominant band (+1.36); R is inconclusive at rank 16.

C1. Phase 1: where free LoRA writes (analysis without training). Factors of free LoRA at rank 1, seeds 2–6: on the format task with Llama-3.2-1B at 10^{-3} , the published runs; on OpenBookQA with Qwen2.5-1.5B at $2 \cdot 10^{-3}$, five runs retrained with saved factors (0.5452 against the published 0.5456). For each module, the share of energy of B 's column in the dominant and minor bands of R ($k = 2$), and of A 's row in the bands of V , divided by the null reference k/d ; median over layers, mean over seeds. Rule, committed before the OpenBookQA analysis and applied a posteriori to the format task: outside the tested

bands if below $5\times$ the null reference in both bands; otherwise in the band of larger ratio if its energy share reaches 0.10; otherwise above chance but diffuse. OpenBookQA (pre-registered reading): B is outside the tested bands for six module types of seven (**q**, **v**, **o**, **gate**, **up**, **down**); for **k_proj** it is above chance but diffuse toward the minor band (ratio 7.80, 6.1 % of its energy); no type is in a band. A is outside for six types; **gate** is diffuse toward the dominant band (19.1, 2.5 %). Format task (post hoc): the same picture for B (six of seven outside; **k_proj** diffuse toward the minor band, 6.42, 2.5 %); A is diffuse toward the dominant band for **q**, **k**, **gate** and **up** (ratios 10.7 to 37.3; shares 1.0 to 3.6 %), outside for **v**, **o** and **down**. Spectral profile (share per tenth of the singular vectors): B 's energy is spread almost uniformly over the spectrum of W_0 (about 10 % per tenth); **k_proj** puts 22–25 % in the last tenth; for **gate** and **up**, 75–83 % falls outside the image of W_0 , which is what chance gives when d_{out} exceeds the rank. A reads slightly more in the dominant input directions (first tenth 12–22 % for **q**, **gate**, **up**). For six module types of seven, the direction in which free LoRA writes lies outside the tested bands, and its energy is spread almost uniformly over the spectrum of W_0 .

C2. Phase 2: B frozen on the direction learned by free LoRA (OpenBookQA, Qwen2.5-1.5B, rank 2). Addendum ANALYSE_writes_additif.md. Trigger, declared: the committed trigger did not say how to aggregate the seven module types; with six of seven outside and **k_proj** diffuse, the strict reading is not met and the majority reading is. A dated note (28/09 09:22, before the first phase-2 run at 17:31) records that phase 2 was launched and declared as such, its test and reading unchanged; the strict trigger was not met. Set-up: direction from a free LoRA at rank 2, seed 0, $2 \cdot 10^{-3}$ (seed 0 enters no reported score); per module, the orthonormal basis of the column space of its B by QR (196 modules). New arm **learned**: B frozen on that basis, no Σ , A trained from zero, the same budget as **top** ($r \cdot d_{\text{in}}$); the new mode was tested before any run (B equal to the file, B unchanged during training, $\Delta W = 0$ at initialisation, budget; run IDs of all other modes unchanged). Rates: five, selection on seeds 0–1, report on seeds 2–6. References: **top** 0.4652 and **random_ortho** 0.4592 are the published values; the comparisons with **bottom** and free LoRA ($r = 1$) use references retrained on one GPU model, 0.4276 and 0.5484 (published: 0.4220 and 0.5456). Reading fixed in advance on the 95 % interval of learned – **top**, as for R : most of the cost comes from the location if its lower bound reaches $0.7G = 5.63$ ($G = 8.04$ fixed in advance); the location matters little if its upper bound stays below $0.3G = 2.41$, whether or not the test is established; otherwise inconclusive; $t > 2.776$ reported alongside. Result: **learned** at $2 \cdot 10^{-3}$ (inside the grid) scores 0.5588; learned – **top** = +9.36 [+8.42, +10.30], $t = 27.77$, established; $P = 1.16$ [1.05, 1.28]. Fixed reading: most of the cost of freezing B comes from where B is frozen. Without test: learned – **bottom** +13.12 [+11.42, +14.82]; learned – **random_ortho** +9.96 [+7.87, +12.05]; learned – free LoRA ($r = 1$) +1.04 [−0.60, +2.68]. Caveats: it is a diagnostic, not a method (the direction comes from a free LoRA trained on the same data, so the learned arm benefits from prior training on the task and is not compared with **top** at equal information); the direction run is a free LoRA at rank 2; one task–model pair and one direction seed; the trigger was ambiguous and the launch is declared.

C2, replications. Each is read by the rule of phase 2 (most of the cost comes from where B is frozen if the lower bound of the 95 % interval of learned – **top** reaches $0.7G$); both were pre-registered, their protocol committed before their first run. Replication A, OpenBookQA with Qwen2.5-1.5B, direction seed 7: learned 0.5528; learned – **top** +8.76 [+7.66, +9.86], $t = 22.1$; $0.7G = 5.63$; $P = 1.09$ [0.95, 1.23]; learned – free LoRA +0.44 [−1.31, +2.19], without a test. Reading: most of the cost comes from where B is frozen. Replication B, the format task with Llama-3.2-1B, direction seed 0: learned 0.8081; learned – **top** +16.51 [+15.15, +17.88], $t = 33.7$; $0.7G = 8.69$; $P = 1.33$ [1.22, 1.44]; learned – free LoRA +3.94 [+0.44, +7.45], without a test, reported as indicative only. Reading: most of the cost comes from where B is frozen. References: on OpenBookQA, replication A reuses the references retrained on one GPU model for **bottom** and free LoRA (0.4276 and 0.5484), and the published values for **top** and **random_ortho**; on the format task, **top** and free LoRA are the retrained 0.6430 and 0.7687. The retrained references fall within one point of the published ones (0.6430 against 0.6443, 0.7687 against 0.7684; **random_ortho** moves most, by +0.98). G recomputed on one GPU model is 8.32 and 12.57, against 8.04 and 12.41 fixed in advance; the reading uses the values fixed in advance. Caveat: on the format task, B frozen on the directions of a free LoRA at rank 2 scores 0.8081, about the score of that free LoRA itself (0.8033): the learned arm recovers the level of the adapter its directions come from, which is why it is a diagnostic and not a method.

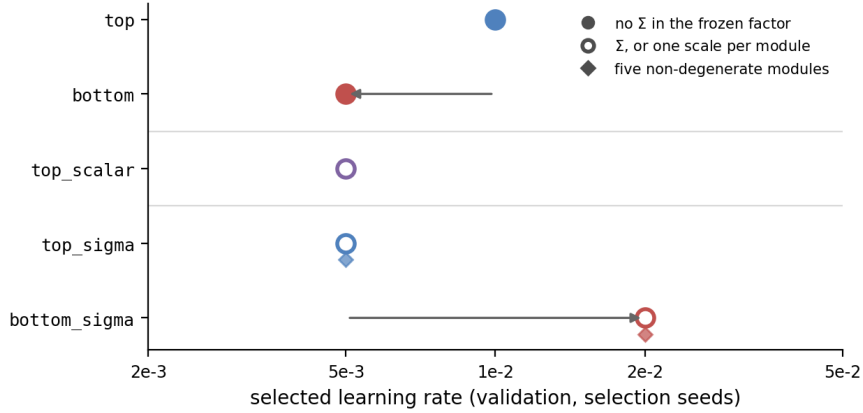


Figure S8: **The rate each arm selects, on Llama at rank 2** (selection seeds, validation). The arrows go from the dominant band’s rate to the minor band’s: without Σ in the frozen factor the minor band selects a lower rate, with Σ ($B = U_r \sqrt{\Sigma_r}$) a four times higher one, with all seven modules and with the five non-degenerate ones. **top_scalar**, which gives each module a single scale of the same norm as $\sqrt{\Sigma_r}$, selects the rate of **top_sigma**.

Square modules. On the format task with Llama-3.2-1B at rank 2, the comparison at exactly equal budget is inconclusive on the dominant band, by the rule fixed in advance. Adapting only **q_proj** and **o_proj** gives every arm the same trainable count, but the dominant B-frozen arm then collapses below the pre-set threshold (Table 12 of the main paper). A-frozen minus B-frozen would then measure that collapse rather than the budget. On the minor band, degenerate on these modules (§S6), freezing *A* scores 33.6 points higher than freezing *B*. That A-frozen arm selected the highest rate of its grid, a limit in the conservative direction. The gap between the bands changes sign, so we do not read it as a shrinkage.

S16 The minor band needs a lower learning rate

At rank 2, bottom selects half the learning rate of the other frozen arms on both models, at the rates of the two-seed selection; on Llama, the five-seed selection places the shift between a factor of two and five. The shift survives the removal of gradient clipping (§S23.2.1). Curvature predicts its direction, but not its size (§S23.2.2).

S16.1 bottom selects a lower rate on both models

At rank 2, bottom’s selected rate is lower than that of every other frozen arm, on Llama and on Qwen. On Llama, **bottom** selects $5 \cdot 10^{-3}$, against 10^{-2} for the other three arms; a tie with $2 \cdot 10^{-3}$ on the two selection seeds was settled on three added ones (§S17.1). On Qwen, **bottom** selects $5 \cdot 10^{-3}$ against 10^{-2} for **top**, **random** and **random_ortho**, a factor of two in the same direction. The direction is unambiguous, and at the selected rates the magnitude is a factor of two; on Llama the five-seed selection places it between two and five (§S16.2). Any common-rate comparison therefore evaluates at least one arm away from its optimum.

Status: [Post hoc · Descriptive]. These rates come from validation on the selection seeds, and §S23.2.1 tests the shift.

Caveats. At the selected rates the magnitude is a factor of two on both models (§S17.1). Several of these selections sit on the boundary of the grid, which can only understate the shift; six boundary cells tested at $2 \cdot 10^{-2}$ are interior maxima (Appendix E of the main paper).

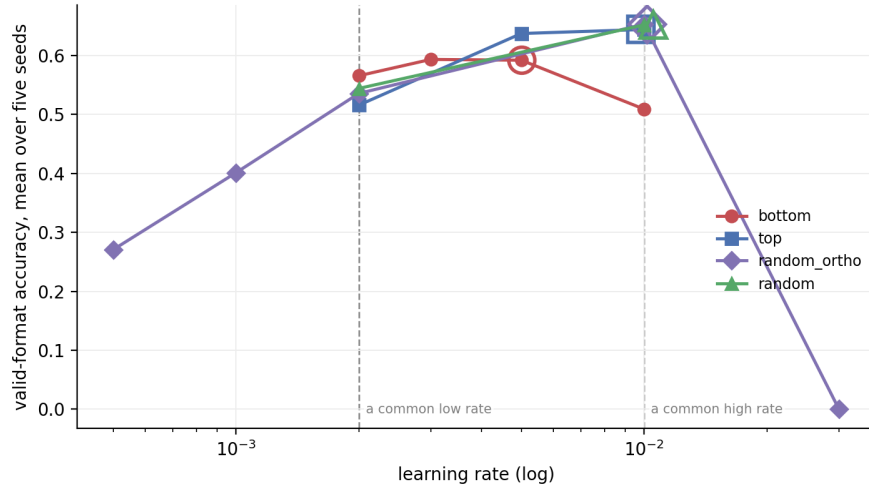


Figure S9: **Valid-format accuracy of the B-frozen arms against the learning rate, on Llama at rank 2:** mean over the five reported seeds, 614 items; rings mark the rate each arm selects on the selection seeds (0 and 1, plus 30 to 32 for **bottom**), not the best point on the reported seeds. **bottom** leads at $2 \cdot 10^{-3}$ and is last at 10^{-2} : which B-frozen arm looks best depends on the common rate imposed. On these reported seeds, **bottom**'s point at $3 \cdot 10^{-3}$ does not fall below its ring at $5 \cdot 10^{-3}$; the ring marks the rate chosen on the selection seeds, not the best point on the seeds drawn here.

S16.2 Across campaigns and on five seeds

The shift without Σ . On Llama at rank 2, **bottom** selects $5 \cdot 10^{-3}$ and the three other frozen arms 10^{-2} ; on Qwen2.5-0.5B the minor band also selects half the others' rate (§S16.1). A five-seed selection, under reading rules committed before any result was read, confirms the direction at rank 2 on the format task, on four seeds of five. It moves the minor band to $2 \cdot 10^{-3}$, so the size of the shift lies between a factor of two and five, the minor band's optimum being flat between $2 \cdot 10^{-3}$ and $5 \cdot 10^{-3}$. On OpenBookQA the same selection places the minor band below **random_ortho** only, on nearly flat validation curves. Predicted before the runs, the shift appears on the format task at rank 16 and at twice the training budget, and on OpenBookQA at rank 2 in the two-seed selection; it is partial at rank 8. At rank 16 on Llama it lies within the noise of the reported seeds, where **bottom** scores slightly higher at $2 \cdot 10^{-3}$ than at its selected $5 \cdot 10^{-4}$ (0.7655 against 0.7554). In two-seed selections read as observations, the minor band also selects the lowest rate on the format task with Qwen2.5-1.5B at ranks 2 and 16, and on ARC-Challenge at ranks 2 and 32. It is not reproduced on ARC at rank 16, where the selection curves differ by about a point. The reported tests keep the rates selected on seeds 0 and 1 (below). Curvature predicts the direction of the shift, not its size (§S23.2.2).

The rate shift on five seeds. To test the shift against the noise of a two-seed selection, the four frozen arms ran on three fresh seeds (110 to 112) at five rates in the two cells that carry the claim; with seeds 0 and 1, the selection uses five seeds, on validation only. The reading rules (which seeds count, rates without all five seeds excluded, ties at the fourth decimal) were committed three minutes after launch, before any result was read. On the format task, $2 \cdot 10^{-2}$ is excluded (**bottom** has three seeds of five there); **bottom** selects $2 \cdot 10^{-3}$, the other three 10^{-2} : **confirmed**, on four seeds of five. On OpenBookQA, **bottom** selects $5 \cdot 10^{-3}$, below **random_ortho** (10^{-2}) on every seed, but level with **random** and within **top**'s tie ($5 \cdot 10^{-3}$ and 10^{-2}): **partial**, on validation curves that span 0.38 to 0.41. In both cells the five-seed selection moves **bottom**'s rate ($5 \cdot 10^{-3}$ to $2 \cdot 10^{-3}$ on the format task, where the two nearly tie; $2 \cdot 10^{-3}$ to $5 \cdot 10^{-3}$ on OpenBookQA); the reported tests keep the rates selected on seeds 0 and 1. On the format task, the size of the shift therefore lies between a factor of two and five, because **bottom**'s optimum is flat between $2 \cdot 10^{-3}$ and $5 \cdot 10^{-3}$. Status: [Pre-registered · Established] on the format task; partial on OpenBookQA.

The rate shift. Written as a prediction before the runs (**bottom** selects a strictly lower rate than **top**, **random** and **random_ortho**), in the two-seed selection it is confirmed at rank 16, on OpenBookQA at rank 2 and at twice the budget, and partial at rank 8 (Table S21); on five seeds it is confirmed on the format task and partial on OpenBookQA (above).

population	top	bottom	random	r_ortho	free
format, $r=2$	10^{-2}	$5 \cdot 10^{-3}$	10^{-2}	10^{-2}	10^{-3}
format, $r=8$	$2 \cdot 10^{-3}$	$2 \cdot 10^{-3}$	$5 \cdot 10^{-3}$	$5 \cdot 10^{-3}$	$2 \cdot 10^{-3}$
format, $r=16$	$2 \cdot 10^{-3}$	$5 \cdot 10^{-4}$	10^{-3}	10^{-3}	$2 \cdot 10^{-3}$
format, $r=2, 2\times$	10^{-2}	$5 \cdot 10^{-3}$	10^{-2}	10^{-2}	$2 \cdot 10^{-3} *$
OpenBookQA, $r=2$	$5 \cdot 10^{-3}$	$2 \cdot 10^{-3}$	$5 \cdot 10^{-3}$	$2 \cdot 10^{-2}$	$2 \cdot 10^{-3}$
ARC-Easy, $r=16$	10^{-4}	$5 \cdot 10^{-4}$	—	$5 \cdot 10^{-4}$	10^{-4}
ARC-Challenge, $r=16$	$5 \cdot 10^{-4}$	10^{-3}	—	10^{-3}	$5 \cdot 10^{-4}$
ARC-Challenge, $r=2$	$5 \cdot 10^{-3}$	$2 \cdot 10^{-3}$	—	$5 \cdot 10^{-3}$	10^{-3}
ARC-Challenge, $r=32$	$5 \cdot 10^{-4}$	$2 \cdot 10^{-4}$	—	$5 \cdot 10^{-4}$	$5 \cdot 10^{-4}$
format (Qwen), $r=2$	10^{-2}	$5 \cdot 10^{-3}$	—	$2 \cdot 10^{-2}$	$5 \cdot 10^{-3}$
format (Qwen), $r=16$	$2 \cdot 10^{-3}$	10^{-3}	—	$5 \cdot 10^{-3}$	$2 \cdot 10^{-3}$

Table S21: Selected rates (selection seeds, validation). The free reference runs at half the rank on the format task at ranks 2 to 16 and on OpenBookQA, at rank 7 on ARC at rank 16, at rank 1 on ARC-Challenge at rank 2, at rank 14 on ARC-Challenge at rank 32, and at ranks 1 and 7 on the format task with Qwen2.5-1.5B. *At twice the budget, a control with three rates per arm keeps every frozen arm’s rate; only **free** moves, to the upper edge of its three rates, and its tests keep the paper’s 10^{-3} .

S16.3 Learning rates in detail

As a secondary result, the best learning rate depends on the band (§S16.1). Without Σ in the frozen factor, the minor band selects a rate one grid step lower than the other B-frozen arms. This shift is confirmed on five seeds on the format task only. With $B = U_r \sqrt{\Sigma_r}$ frozen, the direction reverses, and one scale per module accounts for Σ (below). The inversion of rankings at a common rate is LoRA-XS’s result (Bałazy et al., 2024). In our data, both of its halves are established only on Qwen2.5-0.5B, whose compared cell predates its protocol, and post hoc on Llama at rank 2 against **top** alone.

A five-seed selection confirms this direction on the format task, and supports it only in part on OpenBookQA. The rate also depends on Σ . With $B = U_r \sqrt{\Sigma_r}$ frozen, the direction reverses (Fig. S8). Under Adam, the scale of a frozen factor acts as a per-module learning rate, which predicts this reversal (§S10). The reversal agrees with LoRA-XS, with the lower rate PiSSA needs in Lee et al. (2026), and with the curvature of principal directions reported under PiSSA in RLVR (Zhu et al., 2025). One scale per module accounts for Σ . A single scalar per module reproduces the arm with Σ within one point on 40 seeds, an established equivalence (§S9.3). Moving that scalar into a per-module learning rate gives a marginally established difference instead. At a common rate, the shift can invert which subspace looks best, on the format task at ranks 2 and 4, as LoRA-XS (Bałazy et al., 2024) found (Fig. S9). The inversion is not reproduced at rank 16, at twice the training budget, or on the public benchmarks (§S23.4). The shift is not an artefact of gradient clipping (§S23.2.1). Curvature predicts the direction of the shift, not its size (§S23.2.2). Finally, the energy of the update on the side of a frozen factor equals one by construction, so it is not evidence about the band (§S7).

S17 Method details, continued

S17.1 Selection details

We require every selected rate to be an interior maximum of its sweep. Appendix E of the main paper lists the cells where it is not: several arms select 10^{-2} , the top of the main grid, and at rank 64 **bottom** selects $5 \cdot 10^{-4}$, the lowest rate it was run at (§S5.2).

Exceptions. Two selections are not independent of the reported seeds. **random_ortho** at rank 2 takes its rate from the seed grid itself: 10^{-2} , a clear optimum (0.674 against 0.571 for the runner-up). The common rate of the rank-64 row’s secondary line comes from a validation split of runs that include the reported seeds; the row’s per-arm rates are selected on seeds 0 and 1 (§S5.2). One selection also breaks the eight-field rule. **top** at rank 4 takes its rate from a sweep on 400 validation items, while its reported cells use 614. The 614-item position sweep makes its own selection for the same arm ($\beta = 0$). We report both: the *400-item selection* ($2 \cdot 10^{-3}$, Table S14) and the *614-item selection* ($5 \cdot 10^{-3}$, Table S12).

The rate of bottom. In the first selection, on seeds 0 and 1, **bottom** scored 0.6312 at $2 \cdot 10^{-3}$ and 0.6325 at $5 \cdot 10^{-3}$, a tie far inside the noise, while the position sweep selected $5 \cdot 10^{-3}$ for the same band ($\beta = 1$) by a clear margin. We added three selection seeds, 30 to 32, disjoint from every seed reported anywhere in the paper, under a rule written before the runs: the rate that maximises validation over seeds 0, 1 and 30 to 32 is retained if it leads by more than 0.005. It does: $5 \cdot 10^{-3}$ scores 0.6330, against 0.6057 at $2 \cdot 10^{-3}$ and 0.6040 at $3 \cdot 10^{-3}$ (three seeds), a margin of 0.027. $5 \cdot 10^{-3}$ is therefore the reported rate. The numbers at $2 \cdot 10^{-3}$, **bottom**’s former rate, are kept in Table S22.

The size of the shift. The selection above fixes **bottom**’s rate at $5 \cdot 10^{-3}$. It agrees with the evidence without clipping, where **bottom** also peaks at $5 \cdot 10^{-3}$ and falls at 10^{-2} (Table S30). At the selected rates, the rate shift is therefore a factor of two. The five-seed selection of §S16.2 moves **bottom** to $2 \cdot 10^{-3}$, and the size of the shift lies between a factor of two and five, because the minor band’s optimum is flat between $2 \cdot 10^{-3}$ and $5 \cdot 10^{-3}$. The gap to free LoRA exceeds the spread among frozen arms at either rate of **bottom**: 1.3 times at $2 \cdot 10^{-3}$, 1.9 times at $5 \cdot 10^{-3}$ (Table S22).

bottom at	score	spread	gap to free LoRA
$2 \cdot 10^{-3}$ (former)	0.5658	8.76	11.50
$5 \cdot 10^{-3}$ (selected)	0.5922	6.12	11.50

Table S22: **bottom** on Llama at rank 2, at its former rate and at the rate the selection retains: valid-format accuracy, the spread among frozen arms and the gap to a budget-matched free LoRA, in points, five seeds each. No run is excluded.

S17.2 Statistics in detail

α became a field after one run of another configuration had been pooled into a reported cell; §S23.2.1 gives the details.

Why a common rate is biased. The frozen arms ($A = 0$ with a frozen orthonormal B) and the free arm ($B = 0$ with Kaiming A) do not share an optimum. Any common-rate comparison is therefore biased against at least one arm. Status: [Post hoc · Descriptive].

Batch size. The learning rate is not the only hyperparameter a spectral-position comparison can hide behind. Lee et al. (2026) show that the learning rate alone can close reported gaps between LoRA variants. Lee & Lee (Lee and Lee, 2026) show a parallel effect for batch size, and argue that it must be tuned jointly with the learning rate. We hold batch size fixed across every arm, so an interaction between the two is not ruled out.

Magnitude. We record the *update norm*, $\|(\alpha/r)BA\|_F$, per module on every run. We also record the *relative perturbation*: the Frobenius norm of the update divided by the Frobenius norm, not squared, of W_0 restricted to the adapted band.

Optimisation budget. A budget of *supervised tokens*, not a number of epochs, sets the number of steps, so every arm sees the same amount of supervision (Appendix E of the main paper).

Seeds. Rates are selected on the selection seeds, 0 and 1; three added selection seeds, 30 to 32, settle **bottom**’s rate at rank 2 on Llama (above). Results are reported on seeds 2 to 6, or 2 to 8 in the position sweep. The pre-registered extensions add the *fresh seeds*, 9 to 16 on Llama (seed 9 had already been run at part of the positions; §S3.3), and the *added seeds*, 7 to 16 on Qwen. The curvature trajectory runs use seeds 17 to 21. Cells have five seeds unless a table states otherwise, and every table gives the count.

Noise. Differences are computed on unrounded means, so they can differ by 0.01 from the difference of the printed values. Each format score is a proportion over 614 shared items, with a binomial standard error of about two points per run. This error cancels in part in paired comparisons between arms, but not in absolute statements. Training is not bitwise deterministic on our GPUs, as the fifteen fixed-seed re-runs of Appendix C.3 of the main paper show. Pairing by seed therefore shares the data order and the initial draw, not the whole trajectory. A comparison between two seed populations carries that spread as well. We report two distinct uncertainties: the between-seed standard deviation, and the binomial sampling error of the metric on the test set. Two seeds can agree perfectly and still both sit two points from the truth if the test set is small.

Replicates. Two cells were trained twice, **bottom** at $r = 2$ and **random** at $r = 1$, both on seed 2, and we report the mean of both runs.

Corrections and pooling. We apply the correction within each family of comparisons that answers one question, never across the paper as a whole. Every family is reported in full, so a reader who wants a paper-wide correction can apply it to the tables as printed. We do not pool seeds across populations that answer different questions. Where seeds are pooled, in the secondary tests of §S3, the text says so. The sign test bottoms out at 0.0625 for $k = 5$ and can establish nothing on its own.

S17.3 The rank regime

We chose the rank regime from a measurement of the free arm. At its selected rate, over five seeds, **free** scores 0.7684 ± 0.0198 at $r = 1$, 0.8033 ± 0.0031 at $r = 2$ and 0.8186 ± 0.0139 at $r = 4$ (Table S14). These ranks train 704,512, 1,409,024 and 2,818,048 parameters. Doubling the rank buys 3.5 points, then 1.5. The 11.5-point gap to a budget-matched free LoRA at rank 2 (§S11) therefore does not come from a shortage of directions. Status: [Post hoc · Descriptive].

free at $r = 1$ runs at 10^{-3} , the rate the selection rule picks. At $2 \cdot 10^{-3}$ it loses in validation (0.782 against 0.813).

This is the regime where a subspace constraint is likely to bite, so we sweep $r \in \{1, 2, 4\}$ there. Later campaigns check the main results at ranks 8 to 32, in the regime of the methods we study (§S23.5 and Appendix D of the main paper).

S18 The late campaigns: further records

Fresh seeds. On fresh seeds and one GPU model, the comparison of the introduction replicates at the rates already selected, and R and the cost of freezing B stay established on both bands (Table 12 of the main paper).

Four times more training. With four times the training tokens and every rate selected again, the cost of freezing B at equal rank narrows but stays established on both bands. Longer training helps the B-frozen

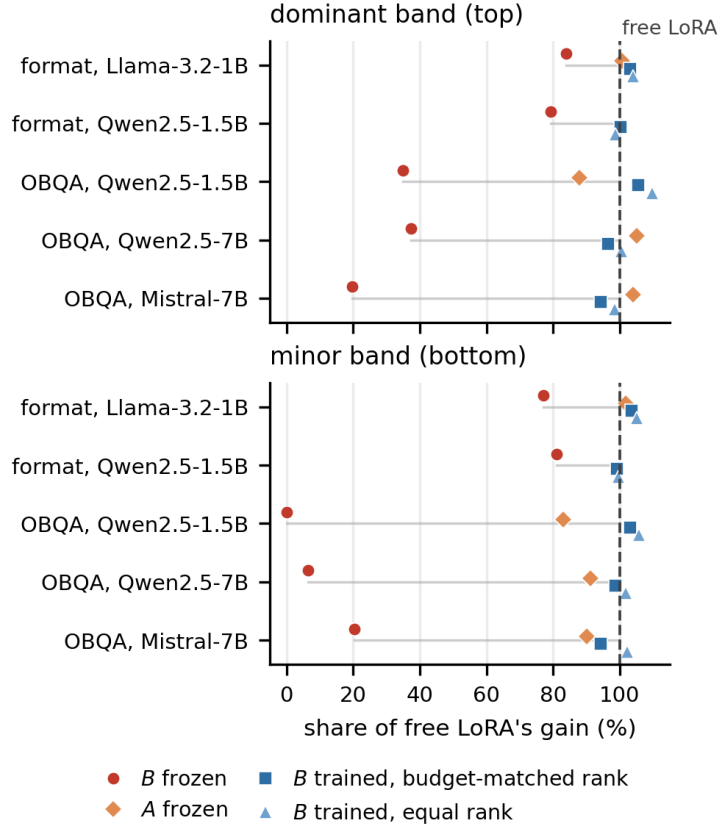


Figure S10: **Each arm’s share of free LoRA’s gain over the base model, at rank 2**, on the five task–model pairs of Table 9 of the main paper. Top, the dominant band; bottom, the minor band. B frozen, A frozen (not run on the format task with Qwen2.5-1.5B), and B trained at the budget-matched rank and at equal rank; the dashed line marks free LoRA. Share = $(\text{arm} - \text{base})/(\text{free} - \text{base})$, from the values of Table 9 of the main paper; a descriptive summary with no test.

arms more than the arms with B trained (Table 12 of the main paper). The gap between the bands still shrinks once B trains. Two selected rates lie at the edge of their grids. The rate of `top_unfrozen` could understate the cost of freezing the dominant band and overstate the reduction of the band gap. The rate of free LoRA could understate G .

The two 7-billion models. On both, the dominant A -frozen arm scores above free LoRA in the means, without a test. On Qwen2.5-7B, the minor band’s B -frozen arm selected the lowest rate of its grid, which may inflate every quantity that involves it. The dominant band, at an interior rate, carries the cost of freezing B and the comparison with freezing A on its own. On Mistral-7B, the dominant A -frozen arm selected the lowest rate of its grid, a limit in the conservative direction for this comparison.

S19 The rank-16 campaigns in detail

Table 10 of the main paper gives the numbers of the three rank-16 isolations; this appendix gives the record of each protocol.

The isolation at rank 16. The protocol of Table S17 runs at rank 16: the frozen references at their selected rates (format: `top` 0.7756 at $2 \cdot 10^{-3}$, `bottom` 0.7554 at $5 \cdot 10^{-4}$, free LoRA at rank 8 0.8205; ARC-Challenge: 0.4970 at $5 \cdot 10^{-4}$, 0.4452 at 10^{-3} , free LoRA at rank 7 0.4952), the unfrozen arms at the free LoRA’s

rank and at rank 16, with G fixed in advance (4.49 and 6.51; -0.18 and 5.00); test (1) applies only where $G \geq 2$. *Format task*: R is 5.05 for **top** ($[+3.46, +6.63]$, above $0.7G$: the freeze explains most of the gap) and 5.15 for **bottom** (inconclusive); cost at equal rank, 2.18 ($t = 3.80$, a marginal pass against 3.495) and 6.19 ($t = 16.51$); the gap between the bands goes from -2.02 to $+1.99$, a change of 4.01 ($t = 7.20$) that the oracle rate brings to 0.85 ($t = 2.36$): a sign flip, which we do not read as a shrinkage. **top_unfrozen** at rank 16 selects $2 \cdot 10^{-3}$, next to rates that diverge ($5 \cdot 10^{-3}$ and 10^{-2}), and falls to 0.7974 on the reported seeds, which inflates the unfrozen gap. *ARC-Challenge*: the cost at equal rank is nothing for **top** ($+0.32$) and 4.60 for **bottom** ($t = 12.08$); R for **bottom** is 3.78 (inconclusive; 4.86 at the oracle rate, above $0.7G$); the dominant band trained at rank 7 scores 0.5030 ($R = +0.60$, not tested, since $G < 2$); the gap between the bands goes from -5.18 to -0.90 , a reduction of 4.28 ($t = 13.93$; oracle 4.78). The GPU-model control of test (3) was computable on one and two seeds, which triggered the step the protocol planned: the four frozen references retrained, pinned to the same GPU model as the new runs. With them every compared evaluation runs on one GPU model: ARC-Challenge holds (**bottom**’s cost at equal rank 4.70, $t = 12.22$; reduction 4.46, $t = 8.91$); on the format task **bottom**’s cost at equal rank holds (6.61, $t = 12.32$), **top**’s is not established (3.42, $t = 3.01$), and the band change is a marginal pass (3.19, $t = 2.89$). The retrained references reproduce the originals within 0.1 point on ARC and 1.2 points on the format task. In this step the reading recomputed G from the retrained references rather than using the values fixed in advance; only one verdict depends on it, **top**’s R on the format task (6.29, $[+3.82, +8.75]$), which reads “the freeze explains most of the gap” with the pre-registered G and “inconclusive” with the recomputed one. Status: **[Pre-registered · Established]** on ARC-Challenge, robust to the oracle rate and to one GPU model; on the format task, the cost at equal rank is established for **bottom** and fragile for **top**, and the band change is a sign flip and is not robust.

The isolation at rank 16 on OpenBookQA. The protocol (PROTOCOL_isolation_obqa16.md) trains **top_unfrozen** and **bottom_unfrozen** at ranks 7 and 16, seven rates each, on seeds 0 to 6, on the configuration of the rank-16 campaign’s own runs; the references, including free LoRA at rank 7 at its only rate with five report seeds, $2 \cdot 10^{-3}$, and G were committed before any run, and the three references were re-trained on one GPU model as a control. All four selected rates are 10^{-3} , interior. Scores: 0.5552 at rank 7 and 0.5724 at rank 16 for the dominant band, 0.5552 and 0.5528 for the minor one, against 0.5408, 0.4984 and 0.5616 for the references.

The A-frozen arms at rank 16. The protocol (PROTOCOL_freezeA_rank16.md) and its addendum, committed before any result, add **dual_top** and **dual_bottom** at rank 16 on Qwen2.5-1.5B, seven rates each, on seeds 0 to 6, pinned to one GPU model; the references are the frozen arms of the rank-16 campaigns. By the addendum, the A-frozen arms hold 26.8% more parameters than the B-frozen ones (10,321,920 against 8,142,848), mostly through **gate** and **up**, and 27.8% more than free LoRA at rank 7. Selected rates, all interior: on ARC-Challenge $5 \cdot 10^{-4}$ (**dual_top**, 0.4922) and $2 \cdot 10^{-4}$ (**dual_bottom**, 0.4982), against the references 0.4970 and 0.4452; on OpenBookQA $2 \cdot 10^{-3}$ (0.5544) and 10^{-3} (0.5364), against 0.5408 and 0.4984. The original references ran on several GPU models; on ARC-Challenge, the control against the references re-trained on one GPU model gives the same verdicts (-0.56 , not established; $+5.40$, $t = 10.12$; band gap $+5.96$, $t = 7.14$). Against free LoRA at rank 7, reported without a test: -0.30 and $+0.30$ points on ARC-Challenge, -0.72 and -2.52 on OpenBookQA.

S20 Related work in full

No work in Table S23 compares positions with the rate tuned per position. The nearest antecedent is LoRA-XS (Balazy et al., 2024), which freezes both singular bases and trains an $r \times r$ core. Its ranking of the top and bottom bases inverts between two rates, so an inversion under a common rate is not new. We add a single frozen factor, per-position rates selected on disjoint seeds, and a budget-matched free reference. Our reading is that a known asymmetry between the factors (Zhu et al., 2024; Zhang et al., 2023) can confound comparisons across methods that freeze different factors, such as MiCA and PiCa. It can also confound the attribution of a frozen method’s gap to LoRA to its band. A band ablation within one method holds the factor fixed. LoRA+ (Hayou et al., 2024b) finds that B needs the larger learning rate, and freezing B is the limit of a zero rate on it, which is consistent with its cost. Hayou et al. (2024a) show that which factor starts at zero changes the dynamics. The LoRA paper (Hu et al., 2022) found that the learned update amplifies

	subspace	rate/arm
PiSSA (Meng et al., 2024)	dominant, init.	no
PiCa (Hwang et al., 2026)	dominant, A frozen, B	per method
	shared	
MiLoRA (Wang et al., 2024b)	minor, init.	no
MiCA (Rüdiger and Raschka, 2026), main	minor, B frozen	per method
MiCA, ablation	dominant, minor, random	no
LoRA-XS (Balazy et al., 2024)	spectral with Σ , frozen	grid ^b
KaSA (Wang et al., 2025)	minor, truncated + learned Σ	n.r.
SVFT (Lingam et al., 2024)	U and V of W_0 , frozen	no
SoMA (Yun et al., 2025)	minor, init.	n.r.
OPLoRA (Xiong and Xie, 2026)	complement of dominant, projected	—
Quercia et al. (2026)	swept, init.	no
Lee et al. (2026)	PiSSA, MiLoRA, ..., init.	yes
Lee & Lee (Lee and Lee, 2026)	LoRA variants	batch size
ROSA (Hameed et al., 2024)	resampled	yes
Zhu et al. (2024)	random, and B on the dominant band, frozen A or B	—
Gu et al. (2026)	Stiefel	—
this work	top/mid/bottom, frozen	yes

Table S23: **Works that fix or initialise a spectral subspace, and whether they tune the learning rate per arm.** LoRA-XS compares top and bottom bases over a grid of rates, and its ranking inverts with the rate; none combines a per-position rate selected on separate seeds with a budget-matched free reference.
^b Three adapter by three classifier rates, every cell reported (their Appendix G), without separate selection seeds or a free reference. In the rate column, *no* means the paper reports one rate for every arm it compares, *n.r.* that we could not find a per-variant rate in it, either way, and — that the column does not apply.

directions that W_0 does not emphasise, and Shuttleworth et al. (2024) name them intruder dimensions, both anticipating our first phase. Data-derived subspaces are the non-oracle counterpart of our second phase. LoRA-GA (Wang et al., 2024a) initialises A and B from the singular vectors of the full gradient at the first step, and CorDA (Yang et al., 2024) and EVA (Paischer et al., 2024) derive the subspace from task data. LoRA-One (Zhang et al., 2025a) proves that LoRA adapters align with singular subspaces of the one-step full fine-tuning gradient. Our frozen B on those directions does not recover the gap, even with the gradient accumulated. LoRA-SB (Ponkshe et al., 2024) freezes B and A on the SVD of an approximate first full update and trains an $r \times r$ core, the nearest method to a frozen B at a learned place. GOAT (Fan et al., 2025) initialises LoRA experts on different SVD segments and finds the useful segment task-dependent. Our learned-direction arm is a diagnostic close in spirit to LoRA-SB and LoRA-One, and the paper does not propose a method. LoRA-FA (Zhang et al., 2023) freezes a random A and trains B , close to LoRA, a direct precedent for the low cost of freezing A . Unequal tuning has reordered published rankings before (Melis et al., 2018; Lucic et al., 2018; Dodge et al., 2019; Bouthillier et al., 2021). Lee et al. (2026) find that per-variant tuning closes the gaps between LoRA variants that initialise on a band. For variants that freeze B on a band at rank 2, it does not (§3 of the main paper). The rest of the related work is in §S14.

S21 Setup in full

We adapt seven modules per layer of Llama-3.2-1B (Grattafiori et al., 2024), of Qwen2.5-0.5B, 1.5B and 7B (Qwen Team, 2024), and of Mistral-7B-v0.3 (Jiang et al., 2023). A layer has frozen weights $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ with reduced SVD $W_0 = U \text{diag}(\sigma_1, \dots, \sigma_m) V^\top$. LoRA replaces $W_0 x$ with $W_0 x + \frac{\alpha}{r} B A x$. The *dominant band* is the r leading singular directions. The *minor band* is the r trailing ones with nonzero singular value. Table 1 of the main paper lists the arms and their budgets.

A *B-frozen arm* fixes B on a band, without Σ , and trains A from zero. Our *bottom* arm is MiCA’s parametrisation as implemented in PEFT (Appendix C.2 of the main paper). Two further arms isolate the frozen factor. One trains B from the band’s basis, at equal rank or at the budget-matched rank. It is not PiSSA or MiLoRA, since A starts at zero and nothing is subtracted from W_0 . The other freezes A on the same band and trains B from zero. The budget-matched reference is free LoRA at half the B-frozen arms’ rank. Every conclusion about the subspace is read among the B-frozen arms, which hold identical budgets.

The format task applies an invented output format to CoNLL-2003 sentences (Tjong Kim Sang and De Meulder, 2003), scored by valid-format accuracy. OpenBookQA (Mihaylov et al., 2018) and ARC (Clark et al., 2018) are trained on their own training splits and scored by multiple-choice accuracy, computed as described in Appendix C.3 of the main paper. A knowledge task is reported in §S5. Each arm’s learning rate is selected on a validation set and on two seeds that no reported number uses. Comparisons are paired by seed, with a Bonferroni correction within each family of comparisons, and gaps are in points. Appendix C of the main paper gives the models, the tasks, the selection and the status tags in full.

S22 Every test in one place

Table S24 lists every test the paper reports. The main paper’s count of 165 tests, 127 of them pre-registered, is the 171 rows of Tables S24 to S28, minus the four descriptive rows (D) and the two selection outcomes (C). The one pre-registered row that was not read, on **q** and **o** only, is counted among the 127. Each row gives the comparison, the seeds the rate was selected on and those the result is reported on, the family the Bonferroni correction is taken over, the threshold that follows, the observed statistic, and the status tag of Appendix C.4 of the main paper. Rows that stand for a whole family point to Table S3, which gives each comparison. A test is called established only when it clears its threshold; the paper reports the others as well, and says so where they fall short. Thresholds are those stated in the text, and a cell says so where the text states none.

Exact ties. Four pairs of arms selected at the same rate have identical mean scores: on the format task at rank 8, **top** and **bottom**; on OpenBookQA at rank 2, **top** and **random**; with Mistral-7B at rank 1 and on OpenBookQA at rank 7, **top_unfrozen** and **bottom_unfrozen**. Each pair consists of distinct runs, with different losses and per-seed scores; only the total count of correct answers over five seeds coincides, which integer scores make likely across the dozens of close pairs we compare. On Qwen2.5-7B, **top** also scores 0.5464 at rank 3 ($2 \cdot 10^{-3}$) and at rank 4 ($5 \cdot 10^{-3}$), a coincidence of totals like the others. On the format task with Llama-3.2-1B, *R* on the dominant band is 14.85 on seeds 2 to 6 and on fresh seeds 120 to 124. Both the frozen arm and the arm that trains *B* at rank 1 moved by exactly +1.50 points.

Marginal passes. A pass is *marginal*, marked ^m, when its statistic lies within 15% of its threshold: for a *t* test, when $|t|$ is below 1.15 times the threshold; for an interval criterion, when the distance from the estimate to the criterion is below 1.15 times the distance from the estimate to the bound of the interval that faces the criterion; for an exact test, when *p* is above half its threshold. The rule is applied to every established result in Tables S24 to S28 and in Table S3, and the text calls a result marginal only where the mark applies.

S23 Further results, not used by the main paper

These results are recorded for completeness. No claim of the main paper rests on them.

S23.1 Secondary: the direction of forgetting follows the learning rate

*Unless marked as a task score, numbers in this section are WikiText perplexities. Three values of Table S29, **random**, **random_ortho** and **top** at their selected rates, were first measured before the forgetting script was corrected; all three have been re-measured with the corrected script, which moved them by at most 0.06, and the re-measurement also moved **free** at rank 2 from 24.06 to 23.81. The table and the tests use the re-measured values; every other value, including the base model’s 14.09, comes from the corrected script. The perplexities of §S13 come from the evaluation run at the end of training, and the ‘three more units’ of §S4.3 compare values at the common rate: neither depends on the correction.*

The order of forgetting inverts with the learning rate, as the order of the score does: the direction holds on every seed, but at five seeds it does not pass a corrected threshold. The premise that LoRA forgets less than full fine-tuning (Biderman et al., 2024) motivates using a frozen subspace to shape *which* directions an adapter can move in at all. Quercia et al. find forgetting U-shaped across the spectrum on initialisations, highest at both ends. We measure it on frozen subspaces in the two regimes this paper keeps apart, with one procedure and the same five seeds throughout (Table S29).

comparison	seeds: rate / reported	family	threshold	observed	status
<i>position sweep, Llama rank 2</i> (§S3.3, §S3.5)					
$\beta=0.75$ vs the four other positions, first seven seeds	0,1 / 2-8	4	$t > 3.52$, 6 d.f., 2-sided	Table S3; the pass vs $\beta=0.25^m$	PH; Table S3
the same, sign and exact Wilcoxon tests	0,1 / 2-8	4	$p < 0.0125$	$p \geq 0.016$	PH · NE
$\beta=0.75$ vs random_ortho , first seven seeds	0,1 / 2-8	added to the 4	$t > 2.45$ alone; $t > 3.71$ in the family	$t = 2.94$	PH · NE
extension: $\beta=0.75$ vs the four other positions, fresh seeds	0,1 / 9-16	5	$t > 2.998$, 7 d.f., 1-sided	Table S3; the pass vs the top ^m	PR; Table S3
extension: $\beta=0.75$ vs random_ortho , fresh seeds	0,1 / 9-16	same 5	$t > 2.998$	$t = 2.75$	PR · NE
extension: $\beta=0.75$ vs the top, exact Wilcoxon	0,1 / 9-16	5	$p < 0.01$	$p = 0.008$	PH · E ^m
extension: $\beta=0.75$ vs the top, sign test	0,1 / 9-16	5	$p < 0.01$	$p = 0.035$	PH · NE
$\beta=0.75$ vs random_ortho , fifteen seeds pooled (the protocol's second test)	0,1 / 2-16	5	$t > 2.624$	$t = 3.94$	PR · D
<i>position sweep, Llama rank 4</i> (§S3.2)					
$\beta=0.75$ vs the top, and vs $\beta=0.5$	0,1 / 2-8	—	none fixed in advance	$t = 0.60$, 0.33	PH · D
<i>position sweep, Qwen2.5-0.5B</i> (§S3.4, §S3.5)					
first five seeds, $\beta=0.75$ vs the four other positions	0,1 / 2-6	4	not stated	$t = 1.08$ to 1.79	PH · NE
fifteen-seed pool	0,1 / 2-16	4	$t > 2.51$, 14 d.f., 1-sided	Table S3	PR · D
ten added seeds alone	0,1 / 7-16	4	$t > 2.685$, 9 d.f., 1-sided	Table S3	PH; Table S3
$\beta=0.75$ vs random_ortho , ten added seeds	0,1 / 7-16	1	$t > 1.833$, 9 d.f., 1-sided	$t = 2.08$	PR · E ^m
the same, fifteen seeds (the protocol's secondary test)	0,1 / 2-16	4	$t > 2.51$, 14 d.f., 1-sided	$t = 0.75$	PR · NE
<i>the minor band's deficit, rank 2</i> (§S4)					
bottom vs the three other main arms, Llama	0,1,30-32 / 2-6	3	$t > 3.96$, 4 d.f., 2-sided	$t = 4.39$, 5.97, 7.42	PH · E (vs top ^m)
bottom vs the nearest other main arm, Llama, five non-degenerate modules	0,1,30-32 / 2-6	—	t interval excludes 0	5.85 [2.9, 8.8]	PH · E
the same on Qwen2.5-0.5B	0,1 / 2-6	3	$t > 3.96$	vs random_ortho : 3.77 at $2 \cdot 10^{-3}$, 0.96 at $5 \cdot 10^{-3}$	PH · NE
<i>common-rate inversion (no rate selection): paired lead at $2 \cdot 10^{-3}$ and deficit at 10^{-2}</i> (§S23.4)					
Llama rank 2, lead	— / 2-6	1	$t > 2.776$	$t = 1.12$	PH · NE
Llama rank 2, lead against top alone	— / 2-6	—	not stated (the halves use $t > 2.776$)	+4.98 pt, $t = 5.68$	PH · E
Llama rank 2, deficit	— / 2-6	1	$t > 2.776$	$t = 17.14$	PH · E
Llama rank 4, lead	— / 2-6	1	$t > 2.776$	$t = 2.42$	PR · NE
Llama rank 4, deficit	— / 2-6	1	$t > 2.776$	$t = 13.48$	PR · E
Qwen2.5-0.5B, lead	— / 2-6	1	$t > 2.776$	$t = 5.16$	PR [†] · E
Qwen2.5-0.5B, deficit	— / 2-6	1	$t > 2.776$	$t = 9.08$	PR · E
Qwen2.5-1.5B, lead	— / 2-6	1	$t > 2.776$, 4 d.f., 2-sided	$t = 8.70$	PR · E
Qwen2.5-1.5B, deficit	— / 2-6	1	$t > 2.776$	$t = 2.14$	PR · NE
arm × rate interaction, each configuration against each fixed competitor	— / 2-6	11	$t > 5.75$, 4 d.f., 2-sided	$t \geq 8.0$, positive on every seed	PH · E
<i>rank 16, OpenBookQA, Qwen2.5-1.5B, pre-registered</i> (§S23.5)					
the floor: free above the base model	0,1 / 2-6	—	5 points, fixed in advance	+14.0 pt, 5/5	PR · E
bottom vs top at 10^{-3}	0,1 / 2-6	—	$t > 2.776$, 4 d.f., 2-sided	$t = -10.75$, 0/5 ahead	PR · E
bottom vs the best other arm at $2 \cdot 10^{-3}$	0,1 / 2-6	—	$t > 2.776$, 4 d.f., 2-sided	$t = 11.07$, 5/5	PR · E
gap minus spread, bootstrap	0,1 / 2-6	—	interval excludes 0	-2.16 [-4.60, +0.64]	PR · NE
<i>gap minus spread: paired bootstrap, 95% percentile intervals, approximate at five seeds</i> (§S11)					
rank 2, main arms	0,1,30-32 / 2-6	—	interval excludes 0	+5.4 [+2.3, +7.0]	PH · E
rank 4, main arms	0,1 / 2-6	—	interval excludes 0	+5.9 [+2.7, +7.8]	PH · E
rank 4, along the spectrum	0,1 / 2-6	—	interval excludes 0	+4.9 [+0.6, +5.8]	PH · E ^m
rank 2, along the spectrum	0,1 / 2-6	—	interval excludes 0	-0.1 [-4.3, +4.0]	PH · NE
rank 4, main arms with top_sigma , seven seeds (§S11.4)	0,1 / 2-8	—	interval excludes 0	+5.44 [+2.98, +7.42]	PR · E
<i>rate shift without clipping, Llama rank 2, on a rate grid</i> (§S23.2.1)					
bottom's gain from 10^{-2} to $5 \cdot 10^{-3}$	— / 2-6	1	$t > 2.78$	$t = 2.56$	PH · NE
the same move, contrast with top	— / 2-6	2	$t > 3.50$, 4 d.f., 2-sided	$t = 3.77$	PH · E ^m
the same move, contrast with random	— / 2-6	same 2	$t > 3.50$	$t = 4.74$	PH · E

Table S24: Every test the paper reports (continued in Table S25), with the observed statistic and its status (Appendix C.4 of the main paper): PR pre-registered, PH post hoc; E established, NE not established, D descriptive; C a selection outcome written as a prediction before the runs and confirmed, not a statistical test. Rate seeds and reported seeds are disjoint, except for the two selections of Appendix C.4 of the main paper. In the *post hoc*, *no family* rows the rate was selected as usual, but the comparison was decided after seeing the reported seeds, so it carries no confirmatory weight. Every pre-registration was audited against the repository history. [†]One of the compared cells predates its protocol, which listed it as new (§S23.4.3). ^m: a marginal pass, whose statistic lies within 15% of its threshold (rule in the text above).

comparison	seeds: rate / reported	family	threshold	observed	status
<i>forgetting</i> , Llama rank 2 (§S23.1) common rate, each arm vs top [†]	— / 2-6	4	$t > 4.31$, 4 d.f., 2-sided	random_ortho -4.85, bottom 3.91, free 17.57, random -3.25	PH; E for random_ortho ^m and free, NE otherwise PH; E for random and free
selected rates, each arm vs bottom at $5 \cdot 10^{-3}$ [†]	0, 1 / 2-6	4	$t > 4.31$	random -6.73, free -7.10, random_ortho -0.83, top 1.84	PH · NE
top vs random at 10^{-2} [†]	0, 1 / 2-6	1	$t > 2.776$, 4 d.f., 2-sided	$t = 2.18$	PH · NE
knowledge, rank 64 (§S5.2) free vs top, random and bottom, each at its selected rate	0, 1 / 2-6	3	$t > 3.96$, 4 d.f., 2-sided	$t = 4.90, 7.08, 8.74$	PH · E
the same at a common 10^{-3} , the secondary line	Appendix C.4 of the main paper / 2-6	3	$t > 3.96$	$t = 4.90, 4.92, 6.28$	PH · E
gradient probe, before training (§S4.3) knowledge vs format, paired by (layer, module)	—	1	not stated	$p = 0.15$ (47 of 80)	PH · NE
post hoc, no family (§S9) top_sigma vs top	0, 1 / 2-6	—	$t > 2.776$, 4 d.f., 2-sided	$t = 4.22$	PH · E
top_sigma vs random_ortho	0, 1 / 2-6	—	$t > 2.776$	$t = 12.79$	PH · E
top_sigma vs $\beta=0.75$	0, 1 / 2-6	—	$t > 2.776$	$t = 0.27$	PH · NE
top_scalar at rank 2, pre-registered (§S9.3) top_scalar vs top, the gain g	0, 1 / 2-6	1	$t > 2.776$, 4 d.f., 2-sided	$t = 3.32$	PR · E
top_sigma vs top_scalar, the remainder	0, 1 / 2-6	—	$t > 2.776$	$t = 2.21$	PR · NE
late campaigns, pre-registered (§S16 and Appendix D of the main paper)					
format $r=16$: interaction, 3 arms	0, 1 / 2-6	3	$t > 3.96$	$t = 0.69-2.81$	PR · NE
format $r=16$: gap - spread	0, 1 / 2-6	—	bootstrap 95 %	$[-0.81, +3.39]$	PR · NE
format $r=8$: gap - spread	40, 41 / 42-46	—	bootstrap 95 %	$[+3.09, +7.33]$	PR · E
2× budget: interaction, 3 arms	0, 1 / 2-6	3	$t > 3.96$	$t = 3.06-3.83$	PR · NE
2× budget: gap - spread	0, 1 / 2-6	—	bootstrap 95 %	$[+1.66, +4.36]$	PR · E
OpenBookQA $r=2$: lead of bottom, 2 pairs	0, 1 / 2-6	2	$t > 3.495$	$t = -4.06$	PR · NE
OpenBookQA $r=2$: gap - spread	0, 1 / 2-6	—	bootstrap 95 %	$[+0.32, +5.80]$	PR · E ^m
ARC-Easy: lead of bottom	0, 1 / 2-6	—	$t > 2.776$	$t = -10.73$	PR · NE
ARC-Easy: gap - spread	0, 1 / 2-6	—	bootstrap 95 %	$[-3.16, -1.50]$	PR · E (reversed)
ARC-Challenge: gap - spread	0, 1 / 2-6	—	bootstrap 95 %	$[-7.48, -3.24]$	PR · E (reversed)
bottom_sigma above top_sigma (7 and 5 modules)	0, 1	—	selected rate	$2 \cdot 10^{-2}$ vs $5 \cdot 10^{-3}$	PR · C
top_sigma vs top_scalar, 15 seeds	— / 50-64	—	$t > 2.145$; $IC_{90} \subset \pm 1$	$[-1.16, +0.61]$	PR · NE
top_sigma vs top_scalar, 40 seeds	— / 65-104	—	$t > 2.023$; $IC_{90} \subset \pm 1$	$[-0.57, +0.60]$	PR · E (equivalence)
top_unfrozen: R against 0.7 G	0, 1 / 2-6	2	IC_{95} above 0.7 G	$[+13.14, +16.57]$; 8.69	PR · E
bottom_unfrozen: R against 0.7 G	0, 1 / 2-6	2	IC_{95} above 0.7 G	$[+18.47, +22.18]$; 12.34	PR · E
cost at equal rank, top ($r=2$)	0, 1 / 2-6	2	$t > 3.495$	$t = 19.17$	PR · E
cost at equal rank, bottom ($r=2$)	0, 1 / 2-6	2	$t > 3.495$	$t = 30.86$	PR · E
ARC-Challenge $r=2$: gap - spread	0, 1 / 2-6	—	bootstrap 95 %	$[-1.74, +2.34]$	PR · NE
ARC-Challenge $r=32$: gap - spread	0, 1 / 2-6	—	bootstrap 95 %	$[-4.98, -0.60]$	PR · E (reversed)
format: bands, B trained vs frozen	— / 2-6	1	$t > 2.776$	$t = 4.96$	PH · E
freezing A vs B , top ($r=2$)	0, 1 / 2-6	2	$t > 3.495$	$t = 11.24$	PR · E
freezing A vs B , bottom ($r=2$)	0, 1 / 2-6	2	$t > 3.495$	$t = 35.99$	PR · E
format: bands, A frozen vs B frozen	— / 2-6	1	$t > 2.776$	$t = 3.91$	PH · E
OpenBookQA, top_unfrozen: R against 0.7 G	0, 1 / 2-6	2	IC_{95} above 0.7 G	$[+7.73, +9.71]$; 5.63	PR · E
OpenBookQA, bottom_unfrozen: R against 0.7 G	0, 1 / 2-6	2	IC_{95} above 0.7 G	$[+10.21, +15.31]$; 8.65	PR · E
OpenBookQA, cost at equal rank, top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 12.79 / 19.15$	PR · E
OpenBookQA: bands, B trained vs frozen the same, on one GPU model (follow-up)	0, 1 / 2-6	1	$t > 2.776$	$t = 4.12$	PR · E
OpenBookQA: freezing A vs B , top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 9.13$	PR · E
OpenBookQA: bands, A frozen vs B frozen	0, 1 / 2-6	1	$t > 2.776$	$t = 9.66 / 11.39$	PR · E
top_lrsize vs top_scalar, 40 seeds	— / 65-104	1	$IC_{90} \subset \pm 1$; $t > 2.023$	$t = 4.81$ $[-1.50, -0.16]$; $t = -2.08$	PR · E PR · E ^m (difference)
format (Qwen) $r=2$: gap - spread	20, 21 / 22-26	—	bootstrap 95 %	$[+11.30, +14.20]$	PR · E
format (Qwen) $r=16$: gap - spread	20, 21 / 22-26	—	bootstrap 95 %	$[-0.52, +6.09]$	PR · NE
rate shift on five seeds, format / OpenBookQA	0, 1, 110-112 / —	—	lower than all three	confirmed / partial	PR · C / partial
format $r=16$: R against 0.7 G , top / bottom	0, 1 / 2-6	2	IC_{95} above 0.7 G	$[+3.46, +6.63]$ / $[+3.38, +6.91]$	PR · E / inconclusive
format $r=16$: cost at equal rank, top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 3.80 / 16.51$	PR · E (top ^m)
format $r=16$: bands, B trained vs frozen (sign flip)	0, 1 / 2-6	1	$t > 2.776$	$t = 7.20$ (oracle 2.36)	PR · E, fragile
both, on one GPU model (planned step): cost at equal rank, top; bands	0, 1 / 2-6	2; 1	$t > 3.495$; $t > 2.776$	$t = 3.01$; 2.89	PR · NE; PR · E ^m
ARC-Challenge $r=16$: cost at equal rank, bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 12.08$	PR · E
ARC-Challenge $r=16$: bands, B trained vs frozen	0, 1 / 2-6	1	$t > 2.776$	$t = 13.93$ (oracle 13.72)	PR · E
both, on one GPU model (planned step)	0, 1 / 2-6	—	as above	$t = 12.22$; 8.91	PR · E

Table S25: Every test the paper reports, continued from Table S24; same notation, ^m included. [†]Forgetting, on the corrected re-measurement (§S23.1).

comparison	seeds: rate / reported	family	threshold	observed	status
format $r=2$, Qwen2.5-1.5B: R against 0.7 G	20, 21 / 22-26	2	IC ₉₅ above 0.7 G	[+14.08, +18.69] / [+11.72, +16.68]	PR · E
format $r=2$, Qwen2.5-1.5B: cost at equal rank, top / bottom	20, 21 / 22-26	2	$t > 3.495$	$t = 14.17 / 12.54$	PR · E
format $r=2$, Qwen2.5-1.5B: bands, B trained vs frozen	20, 21 / 22-26	1	frozen gap ≥ 2	frozen gap 1.34	PR · below floor
format $r=2$, 4 $\times$ tokens: cost at equal rank, top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 12.26 / 16.57$	PR · E
format $r=2$, 4 $\times$ tokens: bands, B trained vs frozen	0, 1 / 2-6	1	$t > 2.776$	$t = 5.90$	PR · E
format $r=2$, q and o only: A-frozen minus B-frozen, top / bottom	130, 131 / 140-144	2	$t > 3.495$	$t = 18.78 / 33.54$	PR · not interpretable (collapse) / E
format $r=2$, q and o only: bands, A frozen vs B frozen	130, 131 / 140-144	1	$t > 2.776$	sign change	PR · not read
OBQA $r=16$: cost at equal rank, top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 5.09 / 9.20$	PR · E
OBQA $r=16$: R against 0.7 G , top / bottom	0, 1 / 2-6	2	IC ₉₅ above 0.7 G	[-0.83, +3.71]; 1.46 / [+3.83, +7.53]; 4.42	PR · inconclusive
OBQA $r=16$: bands, B trained vs frozen	0, 1 / 2-6	1	$t > 2.776$	$t = 3.94$	PR · E
ARC-C $r=16$: A-frozen minus B-frozen, top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = -0.78 / 12.42$	PR · NE / E
OBQA $r=16$: A-frozen minus B-frozen, top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 1.63 / 7.89$	PR · NE / E
ARC-C $r=16$: bands, A frozen vs B frozen	0, 1 / 2-6	1	$t > 2.776$	$t = 8.25$	PR · E
OBQA $r=16$: bands, A frozen vs B frozen	0, 1 / 2-6	1	$t > 2.776$	$t = 2.99$	PR · E ^m
OBQA $r=2$, Qwen2.5-7B: R against 0.7 G , top / bottom	0, 1 / 2-6	2	IC ₉₅ above 0.7 G	[+7.37, +10.95]; 6.78 / [+11.74, +16.74]; 10.11	PR · E
OBQA $r=2$, Mistral-7B: R against 0.7 G , top / bottom	0, 1 / 2-6	2	IC ₉₅ above 0.7 G	[+10.32, +12.88]; 8.74 / [+9.26, +13.70]; 8.65	PR · E
OBQA $r=2$, Mistral-7B: cost at equal rank, top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 11.52 / 17.53$	PR · E
OBQA $r=2$, Mistral-7B: A-frozen minus B-frozen	0, 1 / 2-6	2	$t > 3.495$	$t = 18.44 / 26.02$	PR · E
OBQA $r=2$, Mistral-7B: bands, B trained vs frozen	0, 1 / 2-6	1	$t > 2.776$	$t = 0.35$	PR · NE
trajectories of top_lr scale and top_scalar : single cause	— / 200	3	final $D > 10\times$ control	control 0.26: no cause can pass	PR · not conclusive
format $r=2$, fresh seeds: bands, B trained vs frozen	— / 120-124	1	$t > 2.776$	$t = 28.31$	PR · E
format $r=2$, fresh seeds: cost at equal rank, top / bottom	— / 120-124	2	$t > 3.495$	$t = 72.67 / 149.81$	PR · E
format $r=2$, fresh seeds: R against 0.7 G	— / 120-124	2	IC ₉₅ above 0.7 G	[+13.32, +16.39] / [+17.48, +23.37]	PR · E
format $r=2$, fresh seeds: bands, A vs B frozen	— / 120-124	1	$t > 2.776$	$t = 3.43$	PR · E
OBQA $r=2$, Qwen2.5-7B: cost at equal rank, top / bottom	0, 1 / 2-6	2	$t > 3.495$	$t = 17.10 / 26.59$	PR · E
OBQA $r=2$, Qwen2.5-7B: A-frozen minus B-frozen	0, 1 / 2-6	2	$t > 3.495$	$t = 29.48 / 25.53$	PR · E
OBQA $r=2$, Qwen2.5-7B: bands, B trained vs frozen	0, 1 / 2-6	1	$t > 2.776$	$t = 4.99$	PR · E
R against G itself, the five pairs of $r=2$ (a reading, no threshold)	—	—	—	reaches G in the means on the three pairs at 1-1.5B, all bands but one; 93-99% of G at 7B	PH · D
format $r=2$, Llama: A frozen at $r=2$ vs random at $r=4$, taken at $5\cdot 10^{-3}$, its most favourable rate (0.7121); 47,104 against 81,920 parameters per layer, dual_top / dual_bottom	— / 2-6	2	$t > 3.495$	+6.19 [+3.62, +8.50], $t = 4.43 / +6.97$ [+4.46, +8.57], $t = 5.60$	PH · E
factor $\times$ band, format $r=2$, Llama: factor / band / interaction	— / 2-6	—	bootstrap 95 %	[+15.13, +16.86] / [-3.24, -1.22] / [+3.36, +8.60]	PH · E
factor $\times$ band, format $r=2$, Llama, fresh seeds: interaction	— / 120-124	—	bootstrap 95 %	[+4.33, +10.88]	PH · E
factor $\times$ band, OBQA $r=2$, Qwen2.5-1.5B: factor / band / interaction	— / 2-6	—	bootstrap 95 %	[+7.20, +9.64] / [-3.60, -1.60] / [+2.24, +4.80]	PH · E
factor $\times$ band, OBQA $r=2$, Qwen2.5-7B: factor / band / interaction	— / 2-6	—	bootstrap 95 %	[+11.34, +12.38] / [-3.78, -3.14] / [+1.48, +3.64]	PH · E
factor $\times$ band, OBQA $r=2$, Mistral-7B: factor / band / interaction	— / 2-6	—	bootstrap 95 %	[+11.28, +13.04] / [-2.34, +0.44] / [-3.16, -1.40]	PH · E / NE / E
OBQA $r=2$, Qwen2.5-1.5B: learned — top , replication A (direction seed 7)	0, 1 / 2-6	1	lower bound $\geq 0.7 G = 5.63$; $t > 2.776$	[+7.66, +9.86], $t = 22.1$	PR · E
format $r=2$, Llama-3.2-1B: learned — top , replication B (direction seed 0)	0, 1 / 2-6	1	lower bound $\geq 0.7 G = 8.69$; $t > 2.776$	[+15.15, +17.88], $t = 33.7$	PR · E

Table S26: Every test (continued): the campaigns read on 27/09, the square modules, the isolation at rank 16 on OpenBookQA, the A-frozen arms at rank 16, the budget-matched recovery on the two 7-billion models, the rank-2 isolation with four times more training, the central result on Mistral-7B, the trajectories of the per-module rate and the scalar, freezing B isolated on the format task with Qwen2.5-1.5B, the rank-2 comparison on fresh seeds and the central result on Qwen2.5-7B (§3 of the main paper, Appendix D of the main paper). Columns as in Table S24.

comparison	seeds: rate / reported	family	threshold	observed	status
OBQA $r=2$ vs $r=3$, Qwen2.5-1.5B: A frozen ($r=2$) – B frozen ($r=3$), top / bottom	0, 1 / 2–6	2	$t > 3.495$	$t = 9.71 / 10.14$	PR · E
the same, Qwen2.5-7B	0, 1 / 2–6	2	$t > 3.495$	$t = 33.54 / 22.59$	PR · E
OBQA $r=4$, Qwen2.5-1.5B: R against 0.7 G , top / bottom	0, 1 / 2–6	2	IC ₉₅ above 0.7 G	[+7.53, +10.39] / [+7.56, +12.36]	PR · E
OBQA $r=8$, Qwen2.5-1.5B: R against 0.7 G	0, 1 / 2–6	2	IC ₉₅ above 0.7 G	[+4.13, +6.11] / [+6.06, +9.22]	PR · E / inconclusive
OBQA $r=4$: cost at equal rank	0, 1 / 2–6	2	$t > 3.495$	$t = 51.39 / 7.78$	PR · E
OBQA $r=8$: cost at equal rank	0, 1 / 2–6	2	$t > 3.495$	$t = 19.11 / 26.89$	PR · E
OBQA $r=4$: A frozen – B frozen	0, 1 / 2–6	2	$t > 3.495$	$t = 12.30 / 13.77$	PR · E
OBQA $r=8$: A frozen – B frozen	0, 1 / 2–6	2	$t > 3.495$	$t = 11.29 / 7.07$	PR · E
OBQA $r=4$: bands	0, 1 / 2–6	1	floor 2	frozen gap –1.20	PR · below floor
OBQA $r=8$: bands, B trained vs frozen	0, 1 / 2–6	1	$t > 2.776$	$t = 11.06$	PR · E
OBQA $r=8$: bands, A frozen vs B frozen	0, 1 / 2–6	1	$t > 2.776$	$t = 0.47$	PR · NE
OBQA $r=16$, Qwen2.5-7B: R against 0.7 G	0, 1 / 2–6	2	IC ₉₅ above 0.7 G	[+1.74, +2.82]; 2.10 / [+4.43, +7.57]; 4.51	PR · inconclusive
OBQA $r=16$, Qwen2.5-7B: cost at equal rank	0, 1 / 2–6	2	$t > 3.495$	$t = 5.13 / 5.52$	PR · E
OBQA $r=16$, Qwen2.5-7B: A frozen – B frozen	0, 1 / 2–6	2	$t > 3.495$	$t = 4.04 / 6.58$	PR · E
OBQA $r=16$, Qwen2.5-7B: bands, B trained vs frozen	0, 1 / 2–6	1	$t > 2.776$	$t = 1.88$	PR · NE
OBQA $r=16$, Qwen2.5-7B: bands, A frozen vs B frozen	0, 1 / 2–6	1	$t > 2.776$	$t = 3.03$	PR · E ^m
OBQA $r=2$, Qwen2.5-1.5B: learned – top (phase 2, launch declared)	0, 1 / 2–6	1	lower bound $\geq 0.7 G = 5.63$; $t > 2.776$	[+8.42, +10.30], $t = 27.77$	PR · E
OBQA $r=2$, Qwen2.5-1.5B: grad – top , B on the initial gradient over 64 sequences	0, 1 / 2–6	1	P : lower bound ≥ 0.7 suffices, upper bound < 0.3 does not	$P = -0.47$ [–0.63, –0.32]	PR · does not suffice
format $r=2$, Llama-3.2-1B: the same	0, 1 / 2–6	1	the same	$P = 0.18$ [0.08, 0.29]	PR · does not suffice
OBQA $r=2$, Qwen2.5-1.5B: gradall – top , the gradient over the whole training set	0, 1 / 2–6	1	as grad	$P = -0.43$ [–0.58, –0.29]	PR · does not suffice
format $r=2$, Llama-3.2-1B: the same	0, 1 / 2–6	1	as grad	$P = 0.14$ [–0.07, 0.36]	PR · inconclusive
OBQA $r=2$, Qwen2.5-1.5B: warm10 – top , B after 10 % of a free LoRA’s training	0, 1 / 2–6	1	gain; P as grad	$t = 18.70$; $P = 0.58$	PR · E (gain) / part of the gap
format $r=2$, Llama-3.2-1B: the same	0, 1 / 2–6	1	gain; P as grad	$t = 9.61$; $P = 0.63$ [0.45, 0.81]	PR · E (gain) / inconclusive
OBQA $r=2$, Qwen2.5-1.5B, loss on the answer only: A frozen – B frozen, top / bottom	0, 1 / 2–6	2	$t > 3.495$	$t = 7.96 / 10.85$	PR · E

Table S27: Every test (continued): the extension campaigns of Appendix D.2 of the main paper, all pre-registered, on OpenBookQA and, for the placements, on the format task. Phase 2 was launched under a declared trigger (Appendix F of the main paper). Columns as in Table S24.

test	seeds (sel. / rep.)	family	threshold	result	status
format $r=2$, Llama-3.2-1B, one GPU model: G , top	— / 2-6	2	$t > 3.495$	+12.74, $t = 11.39$	PR · E
the same, bottom	— / 2-6	2	$t > 3.495$	+18.66, $t = 13.86$	PR · E
the same: A frozen — B frozen, top	— / 2-6	2	$t > 3.495$	+13.16, $t = 11.49$	PR · E
the same, bottom	— / 2-6	2	$t > 3.495$	+19.67, $t = 14.36$	PR · E
the same: R against 0.7 G , top	— / 2-6	1	IC ₉₅ above 0.7 G	+16.12 [+13.20, +19.04], 0.7 $G = 8.92$	PR · E
the same, bottom	— / 2-6	1	IC ₉₅ above 0.7 G	+21.37 [+19.09, +23.64], 0.7 $G = 13.07$	PR · E
OBQA $r=16$, Qwen2.5-1.5B: free LoRA ($r=7$) — MiCA as released in PEFT, with its recipe	— / 2-6	1	$t > 2.776$	+3.72 [+1.58, +5.86], $t = 4.83$	PR · E
the same, free LoRA with MiCA's recipe — MiCA, added after the previous result	— / 2-6	1	$t > 2.776$	+3.08 [+1.34, +4.82], $t = 4.90$	PR · E
format $r=2$, Llama-3.2-1B, down only: dual_top — top	0, 1 / 2-9	8	IC at 1 — 0.05/8 ($t = 3.855$) above 0	+36.54 [+32.34, +40.75]	PR · E
the same: dual_bottom — bottom	0, 1 / 2-9	8	IC at 1 — 0.05/8 ($t = 3.855$) above 0	+55.46 [+52.19, +58.72]	PR · E
format, Llama-3.2-1B, gate and up only: dual_top ($r=2$) — top ($r=8$)	0, 1 / 2-9	8	IC at 1 — 0.05/8 ($t = 3.855$) above 0	+12.56 [+10.59, +14.53]	PR · E
the same: dual_bottom ($r=2$) — bottom ($r=8$)	0, 1 / 2-9	8	IC at 1 — 0.05/8 ($t = 3.855$) above 0	+10.04 [+7.33, +12.74]	PR · E
OBQA $r=2$, Qwen2.5-1.5B, down only: dual_top — top	0, 1 / 2-9	8	IC at 1 — 0.05/8 ($t = 3.855$) above 0	+3.30 [+1.76, +4.84]	PR · E
the same: dual_bottom — bottom	0, 1 / 2-9	8	IC at 1 — 0.05/8 ($t = 3.855$) above 0	+8.88 [+7.03, +10.72]	PR · E
OBQA, Qwen2.5-1.5B, gate and up only: dual_top ($r=2$) — top ($r=12$)	0, 1 / 2-9	8	IC at 1 — 0.05/8 ($t = 3.855$) above 0	+4.48 [+2.19, +6.76]	PR · E
the same: dual_bottom ($r=2$) — bottom ($r=12$)	0, 1 / 2-9	8	IC at 1 — 0.05/8 ($t = 3.855$) above 0	+6.00 [+4.30, +7.70]	PR · E
OBQA $r=2$, Qwen2.5-1.5B: warm10 — top , direction seed 301	0, 1 / 2-6	4	gain $t > 4.315$, P at 1 — 0.05/4	+4.12, $t = 11.63$, $P = 0.51$ [0.32, 0.70]	PR · E (gain) / inconclusive
the same, direction seed 302	0, 1 / 2-6	4	gain $t > 4.315$, P at 1 — 0.05/4	+4.44, $t = 11.97$, $P = 0.55$ [0.35, 0.75]	PR · E (gain) / inconclusive
format $r=2$, Llama-3.2-1B: warm10 — top , direction seed 301	0, 1 / 2-6	4	gain $t > 4.315$, P at 1 — 0.05/4	+6.78, $t = 6.62$, $P = 0.55$ [0.19, 0.90]	PR · E (gain) / inconclusive
the same, direction seed 302	0, 1 / 2-6	4	gain $t > 4.315$, P at 1 — 0.05/4	+6.09, $t = 8.72$, $P = 0.49$ [0.25, 0.73]	PR · E (gain) / inconclusive

Table S28: Every test (continued): the additions of the TMLR version (§S25), all pre-registered. The equal-recipe test was added after the result of the MiCA test, and pre-registered before its own runs. Columns as in Table S24.

At a common rate. With the rate held equal across arms, the minor band forgets more than the dominant one. But at $2 \cdot 10^{-3}$, **top**, **random** and **random_ortho** are undertrained (§S23.4). **bottom** has learned the most of the frozen arms, and it has moved the furthest.

At each arm's selected rate. The ranking between the two ends reverses: **bottom** forgets less than **top** at its selected rate (27.30 against 40.64), as it did at its former rate, $2 \cdot 10^{-3}$ (21.25). It no longer forgets least of all arms: at $5 \cdot 10^{-3}$, **random**, **random_ortho** and **free** forget less.

Within and across arms. Within an arm, the higher rate forgets more: **bottom** reaches 21.25 at $2 \cdot 10^{-3}$ and 27.30 at $5 \cdot 10^{-3}$. Across arms, the rate alone does not order forgetting: **bottom** at $5 \cdot 10^{-3}$ forgets more than **random** at 10^{-2} . The best-controlled pair is **top** against **random** at their selected rate, the same 10^{-2} , with task scores within a point (0.644 and 0.652). **top**'s perplexity is higher, by 15.7 in perplexity, not in task score ($t = 2.18$, 95 % interval $[-4.26, +35.66]$, not established at five seeds; one seed of **top** reaches 69): the direction is that the dominant band forgets more than a random draw.

A forgetting comparison between frozen subspaces is therefore confounded with the imposed rate, exactly as a comparison of scores is. It says which end of the U an arm sits at only once the rate is fixed.

Tests. At the common rate, against **top**: **random_ortho** -1.27 ($t = -4.85$), **bottom** $+3.10$ ($t = 3.91$), **free** $+5.66$ ($t = 17.57$) and **random** -0.88 ($t = -3.25$), all on five seeds of five. At the selected rates, against **bottom** at its own selected rate, $5 \cdot 10^{-3}$, on five seeds of five: **random** forgets 2.36 less ($t = -6.73$), **free** 3.49 less ($t = -7.10$) and **random_ortho** 0.99 less ($t = -0.83$, one seed in the other direction), while **top** forgets 13.3 more, on every seed, without passing the threshold ($t = 1.84$; on log-perplexity, which damps its seed at 69, $t = 2.30$). The comparison with **bottom** at its former rate, $2 \cdot 10^{-3}$, where several directions are the opposite, is superseded. For **top**, the median over seeds is 35.9, and one seed reaches 69; that seed was evaluated twice and gives the same value. At its selected rate, $5 \cdot 10^{-3}$, **bottom** forgets 27.30 on five seeds: less than **top** and more than **random**, **random_ortho** and **free**. Bonferroni within each family of four comparisons, at 4 d.f., needs $t > 4.31$. Four comparisons pass: **random_ortho**, marginally, and **free** against

	common rate $2 \cdot 10^{-3}$		each arm's selected rate		
	ppl	task	rate	ppl	task
<i>base model</i>	14.09			14.09	
<i>random_ortho</i>	16.87	0.536	10^{-2}	26.31	0.653
<i>top</i>	18.14	0.516	10^{-2}	40.64	0.644
<i>random</i>	17.26	0.544	10^{-2}	24.94	0.652
<i>bottom</i>	21.25	0.566	$5 \cdot 10^{-3}$	27.30	0.592
<i>bottom, former</i>			$2 \cdot 10^{-3}$	21.25	0.566
<i>free $r=2$</i>	23.81	0.803	$2 \cdot 10^{-3}$	23.81	0.803

Table S29: WikiText perplexity after training (lower forgets less) and valid-format accuracy, on Llama at rank 2, five paired seeds, one measurement procedure; bold marks **bottom** at $2 \cdot 10^{-3}$. At a common rate the minor band forgets more than the dominant one, and at each arm's selected rate it forgets less than the dominant one. **random**, **random_ortho** and **top** at their selected rates, and **free**, are re-measured values.

top at the common rate, and **random** and **free** against **bottom** at its selected rate. The sign test cannot go below $p = 0.0625$.

Status: [Post hoc · Descriptive]. The re-measurement with the corrected script moved **top**, **random** and **random_ortho** at their selected rates by at most 0.06 (40.64, 24.94, 26.31), and **free** from 24.06 to 23.81; the table and the tests use it. With it, **random** and **free** forget less than **bottom** at its selected rate, and **top** against **random** is not established. Table S24 gives the status of each comparison.

Caveats. One rank, one task, one model. At five seeds, two comparisons at the selected rates pass the corrected threshold and the others do not. More seeds would have to be fixed in advance, as for Llama's extension, not added until a test passes. Forgetting is measured at the end of training, not at matched task score: the table shows that the two go together, and does not separate them.

S23.2 The rate shift: the clipping control and curvature

This section holds the two controls of the rate shift that the main paper summarises (§S16 keeps the selections themselves).

S23.2.1 The shift is not a clipping artefact

With gradient clipping removed, bottom still peaks one rate below the other frozen arms, so the shift does not come from clipping. All rates were selected with gradient clipping at 1.0. The shift could therefore be an artefact of a hyperparameter we chose. Table S30 repeats the sweep without clipping, and settles it. **bottom** peaks at $5 \cdot 10^{-3}$ and falls at 10^{-2} , where the other three arms are still improving. This is a factor of two, measured with no clipping anywhere.

no clipping	$2 \cdot 10^{-3}$	$5 \cdot 10^{-3}$	10^{-2}	$3 \cdot 10^{-2}$
<i>bottom</i>	0.517	0.566	0.537	0.000
<i>top</i>	0.499	0.587	0.622	0.000
<i>random</i>	—	0.616	0.649	0.000
<i>random_ortho</i>	—	0.621	0.652	0.000

Table S30: Valid-format accuracy with the clipping threshold removed, five seeds per cell ; bold marks each arm's peak and dashes mark cells not run. **bottom** peaks at $5 \cdot 10^{-3}$ and falls at 10^{-2} , where the other three arms are still improving.

The shift is a relative claim, and the relative test is the one that passes. Going from 10^{-2} down to $5 \cdot 10^{-3}$ gains **bottom** 2.9 points and costs **top** 3.5 and **random** 3.3. **bottom**'s own gain does not reach significance on five seeds ($t = 2.56$ against 2.78). The contrast does, on every seed, against the Bonferroni threshold for these two comparisons ($t > 3.50$): +6.45 points against **top** ($t = 3.77$, 5/5, a marginal pass) and +6.19

against **random** ($t = 4.74, 5/5$). **random_ortho** shares three rates with **bottom** here: $5 \cdot 10^{-3}$, 10^{-2} , and the divergence at $3 \cdot 10^{-2}$. From $5 \cdot 10^{-3}$ to 10^{-2} it gains 3.1 points where **bottom** loses 2.9, a contrast of 6.0 points in the means, reported without a paired test. What is detectable without clipping is therefore that **bottom** responds to the rate in the opposite direction from the others, which is what the shift asserts.

Status: [Post hoc · Established], on the two paired contrasts. Table S24 gives the status of the secondary tests.

No divergence in the paper’s configuration. At $5 \cdot 10^{-3}$, the rate its selection retains, **bottom** scores 0.5922 on seeds 2 to 6, and all five runs converge. A single divergent run, with an update norm of 23.0, was once counted in this cell and at first attributed to the clipping threshold. That run was not of this configuration: it dated from the first days of the project, with $\alpha = 16$ at rank 2 ($\alpha/r = 8$), a larger batch and longer sequences, and our aggregation, which separates cells by seven recorded fields, did not count α among them. It is the only such run among the 731 of the reported population; we quarantined it and made α a field that separates cells. Under the paper’s configuration, clipping included, that seed converges (0.5749). On the four seeds measured on both sides, clipping gains 3.2 points, for **top** as well.

Caveats. **random** and **random_ortho** have no cell at $2 \cdot 10^{-3}$, so their peak is read off three rates, against four for the other two arms.

S23.2.2 Curvature predicts the direction of the shift, not its size

Curvature predicts the direction of the shift, not its size.

In brief. Curvature points the same way as the shift. The minor band is the stiffest frozen arm, on all fifteen comparable ratios (§S23.3), which predicts a lower rate for it. It does not predict how much lower. Measured at initialisation and at the end of training, the minor band’s curvature relative to the dominant band’s is 10.2 and 3.36, about a factor of three apart, and both overshoot the factor of two at the selected rates. The product of curvature and selected rate departs from a constant at either rate of the minor band. We read curvature as evidence for the direction of the shift only (below).

Of six candidate mechanisms, four find no support in our measurements and one is left unsettled (Table S32 and Fig. S12, §S23.3). §S8 gives the relative perturbation each arm applies to W_0 . The sixth mechanism survives: curvature. This reading is not new. Among initialised variants, Lee et al. (2026) report that PiSSA needs a learning rate about ten times lower than vanilla LoRA, and, for Qwen3-0.6B on mathematical reasoning (their Figure 1 and Section 4.2), MiLoRA about 3.2 times lower. They attribute these differences to the largest Hessian eigenvalue, about ten times higher for PiSSA, so in their setting the dominant band needs the lower rate. Our curvature reading follows theirs and extends it to frozen subspaces, where it predicts the reversed direction: the minor band is the stiffer one and needs the lower rate. Unlike in their setting, where a curvature about ten times higher matches a learning rate about ten times lower, the curvature ratio here (about ten) overshoots the rate shift at the selected rates (two); §S23.3 discusses where the residual might sit. We measure it as $\lambda_{\max}$, the largest eigenvalue of the raw Hessian on the directions each arm can move in, and call an arm *stiffer* when its $\lambda_{\max}$ is larger.

At initialisation. The minor band is 10 times stiffer than **top** and 11 times stiffer than either drawn arm, **random** or **random_ortho**. The measurement needs care. With a single random start, the power iteration stalls on **bottom**’s subspace and returns values between $5 \cdot 10^{-4}$ and 357 on the same seed, with the convergence flag set. Taking the largest of five restarts fixes it, because the Rayleigh quotient is bounded above by $\lambda_{\max}$, so a stall can only underestimate. This brings the within-cell spread to 0.47 for **bottom** and 0.15–0.25 for **free**, **random** and **random_ortho**, but it stays at 1.68 for **top**. The ratio at initialisation overshoots the rate shift at the selected rates, and the unclipped runs show why curvature cannot be the whole story (§S23.3).

At the end of training. Curvature at initialisation is not curvature along the trajectory. The factors saved at the end of training let us check the difference (Table S31). Both arms flatten, but not by the same amount: **bottom** falls from 333 to 23.33 at its selected rate, and **top** from 32.6 to 6.95. Measuring at the endpoint closes most of the overshoot. The ratio there is 3.36, and it still overshoots the shift of two at the selected

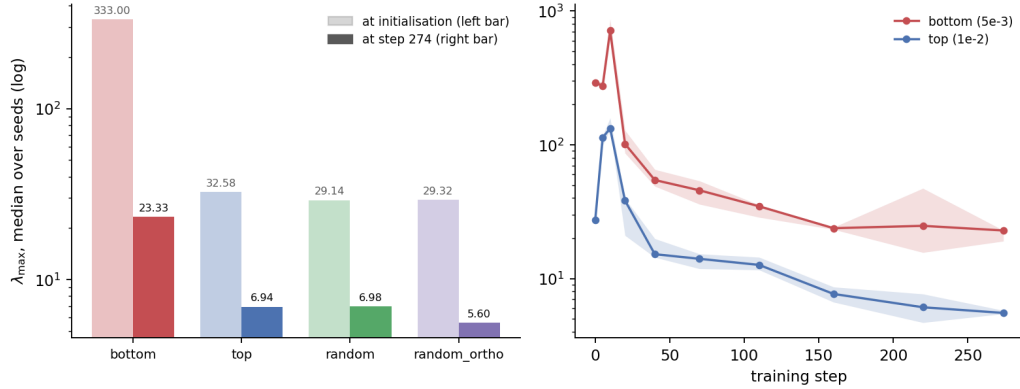


Figure S11: $\lambda_{\max}$ **per arm, median over seeds, log scale**, at initialisation and at step 274 on five seeds (left), and along the path for **bottom** and **top** on seeds 17 to 21, with interquartile bands (right). The minor band is the stiffest arm, at initialisation and at the end of training. **bottom** is at its selected rate, $5 \cdot 10^{-3}$, in both panels; the panels come from different runs and are not compared number for number.

rates, by a factor of 1.7. At the endpoint **top** is well resolved (spread 0.50); **bottom** less so at its selected rate (1.25, against 0.55 at $2 \cdot 10^{-3}$), though still below the 1.68 of **top** at initialisation. Twenty restarts instead of five do not reduce that 1.68. So **top**'s dispersion at initialisation is a property of its subspace, not a stall: the initial measurement is the unreliable one.

	at init	at step 274	spread at 274
bottom	333.0	23.33	1.25
top	32.6	6.95	0.50
ratio	10.2	3.36	

Table S31: $\lambda_{\max}$ of **bottom** and **top** on the same five seeds, at initialisation and on the factors saved after 274 steps. At step 274, **bottom** is read on the factors trained at its selected rate, $5 \cdot 10^{-3}$. The ratio falls from 10.2 to 3.36, against a rate shift of two at the selected rates (§S17.1).

A second campaign measures the two drawn arms in the same way, with the run lookup restricted to the reported population. At step 274, $\lambda_{\max}$ is 6.98 for **random** and 5.60 for **random_ortho** (medians over the same five seeds). The same campaign re-measures **bottom** at $2 \cdot 10^{-3}$ and **top** at 33.55 and 6.94, a ratio of 4.8. At **bottom**'s selected rate, $5 \cdot 10^{-3}$, the same measurement gives 23.33 (seeds from 16.1 to 33.3, all five converged, no batch dropped), a ratio of 3.36, the value the table reports. The 6.95 of Table S31 is **top**'s first measurement on the same runs, and Fig. S11 shows the strict re-measurement. At the end of training, at its selected rate, the minor band is therefore 3.3 to 4.2 times stiffer than every other frozen arm, not only than **top**.

Along the path, the stiffest point comes early. A stability argument needs the stiffest curvature a run meets, not its first or last. We measured $\lambda_{\max}$ at ten steps of five further runs per arm, on seeds 17 to 21. These trajectory runs train **bottom** at its selected rate, $5 \cdot 10^{-3}$, and **top** at 10^{-2} , so their rate shift is two. Both arms peak at step 10, at 716 and 132 (medians over seeds), then fall by more than an order of magnitude. The ratio is 10.7 at initialisation, 5.4 at the peak, and between 2.4 and 4.1 afterwards. The peak ratio overshoots the shift of these runs by a factor of 2.7. The product $\lambda_{\max} \times lr$ at the peak is not constant (3.58 against 1.32), as it was not at initialisation.

Why the endpoint cannot give the size. The endpoint ratio (3.36) comes from runs trained at each arm's selected rate, so it depends on the rate it is meant to explain. A usable rate is also set by the stiffest curvature along the way, not at the end. The peak ratio along the path (5.4) still overshoots the shift. The path measurement thus confirms the reading of this section: curvature gives the direction of the shift, not its

size (Fig. S11). In Fig. S11, the right panel follows these trajectory runs. The left panel reads the runs of the main grid; **bottom** is at $5 \cdot 10^{-3}$ in both panels, but the two come from different runs and are not compared number for number.

Status: [Planned · Descriptive] for the first endpoint measurement, whose script was committed before it ran (19/09, at 14:02 and 19:08); [Post hoc · Descriptive] for every other curvature result in this subsection.

Caveats. **top** is the arm we resolve least well at initialisation, and it is the denominator of the ratios in Table S31. Curvature is measured on Llama only (Appendix B.4 of the main paper). We measure the raw Hessian, not the preconditioned curvature that governs the step AdamW takes (§S23.3). What curvature cannot settle, and where the rest of the shift might sit, is discussed in §S23.3.

S23.3 Curvature: the direction, not the magnitude

Four of the six candidate mechanisms for the rate shift find no support in our measurements and one is unsettled; curvature, the sixth, accounts for its direction, and overshoots its size at the selected rates by a factor of about 1.7 at the endpoint.

The other five mechanisms. Table S32 lists them, with what the measurement says about each. Four are ruled out by the measurement beside them. The Davis–Kahan bound is vacuous for both arms, so it supports no mechanism rather than excluding one.

mechanism	what the measurement says
gradient scale	AdamW normalises it
conditioning	margin ≥ 1.6 over fp32
Davis–Kahan	bound vacuous: nothing settled
gradient SNR	18.05 against 19.27
output share	varies as σ_i^2 <i>by constr.</i>

Table S32: Five of the six candidate mechanisms for the rate shift, with what the measurement says about each; the sixth, curvature, is the subject of §S23.2.2 and of this section. Four are ruled out, and the fifth, the Davis–Kahan bound, settles nothing.

Curvature overshoots the shift. **bottom** is stiffer on *every* one of the fifteen comparable ratios (Fig. S12a). All fifteen exceed the rate shift of two at the selected rates (§S16.1, Table S30), and fourteen exceed even the factor of five that **bottom**’s former rate implied. The curvature does not fall short of the prediction; it overshoots it. **free** is drawn but not counted, because it trains both factors, so its curvature is not commensurable.

The product $\lambda_{\max} \times lr$ is not constant either (Fig. S12b). **bottom**, at its selected $5 \cdot 10^{-3}$, sits at 1.665, where the other three frozen arms are at 0.291 to 0.326: a factor of 5.1 to 5.7, depending on which arm it is compared to. A stability argument alone would require the product to be constant, so this is the part the mechanism does not account for. At $2 \cdot 10^{-3}$, **bottom**’s former rate, the product is 0.666, a factor of 2.0 to 2.3: it departs from a constant at either rate.

What the unclipped runs do not show. What these runs do not show is a difference in stability. In the sweep with the clipping threshold removed (Table S30), all four arms train at 10^{-2} and all four diverge at $3 \cdot 10^{-2}$. In rate space the interval is therefore the same for every arm, and the $\lambda_{\max} \times lr$ products differ only because $\lambda_{\max}$ does. We had drawn the bracket in terms of that product, and read its non-overlap as evidence against a stability threshold. Divided by each arm’s own $\lambda_{\max}$, the bracket says nothing the curvature gap does not already say, so we do not report it. What separates the arms is where each one peaks, not where it diverges.

Where the residual factor might sit. The curvature we measure is that of the Hessian. The step AdamW actually takes is governed by the *preconditioned* curvature (Cohen et al., 2022), and the two differ by

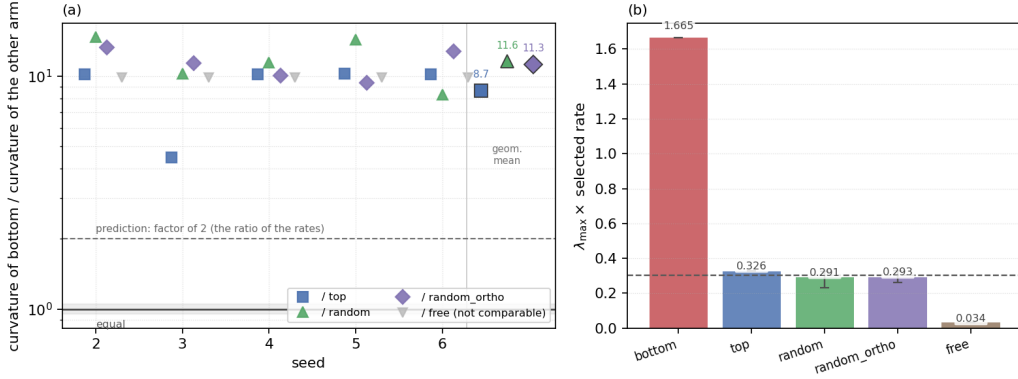


Figure S12: **(a)** Paired curvature ratios of **bottom** to the other arms, per seed, log scale; **free** is drawn but not counted. **(b)** The product $\lambda_{\max} \times lr$ per arm, medians, with $\lambda_{\max}$ at initialisation and **bottom** at its selected $5 \cdot 10^{-3}$. **bottom** is stiffer on all fifteen comparable ratios, and none falls below the factor of two that the ratio of the rates predicts; its product is 5.1 to 5.7 times the others'. In (a) the points are jittered and the three geometric means stand to the right of the rule; in (b) the dashed line is the mean of the three other frozen arms, 0.303, against 1.67 for **bottom**.

whatever the second-moment estimate does to the spectrum. An arm whose directions carry systematically larger gradients is preconditioned more, and its usable rate falls by less than the raw $\lambda_{\max}$ would suggest. We do not measure this. We name it because it is the first thing a reader will think of, and because it predicts the sign of the discrepancy we observe.

Status: [Post hoc · Descriptive].

Caveats. Two limits of the curvature measurement remain. $\lambda_{\max}$ is measured at initialisation, at step 274, and at ten steps of separate trajectory runs, not at the point where a run diverges. **top**'s curvature is the least well resolved of the five arms, and it is exactly the denominator of the ratio we quote. The power iteration can also stall. A plain power iteration would return the eigenvalue of largest modulus, which on an indefinite Hessian need not be the largest one; we iterate instead on $H + cI$, with c estimated from a first unshifted pass so that the largest eigenvalue also has the largest modulus, then subtract c . The value returned is the largest eigenvalue, and every value we report is positive. The iteration stops on a convergence test for both of its passes, and a batch where either pass fails the test is dropped; the residuals themselves are not logged. A stalled iteration underestimates $\lambda_{\max}$, for either arm, so the ratio of ten is an estimate whose error can go either way. Part of the residual factor may sit here.

S23.4 The common-rate inversion, in detail

At ranks 2 and 4, which frozen arm looks best depends on the common learning rate: between two defensible rates, the ranking inverts in the means in all four configurations tested; paired seed by seed, at least one half of the inversion is established in each, and both halves on **Qwen2.5-0.5B**. This result is methodological: at low rank it conditions every common-rate comparison of frozen arms, while at rank 16 on OpenBookQA the ranking does not invert (§S23.5). §S23.4.2 gives the original observation on Llama at rank 2, §S23.4.3 and §S23.4.4 its replications under committed protocols, and §S23.4.5 compares it with the other two effects. Throughout, a lead or a deficit is a difference of means against the best competing arm; a paired test recomputes that arm on each seed. Table S33 summarises which half is established where.

The interaction is established everywhere. A swing adds a half that is established to one that may not be, so it is not by itself a test. The direct test is the interaction between arm and rate: for each seed and each fixed competitor, **bottom**'s gap at $2 \cdot 10^{-3}$ minus its gap at 10^{-2} . It is positive on every seed and past a threshold corrected over the eleven comparisons ($t > 5.75$) in all four configurations: against **top**, $t = 55.7$

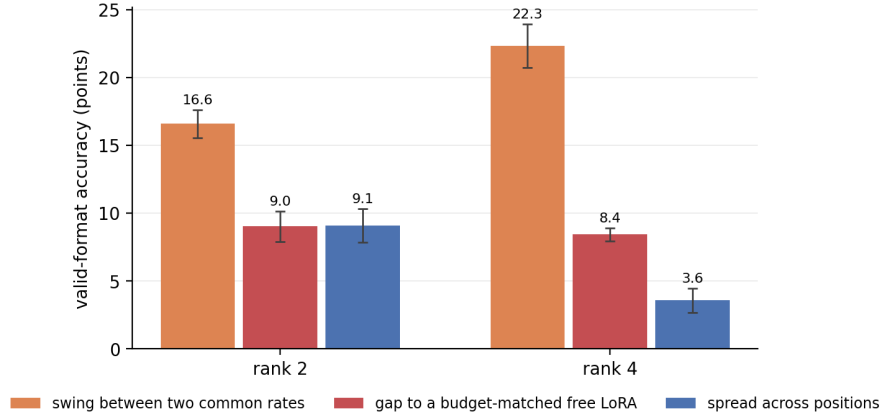


Figure S13: **Three contrasts on Llama-3.2-1B, at ranks 2 and 4**, in points of valid-format accuracy. *The learning rate*: how far **bottom**’s standing swings between the two common rates of Fig. S9. *Gap to a budget-matched free LoRA*: the free reference minus the best frozen arm of that rank, the β sweep included. *The position*: the position effect of the β sweep. The gap and the position are paired on seeds 2 to 6, where the reference exists. Bars are paired bootstrap standard errors over the seeds, the same resample for every arm and the best and worst arm recomputed at each draw, as in Table S12. §S23.4.5 compares the three. The three bars are different contrasts on one scale, a swing between two chosen rates, a gap to a reference and a spread between arms, not a decomposition of one effect or an order of importance; the swing depends on the pair of rates, and at rank 16 the ranking does not invert, on the format task or on OpenBookQA (§S23.5).

configuration	deficit at 10^{-2}	lead at $2 \cdot 10^{-3}$	see
Llama, rank 2	yes	no	§S23.4.2
Llama, rank 4	yes	no	§S23.4.3
Qwen2.5-0.5B, rank 2	yes	yes	§S23.4.3
Qwen2.5-1.5B, rank 2	no	yes	§S23.4.4

Table S33: Whether each half of the inversion is established, paired seed by seed. Llama at rank 2 is post hoc; the other three configurations ran under committed protocols, though one compared cell on Qwen2.5-0.5B predates its protocol (§S23.4.3).

and 9.7 on Llama at ranks 2 and 4, 9.2 and 9.0 on Qwen2.5-0.5B and Qwen2.5-1.5B; against the random bases, $t = 13.5$ and 15.6 on Llama at rank 2, 13.0 at rank 4, 10.1 on Qwen2.5-0.5B and 8.0 on Qwen2.5-1.5B. Taking **top** as a fixed competitor, rather than the best arm recomputed on each seed, the lead at $2 \cdot 10^{-3}$ is also established on Llama at rank 2 ($+4.98$ points, $t = 5.68$). Status: [Post hoc · Established].

S23.4.1 Summary, and where the inversion stops

On the format task at ranks 2 and 4, which frozen subspace looks best depends on the common learning rate. On Llama at rank 2, **bottom** leads the other frozen arms at $2 \cdot 10^{-3}$ by 2.15 points in the means and trails them at 10^{-2} by 14.43 (Fig. S9). At $2 \cdot 10^{-3}$ the minor band is near its optimum while the other arms are undertrained, and at 10^{-2} they are near theirs.

Reading $t = 55.7$. The test is the arm-by-rate interaction, paired by seed: for each seed, **bottom**’s gap to **top** at $2 \cdot 10^{-3}$ minus its gap at 10^{-2} . On Llama at rank 2 it reaches $t = 55.7$ over five seeds. The value is large because the interaction varies little between seeds. It is positive on every seed and carried mostly

by the deficit half: at 10^{-2} the minor band trails on every seed, 14.43 points behind the best other arm in the means, and this half is established on its own ($t = 17.14$). The lead half is weaker. Against the best other arm recomputed on each seed it is not established ($t = 1.12$), and against **top** alone it is (+4.98 points, $t = 5.68$, post hoc). The interaction passes a threshold corrected over eleven comparisons ($t > 5.75$) in all four configurations tested: against **top**, $t = 55.7$ and 9.7 on Llama at ranks 2 and 4, and 9.2 and 9.0 on Qwen2.5-0.5B and Qwen2.5-1.5B at rank 2 (Table S33 gives each half).

Where the inversion stops. Each of the following was tested under a protocol committed before its runs, and none reproduces the inversion (Fig. S14a). At rank 8, the rule that fixes the two common rates yields a single rate, $2 \cdot 10^{-3}$, so the inversion is not testable. At rank 16, **bottom** does not lead at its own rate (-0.16 points against **top**), and the interaction is established against no arm ($t = 0.69$ to 2.81, against 3.96); on OpenBookQA at rank 16 it is last at both common rates (§S23.5). On OpenBookQA at rank 2, **bottom** trails at its own rate, significantly behind **top** (-2.88 points, $t = -4.06$). A tie in selection led us to test two pairs of common rates; on one, the interaction is established against the two random bases ($t = 9.1$ and 9.4), but it describes a gap that widens with the rate, as **bottom** collapses at $2 \cdot 10^{-2}$. On ARC-Easy at rank 16, **bottom** trails at its own rate (-2.90 points against **top**, $t = -10.7$), and on ARC-Challenge the rule yields a single common rate. At twice the training budget on Llama at rank 2, the other arms catch up at $2 \cdot 10^{-3}$, where **bottom** no longer leads (-1.89 points against **top**, not significant); the interaction keeps its sign on all five seeds (+7.3 to +7.6 points) but stays below its threshold ($t = 3.06$ to 3.83, against 3.96).

What the inversion needs. The inversion needs optima that differ between arms and curves steep enough between them. With coinciding optima (rank 8), flat curves (on ARC, where the selection curves differ by about a point) or a longer training that lets the other arms catch up at the low rate, it is not reproduced. We read it as a hazard of the comparison: at low rank, on a task with steep curves, a single rate can decide the ranking, and nothing in the ranking itself signals it.

S23.4.2 Llama at rank 2

On Llama at rank 2, bottom leads the other frozen arms at $2 \cdot 10^{-3}$ in the means and is last at 10^{-2} , a swing of 16.58 points. At $2 \cdot 10^{-3}$, **bottom** leads the three arms measured there by 2.15 points, **random** (0.544) being the closest; at 10^{-2} it is last by 14.43 (Fig. S9). At its selected rate, it is instead 6.12 points behind the best arm (Table S22). The inversion is not a matter of interpretation but an identifiable artefact. At rank 2, $2 \cdot 10^{-3}$ is **bottom**’s own regime and not the other arms’, which are undertrained there. That an arm leads at its own optimum while the others are away from theirs is not a separate finding: it is the mechanism by which a common rate decides the ordering. This is exactly the configuration of the comparisons we discuss: a single rate, and variants whose optima differ by a factor of two at the selected rates.

Paired seed by seed, the two halves are not equally solid. The deficit at 10^{-2} passes by a wide margin ($t = 17.14$, ahead on every seed), while the lead at $2 \cdot 10^{-3}$ does not (+0.98 points, 95 % interval $[-1.4, +3.4]$, $t = 1.12$, against 2.776).

Status: the deficit at 10^{-2} is [Post hoc · Established]; the lead at $2 \cdot 10^{-3}$ is [Post hoc · Not established].

Caveats. Both rates are defensible *a priori*, and only a per-arm sweep says which one each arm needs.

S23.4.3 Rank 4 and Qwen2.5-0.5B, under a committed protocol

The inversion reproduces at rank 4 on Llama and at rank 2 on Qwen2.5-0.5B, under a protocol written before launch. The protocol (Appendix F of the main paper, Table 23 of the main paper) repeats the measurement on seeds 2 to 6. **bottom** leads the arms measured at $2 \cdot 10^{-3}$ by 2.87 and 4.20 points, and is last at 10^{-2} by 19.48 and 20.16. The swings are 22.35 and 24.36 points, against 16.58 at rank 2 on Llama. Paired seed by seed, the deficit at 10^{-2} passes in both ($t = 13.48$ at rank 4 on Llama and 9.08 on Qwen2.5-0.5B, ahead on every seed). The lead at $2 \cdot 10^{-3}$ passes on Qwen2.5-0.5B (+4.20 points, $t = 5.16$) but not at rank 4 on Llama (+2.25, 95 % interval $[-0.3, +4.8]$, $t = 2.42$, against 2.776). What the common rate establishes is therefore the minor band’s deficit at the higher rate, not its advantage at the lower one, except

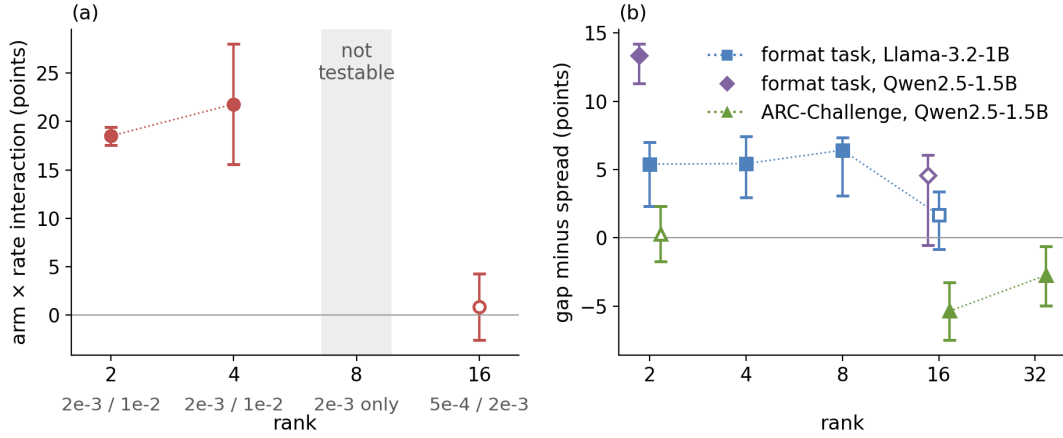


Figure S14: **Dependence on rank.** (a) Format task, Llama-3.2-1B: The arm $\times$ learning-rate interaction (the inversion) against **top** as a fixed competitor: how far **bottom**'s gap to **top** swings between the low and the high common rate, in points, with 95 % intervals. The two common rates are shown under each rank: at ranks 2 and 4 they are those of the original comparison; at ranks 8 and 16 they were fixed by a pre-registered rule (**bottom**'s selected rate and that of the best other frozen arm). At rank 8 the rule yields a single common rate, so the inversion is not testable there: the grey band marks it, and no point is drawn, not even from the selection seeds. (b) Gap minus spread at each arm's selected rate, with 95 % bootstrap intervals: squares, the format task on Llama-3.2-1B, available at every rank (at rank 4, the pre-registered seven-seed measure); triangles, ARC-Challenge on Qwen2.5-1.5B at ranks 2, 16 and 32, each under a pre-registered protocol, shifted right for legibility; diamonds, the format task on Qwen2.5-1.5B at ranks 2 and 16, pre-registered on fresh seeds, shifted left. Squares and triangles differ in task and model and are not compared point by point; within ARC-Challenge, the sign goes from not established at rank 2 to negative at ranks 16 and 32. Diamonds and triangles share a model: at rank 16 the format task does not reverse where ARC-Challenge does, so the reversal follows the task. Lines join only adjacent ranks that were measured. Filled markers: established; open markers: not established.

on Qwen2.5-0.5B. One cell of that lead did not come from this protocol: **bottom** at $2 \cdot 10^{-3}$ on Qwen2.5-0.5B, seeds 2 to 6, had been run five days before the protocol was committed, in the same configuration. The protocol listed it among the runs to launch; the launcher found it done and reused it, and its values may have been seen before the protocol was written.

Status: the deficit at 10^{-2} is [Pre-registered · Established] in both configurations; the lead at $2 \cdot 10^{-3}$ on Qwen2.5-0.5B is established, but not pre-registered in the strict sense, since one of its cells predates the protocol; the lead on Llama at rank 4 is [Pre-registered · Not established].

Caveats. At rank 4 the picture is less one-sided. The rate $2 \cdot 10^{-3}$ is also **top**'s rate under its 400-item selection, and at the selected rates **bottom** is ahead of **top** and 0.65 points behind **random** (Table S14).

S23.4.4 Qwen2.5-1.5B

On Qwen2.5-1.5B, three times the size of Qwen2.5-0.5B and of the same family, the ordering inverts in the means, and only its lower-rate half is established. The measurement ran under a protocol registered before launch (Appendix F of the main paper). **bottom** leads the two other arms at $2 \cdot 10^{-3}$ by 8.44 points (0.6274 against 0.5430 and 0.5378), and is last at 10^{-2} by 3.06 (0.6267 against 0.6440 and 0.6573), a swing of 11.50 points. Paired over the five seeds, the lead is 7.88 points ($t = 8.70$, ahead on every seed). The deficit is 3.13 (95 % interval $[-0.9, +7.2]$, $t = 2.14$, ahead on three seeds of five), below the two-sided threshold of 2.776. The pattern of the two halves is thus reversed at this size. The swing is the smallest of the four, for a reason worth naming: **bottom** barely moves between the two rates (0.6274 and

0.6267), while the other arms gain what they gained before. What inverts the ordering is that the arms respond to the rate differently, not that any one of them fails.

Status: the lead at $2 \cdot 10^{-3}$ is [Pre-registered · Established]; the deficit at 10^{-2} is [Pre-registered · Not established].

S23.4.5 The learning rate against freezing and position

On Llama, the learning-rate effect exceeds both the gap to a budget-matched free LoRA and the position effect, at ranks 2 and 4. Figure S13 sets the three effects side by side, with freezing and position paired on seeds 2 to 6, where the free reference exists. At rank 2, the swing of 16.58 points compares with a gap to a budget-matched free LoRA of 9.0 and a position effect of 9.1. At rank 4, the swing of 22.35 compares with a gap to a budget-matched free LoRA of 8.4, the best frozen arm there being **top** at its 614-item selected rate, and a position effect of 3.6; over the two rate selections of **top**, the gap is 8.4 to 10.1 (Table S12), and over the two seed sets the position effect is 3.3 to 3.6 (§S3.2). The swing adds two gaps, **bottom**’s lead at one rate and its deficit at the other. The larger half alone, the deficit at 10^{-2} , is 14.4 points at rank 2 and 19.5 at rank 4, above freezing and position at both ranks. The ordering of the three effects therefore holds in full at rank 4, while at rank 2 freezing and position are of the same order. At rank 4, a pre-registered test on seven paired seeds, with the Σ arm counted, confirms the step from freezing to position: freezing exceeds the choice among the three original arms by 5.44 points [+2.98, +7.42] (§S11.4).

Status: [Post hoc · Descriptive] for the comparison with the learning-rate effect, which no test makes; the step from freezing to position at rank 4 is [Pre-registered · Established] (§S11.4).

Caveats. The learning-rate effect depends on the two rates chosen. Both are defensible *a priori* (Fig. S9), but a rate at which every arm diverges would give another number. The effect therefore measures what a defensible common rate does to the ordering, not a property of the whole rate axis. The three effects are measured together on Llama only.

The inversion, population by population. At rank 16 on the format task, the common rates are $5 \cdot 10^{-4}$ and $2 \cdot 10^{-3}$; **bottom** does not lead at the first (-0.16 points against **top**, $t = -0.18$), trails at the second, and the interaction is established against no arm ($t = 0.69, 1.86$ and 2.81 against **top**, **random** and **random_ortho**; threshold 3.96). At twice the budget, with the common rates of the original comparison ($2 \cdot 10^{-3}$ and 10^{-2}), **bottom** no longer leads (-1.89 points against **top**, $t = -0.97$), the deficit at 10^{-2} is established against every arm ($+8.7$ to $+10.2$ points, $t \geq 5.49$), and the interaction is not ($t = 3.06, 3.83$ and 3.64). On OpenBookQA at rank 2, **random** and **random_ortho** tie exactly in selection (0.41125), so both pairs of common rates were tested, with thresholds corrected for the two (3.495 for each half, 4.851 for the interaction). On both, **bottom** trails at its own rate, $2 \cdot 10^{-3}$ (-2.88 points against **top**, $t = -4.06$); on the pair $2 \cdot 10^{-3}$ and $2 \cdot 10^{-2}$, the interaction is established against **random** ($t = 9.09$) and **random_ortho** ($t = 9.37$), because **bottom** collapses at the high rate (0.242), not because it leads at the low one. On ARC-Easy, **bottom**’s selected rate is the higher common rate; it trails there (-2.90 points against **top**, $t = -10.73$). The ARC protocol pre-registers no interaction test. At rank 8 and on ARC-Challenge, the rule yields a single common rate.

S23.5 At rank 16, on public benchmarks, the inversion does not reproduce

§§S23.4 and S16 measure the inversion and the rate shift behind it at ranks 1 to 4, on a synthetic task. A protocol written and committed before launch repeats the comparison in the regime of the methods we study: Qwen2.5-1.5B at rank 16 on OpenBookQA (Mihaylov et al., 2018), one of the commonsense tasks PiCa reports, scored by option log-likelihood on its 500 test items, where a single run carries about two points of binomial error. The free reference runs at rank 7, with 8,078,336 parameters against 8,142,848 for a frozen arm at rank 16 (-0.8%). One difference with PiCa is deliberate and stated: we train on OpenBookQA’s own training split (4,957 items), not on the 15K-example commonsense mixture PiCa trains on.

The campaign is informative. **free** reaches 0.562 on seeds 2 to 6, against 0.422 for the base model: 14.0 points above it, on every seed, past the floor of five points fixed in advance.

The rate shift reproduces in the two-seed selection; the inversion does not. On the selection seeds, **bottom** selects 10^{-3} , the lowest rate of the three frozen arms (**top** $2 \cdot 10^{-3}$, **random_ortho** $5 \cdot 10^{-3}$), and at $5 \cdot 10^{-3}$ it falls to 0.316 in validation where **random_ortho** peaks. The rule therefore fixes the two common rates at 10^{-3} and $2 \cdot 10^{-3}$, a factor of two apart, against five at low rank. At both, **bottom** is last: 3.40 points behind **top** at 10^{-3} ($t = -10.75$, no seed of five ahead) and 5.00 behind the best other arm at $2 \cdot 10^{-3}$ ($t = 11.07$, five of five), a deficit that holds against each competitor taken alone. The deficit is established; the ranking does not invert. By the rule registered for this case, the inversion is a low-rank result, and §S16.3 says so.

The gap and the spread do not separate at rank 16. At the selected rates, the gap to free LoRA is 2.08 points, [0.44, 3.72], and the spread among frozen arms 4.24, [2.96, 5.36]; gap minus spread is -2.16 , $[-4.60, +0.64]$, and does not separate from zero. §S11, next, measures the same two quantities at ranks 1 to 4, where the gap to free LoRA exceeds the spread among frozen arms; across ranks, the gap to free LoRA falls from 24 points at rank 1, at $2.15\times$ the budget, to 2 at rank 16, budget-matched (§S11.5).

Status: [Pre-registered · Established] for the floor and the deficit; the inversion is not observed.

ARC-Easy and ARC-Challenge. A second protocol repeats the campaign on ARC (Clark et al., 2018), with the free reference again at rank 7. The floors pass: **free** beats the base model by 8.3 points on ARC-Easy and 5.9 on ARC-Challenge. On ARC-Challenge the rule yields a single common rate, 10^{-3} , so the inversion is not testable; on ARC-Easy, after one extension of the grid at its lower edge, **bottom** trails at its own rate (-2.90 points against **top**, $t = -10.73$), so the ranking does not invert. Neither shows the rate shift: **bottom** selects a rate above **top**’s and equal to **random_ortho**’s, on selection curves that differ by about a point. On both, the spread among frozen arms exceeds the gap to free LoRA, and the difference is established: -2.08 points ($[-3.16, -1.50]$) on ARC-Easy, -5.36 ($[-7.48, -3.24]$) on ARC-Challenge, the best frozen arm coming within a third of a point of free LoRA (Appendix D of the main paper).

Status: [Pre-registered · Established] for the floors and for the reversed order of the two quantities; the inversion is not observed.

S23.6 What was run on OLMo

On OLMo-2-1B, only the rate sweep was run: every arm has an interior optimum on all seven modules, but none on the three non-degenerate ones. We ran the sweep at rank 2 on the two selection seeds: 111 runs, for **top**, **bottom**, **random** and **random_ortho**. It covers all seven modules, and the three that the criterion of §S6 leaves (**gate**, **up**, **down**). On seven modules, every arm has an interior optimum, at $3 \cdot 10^{-2}$ or $5 \cdot 10^{-2}$, with validation scores of 0.50 to 0.52. At 10^{-1} , three arms fall to zero and **top** to 0.42. For those three the training loss rises to five to seven times its starting value before settling above it, so training becomes unstable; their update norms end at 36 to 46, about three times those at $3 \cdot 10^{-2}$. Update norms are logged only at the end of a run, so whether they exploded along the way is not known.

On the three non-degenerate modules, the score still rises at the largest rate swept, $3 \cdot 10^{-2}$, for every arm (validation 0.33 to 0.40), so the sweep found no interior optimum there. No reported seed was run on OLMo, and none of its numbers enters a comparison in this paper. The model enters through its spectrum only (Table S8).

Status: [Post hoc · Descriptive].

S24 Placement and loss campaigns, and the analyses v3 and A1 to A6: full records

This section gives the full records of three pre-registered campaigns, the accumulated gradient, the short warm-up and the loss on the answer only, and of the post hoc analyses that go with them. All ran on RTX 4000 Ada cards, and each protocol was committed before its first run (Table 24 of the main paper). The two placement campaigns test against the published **top**, 0.4652 on OpenBookQA and 0.6443 on the format task, with G fixed in advance at 8.04 and 12.41. Differences against the references retrained on one GPU model

are reported beside them. $P = (\text{arm} - \text{top})/G$ is the share of the gap between frozen B and free LoRA that an arm recovers.

S24.1 The accumulated gradient

The question is whether **grad** fails because 64 sequences give a poor estimate of the gradient. Two precisions were declared before the commit. **grad1024** runs on the format task only, since OpenBookQA has 272 training sequences. The sets are nested by construction: the 64 sequences of **grad**, then a fixed permutation of the others, so that $n = 64$ returns the bases of **grad** byte for byte. The share of the gradient in its top two directions barely moves with the amount of data. It is 0.555 to 0.558 on the format task, from 64 to 3178 sequences, and 0.83 on OpenBookQA, at 256 and 272. Table 19 of the main paper gives the result of **gradall**, the gradient over the whole training set. Without a test, **grad256** – **top** is -3.44 on OpenBookQA and $+0.85$ on the format task, and **grad1024** – **top** is $+3.00$ on the format task. The conclusion fixed in advance, that the accumulated gradient suffices, is drawn on neither pair. On OpenBookQA it does not suffice. On the format task the reading is inconclusive: the upper bound of **gradall** – **top**, 4.42, exceeds $0.3G = 3.72$ but lies far below $0.7G = 8.69$. The result is close to that of **grad**, over 64 sequences.

S24.2 The short warm-up

The question is what fraction of training suffices to place B . A free LoRA at rank 2, direction seed 0, trains at $2 \cdot 10^{-3}$, the rate of phase 2’s direction run, a precision declared before the commit. Its factors are saved after 5, 10 and 25 % of the steps: 3, 5 and 13 of 49 on OpenBookQA, and 14, 28 and 69 of 274 on the format task. B is frozen on each saved B , and A trains from zero for the remaining steps: 46, 44 and 36, and 260, 246 and 205. As a reproduction check, the direction run’s last step returns the bases of phase 2 with an overlap of 0.999 on OpenBookQA and 1.000 on the format task. The same training, rerun another day on another card, learns the same directions. Table 20 of the main paper gives the result of **warm10**, and Table 21 of the main paper the cuts. Without a test, **warm5** – **top** is $+1.04$ on OpenBookQA and $+4.98$ on the format task, and **warm25** – **top** is $+6.24$ and $+11.50$. The conclusion fixed in advance, that a short warm-up suffices, is not drawn. **warm10** recovers close to two thirds of the gap on both pairs, but not most of it in the fixed sense, a lower bound of P at or above 0.7.

S24.3 The loss on the answer only

The protocol repeats the rank-2 arms on OpenBookQA with Qwen2.5-1.5B, with the loss on the answer tokens only. Before any run, the new loader passed its test on the server. Its blocks are identical to the reference’s and in the same order, 272 blocks of 384 tokens. It supervises 4945 answers for 4945 answer marks, and 17300 tokens of 104448 per epoch (16.6 %). The answers decoded in plain text match the data, and no existing run identifier changed. Each run keeps the 49 optimiser steps and supervises a median of 99614 tokens (99513 to 99813), against about 602112 with the full loss. Table 22 of the main paper gives the scores and the tests. The reading fixed in advance is that the asymmetry does not depend on the loss on the question. Without a test, every arm scores higher with the loss on the answer only, with six times fewer supervised tokens. The minor B-frozen arm leaves the level of the base model ($+3.56$ points), where it stayed exactly with the full loss. **dual_bottom** first fell on the upper edge of its grid, 10^{-2} . The engine launched the extension step at $2 \cdot 10^{-2}$ at 18:34 and redid the selection at 18:51, once those runs had finished, and 10^{-2} still won. The optimum is therefore interior to the extended grid, as the protocol requires, and the reading was made at 19:00.

S24.4 Dose and response

Table S34 gives P as a function of the training that fixed the directions of B . Only **grad**, **gradall** and **warm10** carry a fixed reading. The other rows are descriptive, and their intervals must not serve as tests. Read descriptively, P stays flat for the gradient, whatever the amount of data, and rises quickly with training. The useful directions are not among the initial gradient’s top two, and training reaches them within its first few per cent.

directions of B obtained from	OpenBookQA	format task
initial gradient, 64 sequences (grad)	-0.47 [-0.63, -0.32]	0.18 [0.08, 0.29]
gradient, 256 sequences	-0.43 [-0.57, -0.29]	0.07 [-0.22, 0.35]
gradient, 1024 sequences	—	0.24 [-0.05, 0.54]
gradient, whole training set (gradall)	-0.43 [-0.58, -0.29]	0.14 [-0.07, 0.36]
5 % of training (warm5)	0.13 [0.08, 0.18]	0.40 [0.18, 0.62]
10 % (warm10)	0.58 [0.49, 0.66]	0.63 [0.45, 0.81]
25 % (warm25)	0.78 [0.67, 0.88]	0.93 [0.85, 1.01]
100 % (learned , phase 2 and replications)	1.16 (seed 0), 1.09 (seed 7)	1.33

Table S34: **Share of the gap, P , by the training that fixed the directions of B** (rank 2, seeds 2 to 6). Only **grad**, **gradall** and **warm10** carry a fixed reading. The other rows are descriptive.

rank	population	factor	band	interaction	$ f / b $
2	format, Llama, seeds 2-6	+15.96 [+14.67, +17.25]	-2.21 [-3.83, -0.60]	+5.99 [+1.74, +10.25]	7.2
2	format, Llama, seeds 120-124	+15.21 [+14.14, +16.29]	-2.96 [-4.45, -1.48]	+6.84 [+1.31, +12.37]	5.1
2	OBQA, Qwen2.5-1.5B	+8.42 [+6.48, +10.36]	-2.46 [-4.08, -0.84]	+3.72 [+1.57, +5.87]	3.4
2	OBQA, Qwen2.5-7B	+11.78 [+10.95, +12.61]	-3.46 [-4.00, -2.92]	+2.60 [+0.81, +4.39]	3.4
2	OBQA, Mistral-7B	+11.98 [+10.53, +13.43]	-1.02 [-3.20, +1.16]	-2.28 [-3.70, -0.86]	11.7
4	OBQA, Qwen2.5-1.5B	+7.82 [+6.37, +9.27]	-0.38 [-2.42, +1.66]	+1.64 [-0.00, +3.28]	20.6
8	OBQA, Qwen2.5-1.5B	+6.62 [+5.04, +8.20]	-3.10 [-3.96, -2.24]	+0.52 [-2.55, +3.59]	2.1
16	ARC-C, Qwen2.5-1.5B	+2.41 [+1.31, +3.51]	-2.29 [-3.18, -1.40]	+5.78 [+3.84, +7.72]	1.1
16	OBQA, Qwen2.5-1.5B	+2.58 [+1.07, +4.09]	-3.02 [-4.18, -1.86]	+2.44 [+0.18, +4.70]	0.9
16	OBQA, Qwen2.5-7B	+4.00 [+2.81, +5.19]	-1.64 [-2.80, -0.48]	+3.60 [+0.30, +6.90]	2.4

Table S35: **Factor, band and interaction at every rank** (analysis v3, post hoc, descriptive). Differences in points, with paired t intervals on four degrees of freedom.

S24.5 Post hoc analyses v3

These analyses are post hoc and descriptive, and their script was committed before execution. A correction was declared before the second execution: a published value found at several rates is resolved by validation, as in the selection. **dual_bottom** at rank 16 on OpenBookQA scores 0.5364 at 10^{-3} and at $2 \cdot 10^{-3}$, and 10^{-3} is kept, with 0.4862 against 0.4788 in validation.

Factor and band at every rank. Table S35 gives the decomposition, with paired t intervals on four degrees of freedom. The factor is the mean of the A-frozen arms minus that of the B-frozen arms, and the band is the mean of **bottom** minus that of **top**. The interaction is the band difference with A frozen minus that with B frozen. Read descriptively, at rank 2 the factor’s effect exceeds the band’s by 3 to 12 times. At rank 16 it is no larger on Qwen2.5-1.5B (0.9 and 1.1), and modestly larger on Qwen2.5-7B (2.4). At rank 4 the band effect is almost null, which makes the ratio unstable, so the two effects are better reported than their ratio.

Principal angles at rank 2, output side. Table S36 gives the mean of the squared cosines of the two principal angles, 1 for the same subspace and 0 for orthogonal ones, as a median over layers. Chance is $2/d_{\text{out}}$, from 0.0002 to 0.0078. Read descriptively, the learned directions lie outside the spectral bands of W_0 , at chance level except **k_proj** against **bottom** (0.045). This confirms phase 1 with another measure. They are only weakly aligned with the initial gradient (0.05 to 0.18), which is consistent with the failure of **grad**. From one direction seed to the other they overlap in part (0.20 to 0.40, hundreds of times chance), yet both seeds give the same result ($P = 1.16$ and 1.09). Several subspaces therefore serve. The gradient of **k_proj** falls partly in the minor band (0.219, 28 times chance), like B in phase 1.

The bottom arm on OpenBookQA at rank 2. It scores 0.4220, the level of the base model. Its updates are half as large as those of **top**, with a median norm of 0.90 against 1.95, while its gradient before clipping is larger, 0.24 against 0.11. It therefore learns, but little. Neither per-question predictions nor factors were saved on OpenBookQA, so neither a per-question comparison with the base model nor a re-evaluation is possible.

comparison	q	k	v	o	gate	up	down
OBQA: learned, seed 0 / learned, seed 7	0.195	0.276	0.395	0.333	0.334	0.332	0.333
OBQA: learned, seed 0 / gradient	0.071	0.091	0.183	0.137	0.048	0.075	0.078
OBQA: learned, seed 7 / gradient	0.071	0.095	0.163	0.142	0.045	0.071	0.072
OBQA: learned, seed 0 / top	0.003	0.010	0.008	0.001	0.001	0.000	0.002
OBQA: learned, seed 0 / bottom	0.001	0.045	0.006	0.001	0.000	0.000	0.001
OBQA: gradient / top	0.001	0.002	0.011	0.001	0.000	0.001	0.002
OBQA: gradient / bottom	0.003	0.219	0.018	0.001	0.000	0.000	0.002
format: learned / gradient	0.043	0.056	0.178	0.157	0.068	0.081	0.131
format: learned / top	0.001	0.005	0.003	0.001	0.000	0.000	0.001
format: gradient / top	0.000	0.002	0.003	0.001	0.000	0.000	0.002

Table S36: **Principal angles between rank-2 output subspaces** (analysis v3, post hoc, descriptive). Mean of the squared cosines of the two principal angles, median over layers. Chance is $2/d_{\text{out}}$, from 0.0002 to 0.0078.

S24.6 Analyses A1 to A5

These analyses are post hoc and descriptive (PH · D), and their protocol and script were committed before execution. Declared before execution: OpenBookQA kept neither per-question predictions nor factors, so A4 is reduced and A5 covers the format task only. They were corrected before the rerun (29/09 23:42). The comparison of free-LoRA B now keeps only the runs at the rate of the published cell. On the format task the first execution mixed rates, giving 0.13 to 0.33 instead of 0.16 to 0.42, and OpenBookQA is unchanged. A3 reads each population on its reported seeds, 22 to 26 for the format task on Qwen2.5-1.5B (selection on 20 and 21), where the first execution looked for seeds 2 to 6.

A1, principal angles. Table S37 gives the overlaps with their ratio to chance. The largest angles are 88 to 90° between **grad** and the bands, 76 to 88° between **grad** and **learned**, and 71 to 82° between the two direction seeds of **learned**. **grad256**, **grad1024** and **gradall** give the same overlaps as **grad** to the third decimal. By tenth of the spectrum of W_0 , **k_proj** holds 56.6% (OpenBookQA) and 58.0% (format) of **grad** in its last tenth, and **q_proj** 23.7 and 41.3%. **gate** and **up** have 73 to 83% of it outside the image of W_0 , the chance level when d_{out} exceeds the rank. **o**, **v** and **down** are roughly uniform, with a slight excess at both ends. Read descriptively, the initial gradient falls in no band, except in **k_proj** and partly **q_proj**, whose directions lean toward the minor band. It is stable when accumulated, and stays far from the learned directions. The directions of free LoRA and of **learned** recur in part from one seed to another (0.16 to 0.48), with one direction shared and the other not.

comparison	q	k	v	o	gate	up	down
OBQA: grad / top	0.001	0.002	0.011	0.001	0.000	0.001	0.002
OBQA: grad / bottom	0.003	0.219 (30×)	0.018	0.001	0.000	0.000	0.002
OBQA: grad / learned s0	0.071 (56×)	0.091 (13×)	0.183 (25×)	0.137 (107×)	0.048 (196×)	0.075 (304×)	0.078 (61×)
OBQA: gradall / learned s0	0.078	0.094	0.187	0.140	0.052	0.085	0.082
format: grad / bottom	0.004	0.154 (39×)	0.003	0.001	0.000	0.000	0.002
format: grad / learned	0.043 (49×)	0.056 (14×)	0.178 (46×)	0.157 (180×)	0.068 (284×)	0.081 (338×)	0.131 (150×)
format: gradall / learned	0.042	0.053	0.179	0.156	0.068	0.079	0.128
OBQA: learned s0 / s7	0.195 (153×)	0.276 (38×)	0.395 (55×)	0.333 (261×)	0.334 (1358×)	0.332 (1350×)	0.333 (261×)
OBQA: free B , $r = 1$, 5 seeds	0.246	0.184	0.298	0.444	0.479	0.476	0.402
format: free B , $r = 1$, 5 seeds	0.266	0.159	0.419	0.350	0.294	0.334	0.301

Table S37: **Principal angles, with ratios to chance** (analysis A1, post hoc, descriptive). Mean of the squared cosines, median over layers. The last two rows compare the B of free LoRA at rank 1 across five seeds, pairwise, at the published rate ($2 \cdot 10^{-3}$ on OpenBookQA, 10^{-3} on the format task).

A2, update norms by module type. Table S38 gives each module type’s share of the sum of squared update norms, at rank 2 on seeds 2 to 6. The ratio of the A-frozen to the B-frozen norms is 1.3 to 8.4 on **gate** and **up**, and 0.3 to 0.6 on **k**, **v** and **down** (0.9 to 1.6 on the minor band). Per module, the B-frozen, A-frozen and free rank-1 arms train 4096, 16384 and 10240 parameters on **gate** and **up** with Llama, and 3072, 17920 and 10496 with Qwen. On **down** they train 16384, 4096 and 10240, and 17920, 3072 and 10496. Read descriptively, the A-frozen arms put most of their update in **gate** and **up**, like free LoRA, and

arm	OBQA: gate+up	OBQA: down	format: gate+up	format: down
top (B frozen)	0.21	0.45	0.26	0.35
bottom (B frozen)	0.24	0.45	0.30	0.37
dual_top (A frozen)	0.81	0.05	0.72	0.08
dual_bottom (A frozen)	0.81	0.06	0.69	0.09
free LoRA, $r = 1$	0.84	0.09	0.68	0.14

Table S38: **Where each arm puts its update** (analysis A2, post hoc, descriptive). Share of the sum of squared update norms, rank 2, seeds 2 to 6.

the B-frozen arms mostly in **down**. That is also where the A-frozen arms hold most of their parameters, so the analysis localises the difference without separating the side of the factor from the shape of the module, a declared limit.

A3, equivalence of the bands with random_ortho. Table S39 gives, with B frozen at rank 2, the 90 % intervals of each band against **random_ortho**. Read descriptively, intervals over seeds alone exclude an advantage of the best band over **random_ortho** larger than 1.3 to 3.8 points, depending on the population (1.28 to 3.84). On ARC-Challenge, **top** exceeds **random_ortho** by 1.5 points, with its 90 % interval above zero. On the format task with Qwen2.5-1.5B the best band’s interval is also above zero. But the bootstrap picks the best band at each draw, which favours a positive gap, so that interval is read with caution. The statement that no band beats a random basis is qualified accordingly.

population	top – random_ortho	bottom – random_ortho	best band – random_ortho (bootstrap)	margin
format, Llama-3.2-1B	−0.91 [−2.45, +0.63]	−6.12 [−7.88, −4.36]	−0.91 [−2.05, +0.23]	2.05
OBQA, Qwen2.5-1.5B	+0.60 [−0.52, +1.72]	−3.72 [−6.14, −1.30]	+0.60 [−0.28, +1.28]	1.28
ARC-Challenge, Qwen2.5-1.5B	+1.50 [+0.66, +2.34]	−0.24 [−1.67, +1.19]	+1.50 [+0.96, +2.10]	2.10
format, Qwen2.5-0.5B	−2.83 [−7.01, +1.35]	−1.50 [−4.84, +1.84]	−1.50 [−3.84, +1.05]	3.84
format, Qwen2.5-1.5B (seeds 22–26)	+0.20 [−2.47, +2.87]	+1.53 [−0.49, +3.55]	+1.53 [+0.13, +2.90]	2.90

Table S39: **Spectral bands against a random orthonormal basis** (analysis A3, post hoc, descriptive). B frozen, rank 2, differences in points with 90 % intervals over seeds alone. The margin is the largest advantage of the best band that the data do not exclude.

A4, the bottom arm on OpenBookQA. Its relative perturbation, $\|\Delta W\|/\|W_0\|_F$ as a median over layers and seeds, against that of **top**, is 2.10 against 4.38 % on **q**, 3.68 against 10.13 % on **k**, and 4.05 against 9.16 % on **v**. It is 2.06 against 4.25 % on **o**, 0.93 against 1.95 % on **gate**, 0.93 against 1.91 % on **up**, and 1.90 against 4.26 % on **down**. Its update norms are about half as large. Per-question predictions survive for 0 runs of 10, and factors for 0 of 10, so a per-question comparison and a re-evaluation are impossible. Read descriptively, **bottom** learns, about half as much as **top**, on every module type.

A5, a bootstrap over seeds and test items. On the format task with Llama-3.2-1B, per-question correctness was recomputed from the generations of the 5 seeds and checked run by run against the parsed scores. Table S40 resamples both seeds and the 614 test questions, with 10 000 draws. Read descriptively, the band gap with B frozen and the interaction hold when the uncertainty of the questions is also propagated. The small band gap with A frozen, barely established on the seeds alone, no longer is: with A frozen, the band does not count.

statistic	estimate	95 % CI, seeds and questions	95 % CI, seeds only	widening
band gap, B frozen (bottom – top)	−5.21	[−8.96, −1.47]	[−7.20, −3.09]	×1.83
band gap, A frozen (dual_bottom – dual_top)	+0.78	[−1.86, +3.45]	[+0.03, +2.12]	×2.55
interaction	+5.99	[+1.21, +10.62]	[+3.36, +8.60]	×1.80

Table S40: **A two-level bootstrap on the format task** (analysis A5, post hoc, descriptive). Llama-3.2-1B, rank 2, 614 questions, 5 seeds, 10 000 draws. Differences in points. With A frozen, the sign of the band gap is no longer established.

S24.7 Analysis A6: the warm-up cuts, and where their directions fall

This analysis is post hoc and descriptive, and its protocol was committed before execution (29/09 23:44). It trains nothing. The initial gradient of **grad**, over 64 sequences with seed 1000, was recomputed keeping 32 directions, in two gradient jobs. Its first two directions return the bases of **grad**, with a median overlap of 1.0000 on both pairs. Each top- k energy is compared with its own random reference, k/d_{out} , as the protocol states.

The cuts. An exact replay of training, checked on every logged step (5 on OpenBookQA, 28 on the format task), gives the cuts of Table 21 of the main paper. **warm10** saw less than one epoch, 0.59 on OpenBookQA and 0.28 on the format task, at a rate still near its peak.

The energy of the bases. Tables S41 and S42 give the energy of each basis in the initial gradient’s top subspaces and in the two bands, as medians over layers. The top-8 values lie between the top-2 and top-32 values. Read descriptively, at every cut the directions stay outside the bands of W_0 , except in **k_proj**, slightly inside the minor band. They lie far above chance in the initial gradient’s top subspaces, but less and less as training proceeds: **warm5** is the most aligned and **learned** the least. Even the gradient’s top 32 directions capture only 0.14 to 0.47 of the energy of the learned directions. Beside the dose and response of Table S34, with $P = 0.13, 0.58, 0.78$ and 1.16 on OpenBookQA, the directions become useful as they move away from the initial gradient. The gradient points to the right neighbourhood, not to the right place.

basis	subspace	q	k	v	o	gate	up	down
warm5	gradient top-2	0.172 (132×)	0.165 (21×)	0.381 (49×)	0.353 (271×)	0.119 (533×)	0.209 (935×)	0.263 (202×)
	gradient top-8	0.286	0.343	0.550	0.482	0.222	0.310	0.450
	gradient top-32	0.394 (19×)	0.522 (4×)	0.625 (5×)	0.550 (26×)	0.274 (77×)	0.348 (97×)	0.535 (26×)
warm10	gradient top-2	0.138	0.141	0.328	0.286	0.104	0.176	0.171
	gradient top-32	0.375	0.510	0.582	0.507	0.252	0.317	0.423
warm25	gradient top-2	0.093	0.097	0.237	0.192	0.069	0.122	0.099
	gradient top-32	0.346	0.490	0.463	0.414	0.199	0.241	0.307
learned (phase 2)	gradient top-2	0.071 (55×)	0.091 (12×)	0.183 (23×)	0.137 (105×)	0.048 (216×)	0.075 (334×)	0.078 (60×)
	gradient top-32	0.328 (16×)	0.468 (4×)	0.379 (3×)	0.307 (15×)	0.143 (40×)	0.171 (48×)	0.239 (11×)
all	dominant band	0.003	0.008–0.011	0.007–0.010	0.001	0.001	0.000–0.001	0.002–0.003
all	minor band	0.001	0.033–0.045 (4–6×)	0.006–0.011	0.001	0.000	0.000	0.001

Table S41: **Energy of each basis, OpenBookQA with Qwen2.5-1.5B** (analysis A6, post hoc, descriptive). Medians over layers, with the ratio to the random reference k/d_{out} in parentheses.

basis	subspace	q	k	v	o	gate	up	down
warm5	gradient top-2	0.099 (101×)	0.136 (35×)	0.336 (86×)	0.307 (314×)	0.127 (519×)	0.157 (643×)	0.248 (254×)
	gradient top-32	0.396 (25×)	0.497 (8×)	0.570 (9×)	0.606 (39×)	0.395 (101×)	0.481 (123×)	0.557 (36×)
warm10	gradient top-2	0.076	0.103	0.279	0.230	0.103	0.114	0.180
	gradient top-32	0.338	0.451	0.491	0.484	0.314	0.385	0.394
warm25	gradient top-2	0.052	0.073	0.214	0.181	0.084	0.093	0.152
	gradient top-32	0.250	0.358	0.377	0.376	0.251	0.309	0.320
learned (replication B)	gradient top-2	0.043 (44×)	0.056 (14×)	0.178 (46×)	0.157 (160×)	0.068 (279×)	0.081 (332×)	0.131 (134×)
	gradient top-32	0.209 (13×)	0.323 (5×)	0.328 (5×)	0.307 (20×)	0.189 (48×)	0.264 (68×)	0.264 (17×)
all	both bands	at chance level (0 to 2×), except k_proj in the minor band, 3–4×						

Table S42: **Energy of each basis, format task with Llama-3.2-1B** (analysis A6, post hoc, descriptive). Medians over layers, with the ratio to the random reference k/d_{out} in parentheses.

S25 TMLR additions

Five pre-registered experiments and one added test answer the main objections raised for this version, and three post hoc analyses read the existing results. They total 681 runs and 61.3 GPU-hours (Table S43). Each protocol and each addendum was committed and pushed before the first run of its experiment, and no earlier run is reused as a result. Everything not stated is as in the main campaigns: data, 600,000 supervised tokens, evaluation, checker, optimiser, clipping, the main rate grid and its upper extension. Every run of the experiments ran on one GPU model, RTX 4000 Ada. The two main pairs are the

format task with Llama-3.2-1B and OpenBookQA with Qwen2.5-1.5B. Table S44 gives the date and time at which each protocol, addendum and result was pushed.

	question	runs	GPU-hours	main result
A1	Holm over the 145 tests of the paper	—	—	46 of the 109 established results stay established
A2	minimum detectable effect of the band comparisons not established	—	—	3 to 5 points with 5 to 8 seeds, one reversal beyond the lack of power (ARC-Easy)
A3	GPU model behind each reported cell	—	—	86 of the 137 cells entirely on RTX 4000 Ada
E1	the grid boundary at $2 \cdot 10^{-2}$	24	3.4	top at $r=1$ changes rate ($0.5107 \rightarrow 0.4834$), random at $r=4$ stays reported at 10^{-2} (0.7023)
E2	rank 2 on a single GPU model	35	5.0	every established result stays established
E3	MiCA as released, with its recipe, at rank 16	15	0.8	$G_{\text{MiCA}} = +3.72$, established
E3b	free LoRA with MiCA's recipe (added after E3)	5	0.2	G at equal recipe = $+3.08$, established
E4	the frozen factor against the module shape	538	45.9	8 of 8 tests established
E5	the warm-up on two more direction seeds	64	6.0	the dose-response reproduces
total		681	61.3	

Table S43: The TMLR additions. A1 to A3 run no training, and are post hoc and descriptive.

object	pushed (2026)
definitions of analyses A1 to A3	before 02/10, 23:43
A1: results	02/10, 23:43
addendum to the definitions (A2, A3), and A2 and A3 results	03/10, 00:50
code of the experiments, and the protocols of E1, E2, E3 with its first addendum, E4 and E5	03/10, 00:51
E1: three fixes of its engine, before any run of E1	03/10, 01:01, 08:05 and 08:12
E1: grids at $2 \cdot 10^{-2}$	03/10, 08:12
second addendum to E3 (E3b), its code and its grid	03/10, 08:36
second addendum to the analyses (A3b), then its results	03/10, 09:00 and 09:01
third addendum to the analyses (A3c), and its results	03/10, 09:13
E1: reading	03/10, 09:18

Table S44: Pushes of the TMLR additions, as recorded after each successful push.

S25.1 Experiment 1: the grid boundary at $2 \cdot 10^{-2}$

The main grid stops at 10^{-2} , and four cells selected there had never been checked at $2 \cdot 10^{-2}$. They are **top**, **bottom** and **random** at rank 1 and **random** at rank 4, on Llama-3.2-1B with the format task. Each ran at $2 \cdot 10^{-2}$ on seeds 0 and 1, in the configuration of its reported runs, for 8 runs. The paper's selection rule is applied unchanged over the main grid plus $2 \cdot 10^{-2}$, reading the earlier selection runs at the other rates. If the two best rates differ by 0.005 or less, both run on seeds 30 to 32, and a rate that leads the five-seed mean by more than 0.005 is retained. Otherwise the best five-seed mean is retained and the tie is declared unresolved. A cell whose rate changes is rerun at its new rate on seeds 2 to 6. No test is pre-registered: the outcome is the selection itself. The engine was corrected three times after the protocol was committed, before the first run, and the protocol text did not change.

cell	validation at 10^{-2} / $2 \cdot 10^{-2}$	tie	rate	score (seeds 2 to 6)
top $r=1$	0.5325 / 0.5563	no	$10^{-2} \rightarrow 2 \cdot 10^{-2}$	$0.5107 \rightarrow 0.4834$ (-2.73)
bottom $r=1$	0.5062 / 0.1787	no	unchanged (10^{-2})	0.4547
random $r=1$	0.5837 / 0.5825	yes, settled (0.5740 against 0.5590)	unchanged (10^{-2})	0.5277 (published 0.5288)
random $r=4$	0.7125 / 0.0000, and 0.7200 at $5 \cdot 10^{-3}$	no	the selection retains $5 \cdot 10^{-3}$, the reported value stays at 10^{-2}	reported 0.7023 , and 0.7081 at $5 \cdot 10^{-3}$ (complement)

Table S45: Experiment 1, the grid boundary (24 runs, 3.4 GPU-hours).

For **top** at rank 1, the boundary bound the selection. At the new rate the validation score rises but the reported score falls, and $2 \cdot 10^{-2}$ lies at the edge of the tested grid, which the protocol does not extend. **random** at rank 4 diverges at $2 \cdot 10^{-2}$, so the boundary changes nothing there. On its selection runs, the main grid retains $5 \cdot 10^{-3}$ (0.7200 against 0.7125). The paper already reports this cell at 10^{-2} and declares it, since the pre-registered rank-4 measure fixed 10^{-2} in advance. Its reported value stays 0.7023, and 0.7081 at $5 \cdot 10^{-3}$ is given as a complement. For **random** at rank 1, the main configuration gives 0.5277, and the published 0.5288 mixed in an isolated run from another experiment. At rank 1, **top** – **bottom** moves from +5.60 to +2.87 points (Table S14). Rank 1 is a population without a claim, so no conclusion changes.

S25.2 Experiment 2: rank 2 on a single GPU model

The question is whether the paper’s rank-2 results on the format task hold when every run of the comparison runs on one GPU model. Every arm is rerun at its reported rate on seeds 2 to 6, all on RTX 4000 Ada, for 35 runs. The arms are **top**, **bottom**, **dual_top** and **dual_bottom** at rank 2, the budget-matched free LoRA at rank 1, and **top_unfrozen** and **bottom_unfrozen** at rank 1. The tests are the paper’s, repeated per band over seeds 2 to 6. $G > 0$ and A frozen – B frozen > 0 each need $t > 3.495$, with Bonferroni over the two bands. R is read against $0.7G$ as in the paper, established if its whole 95 % interval lies above $0.7G$. Every result the paper establishes there stays established (Table S46), and **dual_top** falls exactly on its published value.

arm	rate	score	published	difference
top $r=2$	10^{-2}	0.6423	0.6443	−0.20
bottom $r=2$	$5 \cdot 10^{-3}$	0.5831	0.5922	−0.91
dual_top $r=2$	10^{-2}	0.7739	0.7739	+0.00
dual_bottom $r=2$	10^{-2}	0.7798	0.7818	−0.20
free LoRA $r=1$	10^{-3}	0.7697	0.7684	+0.13
top_unfrozen $r=1$	$5 \cdot 10^{-3}$	0.8036	0.7928	+1.08
bottom_unfrozen $r=1$	$5 \cdot 10^{-3}$	0.7967	0.7954	+0.13

test	top	bottom	status
G	+12.74, $t = 11.39$ (published +12.41)	+18.66, $t = 13.86$ (published +17.62)	established
A frozen – B frozen	+13.16, $t = 11.49$ (published +12.96)	+19.67, $t = 14.36$ (published +18.96)	established
R against $0.7G$	+16.12 [+13.20, +19.04] against 8.92	+21.37 [+19.09, +23.64] against 13.07	covers most of G

Table S46: Experiment 2, rank 2 on the format task with Llama-3.2-1B, every run on RTX 4000 Ada (35 runs, 5.0 GPU-hours). Valid-format accuracy over seeds 2 to 6, differences in points.

S25.3 Experiment 3: MiCA with its own recipe, and 3b at equal recipe

At rank 16 on OpenBookQA with Qwen2.5-1.5B, our MiCA-like arm (**bottom**) trails the budget-matched free LoRA. Experiment 3 asks whether MiCA as released in PEFT 0.21.0, trained with the recipe of its paper (Rüdiger and Raschka, 2026), trails it too. The MiCA arm uses PEFT’s **mica** initialisation, rank 16 and $\alpha = 16$. Its recipe is that paper’s only rank-16 configuration, with its stated training settings. They are a rate of $5 \cdot 10^{-4}$, a cosine schedule with a warm-up ratio of 0.1, 4 epochs, weight decay 0.01, adapter dropout 0.05, clipping at 0.3 and a per-device batch of 4 without accumulation. The data, evaluation, checker, optimiser family and precision are ours, since that paper does not state them. That paper adapts **q_proj** and **v_proj** only. We adapt our seven modules, so that MiCA and our arm differ only by their recipe, and the free LoRA stays budget-matched. Our **bottom** at rank 16 and the free LoRA at rank 7 are rerun at their selected rates, and all three arms run on seeds 2 to 6. The pre-registered test is $G_{\text{MiCA}} = \text{free LoRA} - \text{MiCA} > 0$, with $t > 2.776$, and $\text{MiCA} - \text{bottom}$ is descriptive. A first addendum, pushed before any run, notes that MiCA’s recipe trains for 4 epochs, against about 5.76 for our 600,000 tokens on OpenBookQA. The comparison is therefore between recipes, not at equal training.

Experiment 3b was decided after the result of experiment 3 was known, and it is pre-registered only before its own runs. It trains PEFT’s standard LoRA, with both factors trained, at rank 7 and $\alpha = 7$, on our seven modules. Its recipe is exactly MiCA’s, through the same code path. Its pre-registered test is $G_{\text{equal}} = \text{free}$

LoRA with MiCA’s recipe – $\text{MiCA} > 0$, with $t > 2.776$. MiCA’s rate was tuned for MiCA, not for LoRA, so the recipe can only disadvantage the free LoRA.

arm	score	supervised tokens	steps
MiCA $r=16$, its recipe	0.5240	417,792	272
our bottom $r=16$	0.4984	602,112	49
free LoRA $r=7$, our recipe	0.5612	602,112	49
free LoRA $r=7$, MiCA’s recipe (3b)	0.5548	417,792	272
$G_{\text{MiCA}} = \text{free LoRA} - \text{MiCA}$	+3.72 [+1.58, +5.86], $t = 4.83$, established		
MiCA – our bottom	+2.56 [+1.30, +3.82], descriptive		
$G_{\text{equal}} = \text{free LoRA, MiCA’s recipe} - \text{MiCA (3b)}$	+3.08 [+1.34, +4.82], $t = 4.90$, established		
free LoRA, MiCA’s recipe – free LoRA, ours	−0.64 [−1.41, +0.13], descriptive		

Table S47: Experiments 3 and 3b, OpenBookQA with Qwen2.5-1.5B at rank 16, seeds 2 to 6 (15 and 5 runs, 0.8 and 0.2 GPU-hours). Differences in points.

G_{MiCA} is established. Our MiCA-like arm underestimates MiCA by 2.56 points at this rank, descriptively. By the reading fixed in advance, MiCA trained with its own recipe still trails budget-matched LoRA trained with ours. G_{equal} is established too, and the free LoRA barely depends on the recipe, descriptively. By the reading fixed in advance, MiCA trails budget-matched LoRA at equal recipe and equal training, on this pair, at rank 16, on our seven modules.

S25.4 Experiment 4: the frozen factor against the module shape

The A-frozen arms put most of their update in **gate_proj** and **up_proj**, the B-frozen arms more of theirs in **down_proj** (analysis A2 of 29/09). The cost of freezing B could therefore come from module shape rather than from the frozen factor, so the factor is tested inside each module type. Group G1 adapts **down_proj** only, and group G2 **gate_proj** and **up_proj** only, in every layer and nothing else. Each group runs **top**, **bottom**, **dual_top** and **dual_bottom** at rank 2. G2 adds **top** and **bottom** at equal trainable parameters with the A-frozen arms: rank 8 on Llama-3.2-1B and rank 12 on Qwen2.5-1.5B. A free LoRA at rank 1 is reported descriptively only, since it is budget-matched in neither group. Table S48 gives the trainable parameters, fixed before any run and checked by the campaign before launch.

group	arm	Llama-3.2-1B	Qwen2.5-1.5B
G1	B frozen, $r=2$	16,384 / 262,144	17,920 / 501,760
G1	A frozen, $r=2$	4,096 / 65,536	3,072 / 86,016
G1	free LoRA, $r=1$	10,240 / 163,840	10,496 / 293,888
G2	B frozen, $r=2$	4,096 / 131,072	3,072 / 172,032
G2	A frozen, $r=2$	16,384 / 524,288	17,920 / 1,003,520
G2	B frozen at equal parameters	$r=8$: 16,384 / 524,288	$r=12$: 18,432 / 1,032,192
G2	free LoRA, $r=1$	10,240 / 327,680	10,496 / 587,776

Table S48: Experiment 4, trainable parameters per module and in total. On Qwen2.5-1.5B, the B-frozen arms of G2 at rank 12 hold 2.9% more than the A-frozen arms.

Rates are selected per arm and per group on seeds 0 and 1, over $5 \cdot 10^{-4}$, 10^{-3} , $2 \cdot 10^{-3}$, $5 \cdot 10^{-3}$, 10^{-2} and $2 \cdot 10^{-2}$. A selected rate must be an interior maximum, and the grid is otherwise extended by one step on the 1-2-5 ladder, with the tie rule of experiment 1. Results are reported on seeds 2 to 9, eight seeds, at the selected rates. A group and pair is declared not testable if its free LoRA gains less than 2 points over the base model. The predictions are that freezing A beats freezing B on both bands in G1 at rank 2 (P1), and in G2 against the B-frozen arms at equal parameters (P2). The tests are paired over seeds 2 to 9, one per pair, band and group, with Bonferroni over these 8 tests. Each is established if its interval at level $1 - 0.05/8$ ($t = 3.855$, 7 degrees of freedom) lies above zero, reversed if below, and inconclusive otherwise. By the reading fixed in advance, if P1 and P2 hold on both pairs, the cost of freezing B comes neither from module shape nor from the parameter count.

The free LoRA gains +73.49 and +78.62 points over the base model on the format task in G1 and G2, and +10.45 and +11.35 on OpenBookQA, so every group and pair is testable. All eight tests are established

pair	group	arm	rate	score	remark
format	G1	top $r=2$	10^{-2}	0.2938	
	G1	bottom $r=2$	10^{-2}	0.1230	
	G1	dual_top $r=2$	$2 \cdot 10^{-2}$	0.6592	grid extended, unresolved tie
	G1	dual_bottom $r=2$	$5 \cdot 10^{-2}$	0.6775	grid extended, at the edge
	G1	free LoRA $r=1$	$5 \cdot 10^{-3}$	0.7349	tie settled
	G2	top $r=2$	$5 \cdot 10^{-2}$	0.3915	grid extended, at the edge
	G2	bottom $r=2$	$2 \cdot 10^{-2}$	0.4053	grid extended
	G2	dual_top $r=2$	$5 \cdot 10^{-3}$	0.7814	
	G2	dual_bottom $r=2$	$2 \cdot 10^{-2}$	0.7553	
	G2	top $r=8$	10^{-2}	0.6557	unresolved tie
	G2	bottom $r=8$	10^{-2}	0.6549	
	G2	free LoRA $r=1$	$5 \cdot 10^{-3}$	0.7862	tie settled
OpenBookQA	G1	top $r=2$	$5 \cdot 10^{-3}$	0.4585	
	G1	bottom $r=2$	10^{-2}	0.4075	
	G1	dual_top $r=2$	$2 \cdot 10^{-2}$	0.4915	grid extended
	G1	dual_bottom $r=2$	$2 \cdot 10^{-2}$	0.4962	grid extended
	G1	free LoRA $r=1$	$5 \cdot 10^{-3}$	0.5265	
	G2	top $r=2$	10^{-2}	0.4208	
	G2	bottom $r=2$	10^{-3}	0.4410	unresolved tie
	G2	dual_top $r=2$	$5 \cdot 10^{-3}$	0.5333	
	G2	dual_bottom $r=2$	$5 \cdot 10^{-3}$	0.5195	
	G2	top $r=12$	$2 \cdot 10^{-2}$	0.4885	grid extended
	G2	bottom $r=12$	$2 \cdot 10^{-2}$	0.4595	grid extended
	G2	free LoRA $r=1$	$5 \cdot 10^{-3}$	0.5355	

Table S49: Experiment 4, the 24 cells (538 runs, 45.9 GPU-hours, parameters as in the protocol). Format task with Llama-3.2-1B, OpenBookQA with Qwen2.5-1.5B, seeds 2 to 9, each arm at its selected rate. G1: `down_proj` only. G2: `gate_proj` and `up_proj` only.

(Table S50). Descriptively, the A-frozen arms of G2 lead the B-frozen arms at rank 2 by +38.99 and +35.00 points on the format task, and by +11.25 and +7.85 on OpenBookQA.

pair, group	band	comparison	difference [corrected interval]	status
format, G1	dominant	dual_top - top $r=2$	+36.54 [+32.34, +40.75]	established
format, G1	minor	dual_bottom - bottom $r=2$	+55.46 [+52.19, +58.72]	established
format, G2	dominant	dual_top - top $r=8$	+12.56 [+10.59, +14.53]	established
format, G2	minor	dual_bottom - bottom $r=8$	+10.04 [+7.33, +12.74]	established
OpenBookQA, G1	dominant	dual_top - top $r=2$	+3.30 [+1.76, +4.84]	established
OpenBookQA, G1	minor	dual_bottom - bottom $r=2$	+8.88 [+7.03, +10.72]	established
OpenBookQA, G2	dominant	dual_top - top $r=12$	+4.48 [+2.19, +6.76]	established
OpenBookQA, G2	minor	dual_bottom - bottom $r=12$	+6.00 [+4.30, +7.70]	established

Table S50: Experiment 4, the eight pre-registered tests. The A-frozen arms run at rank 2, seeds 2 to 9, intervals at level $1 - 0.05/8$.

P1 and P2 hold on both pairs. By the reading fixed in advance, for the MLP modules, the cost of freezing B comes neither from module shape nor from the parameter count. Three caveats apply. `dual_bottom` in G1 on the format task sits at the edge of its grid, and an under-tuning could only weaken the predicted effect. Three ties are unresolved, and the selection was validated on 500 examples instead of 400, a declared deviation. The attention modules were not tested separately.

S25.5 Experiment 5: the warm-up on two more direction seeds

The warm-up campaign froze B on the output directions of a free LoRA after 5, 10 and 25 % of its training, with direction seed 0. Experiment 5 repeats it on both pairs with two new direction seeds, 301 and 302, never used for a direction run before. The direction runs are free LoRAs at rank 2 at $2 \cdot 10^{-3}$, as in the warm-up campaign. Their factors are saved at steps 3, 5 and 13 of 49 on OpenBookQA, and 14, 28 and 69 of 274 on the format task. The bases are built as in the warm-up campaign, and their checksums were committed before the arms ran. Each arm runs on seeds 2 to 6, at the rate the original campaign selected for its placement and pair, for 4 direction runs and 60 arm runs. The test of the warm-up protocol is repeated on `warm10`, for each new direction seed and pair, with Bonferroni over these four tests. The gain `warm10` -

top, against the published **top** (0.4652 and 0.6443), needs $t > 4.315$. P uses the G fixed in advance (8.04 and 12.41) and is read on its interval at level $1 - 0.05/4$. It suffices if the lower bound reaches 0.7, does not suffice if the upper bound stays below 0.3, and is inconclusive otherwise.

pair	placement	seed 0 (original)	seed 301	seed 302
OpenBookQA	warm5	0.13	-0.01 [-0.26, 0.24]	-0.00 [-0.33, 0.32]
	warm10	0.58	0.51 [0.32, 0.70]	0.55 [0.35, 0.75]
	warm25	0.78	0.70 [0.58, 0.82]	0.81 [0.67, 0.94]
format	warm5	0.40	0.45 [0.12, 0.78]	0.23 [0.08, 0.39]
	warm10	0.63	0.55 [0.19, 0.90]	0.49 [0.25, 0.73]
	warm25	0.93	0.90 [0.75, 1.05]	1.07 [0.76, 1.38]
test on warm10		warm10 - top, gain, reading of P		
OpenBookQA	seed 301	+4.12, $t = 11.63$, established, inconclusive		
OpenBookQA	seed 302	+4.44, $t = 11.97$, established, inconclusive		
format	seed 301	+6.78, $t = 6.62$, established, inconclusive		
format	seed 302	+6.09, $t = 8.72$, established, inconclusive		

Table S51: Experiment 5, the share P of the gap recovered on each direction seed (64 runs, 6.0 GPU-hours). Seed 0 is the original campaign. Intervals at level $1 - 0.05/4$ for the new seeds.

The dose-response reproduces on all three direction seeds (Table S51). On every seed and pair, **warm10** gives an established gain and an inconclusive reading of P , as in the original campaign.

S25.6 Analysis A1: Holm across the 145 tests

The definitions of A1 to A3 were committed before execution, but the data existed, so every output is post hoc and descriptive. A1 applies Holm’s step-down procedure at family-wise level 0.05 to the 145 tests of the paper, the 151 rows of Tables S24 to S27 minus the six rows the paper does not count. Each row is converted to one p -value by fixed rules, from its t statistic, its interval or its printed p -value. A row with several components takes the largest p among those the paper calls established, or else the smallest. A criterion that is not a statistical test keeps its place with $p = 1$ and stays outside Holm. Before Holm, each converted p is checked against the row’s own threshold. Only row 88, which joins two tests at two thresholds, does not reproduce its verdict. With five seeds, the first step of Holm needs $p \leq 0.05/145 \approx 3.4 \cdot 10^{-4}$, that is $|t| \geq 11.34$ for a two-sided test with four degrees of freedom (9.48 one-sided). After Holm, 46 of the 109 established results stay established and 63 do not (Table S52). Row 28, a floor fixed in advance, stays outside Holm. Six rows converted from a bootstrap interval lie near the threshold: rows 32, 33, 64, 65, 74 and 114.

claim	holds after Holm	no longer holds
freezing B costs, at equal rank	9 populations: format, Llama (66, 67), OBQA (76), format, Qwen (93), $4 \times$ tokens (95), Mistral (108), fresh seeds (113), Qwen 7B (116), ARC-C $r=16$ (89), OBQA $r=8$ (133)	OBQA $r=4$ (132) and $r=16$ (99), format $r=16$ (86), 7B $r=16$ (140)
A frozen beats B frozen	7: format, Llama (71, 72), Mistral (109), Qwen 7B (117), budget 7B (129), OBQA $r=4$ (134), ARC-C $r=16$ (102), modules q and o (97)	OBQA $r=2$ (79, adjusted 0.062), budget 1.5B (128, 0.062), $r=8$ (135), $r=16$ (103, 141), answer only (151)
R covers most of G	format, Llama (64, 65), OBQA top (74, 0.049)	OBQA bottom (75), format, Qwen (92), 7B and Mistral (106, 107), fresh seeds (114, 0.063), ranks 4, 8 and 16
mechanism	phase 2 (144), replication of B (127), gradient on OBQA (145, 147), warm-up on OBQA (149)	replication of A (126, 0.065), gradient on the format task (146), warm-up on the format task (150, 0.062)
factor $\times$ band	the five decompositions (121 to 125)	—

Table S52: Analysis A1, Holm at family-wise level 0.05 over the 145 tests (post hoc, descriptive). Row numbers in table order, with the adjusted p where it lies near 0.05.

S25.7 Analysis A2: the minimum detectable effect of the band comparisons

A2 takes each band comparison the paper reports as not established, a difference between two bands or band positions, or between their gaps. For a paired t -test on n seeds at the row’s own level, the minimum detectable effect at 80 % power is $\delta_z \sigma_d$. Here δ_z gives a power of 0.80 under the non-central t distribution,

and σ_d is the standard deviation of the paired differences. σ_d comes from the printed mean difference and t where the paper prints both. An addendum, pushed before A2 and A3 ran, declares one deviation. Rows 10 and 110 print t only, so A2 reports their standardised effect d_z only. Rows 35 and 125 print a bootstrap interval, and their detectable effect uses a normal approximation.

row	comparison	n	detectable effect	observed
1	$\beta = 0.75$ against $\beta = 0$ / $\beta = 0.5$ (Llama, 7 seeds)	7	3.36 / 4.51	+2.21 / +0.98
3	$\beta = 0.75$ against random_ortho	7	3.92	+2.42
4, 5	$\beta = 0.75$ against $\beta = 0.5$ / random_ortho (fresh seeds)	8	5.01 / 4.84	+1.38 / +3.36
12	$\beta = 0.75$ against $\beta = 1$ (Qwen 0.5B)	10	4.17	+2.28
14	$\beta = 0.75$ against random_ortho (Qwen 0.5B)	15	2.17	+0.48
17	bottom against random_ortho , $5 \cdot 10^{-3}$ / $2 \cdot 10^{-3}$	5	8.01 / 3.28	+1.50 / +2.41
18, 21	common rate, lead of bottom (Llama, ranks 2 and 4)	5	3.29 / 3.50	+0.98 / +2.25
26	common rate, deficit of bottom (Qwen 1.5B)	5	5.50	+3.13
35	rank 2, along the spectrum (bootstrap)	5	5.93	[−4.30, +4.00]
42	top against random at 10^{-2} , perplexity	5	27.1 (perplexity)	+15.7
48	top_sigma against $\beta = 0.75$	5	3.20	+0.23
56	OBQA rank 2: lead of bottom	5	3.25	−2.88
58	ARC-Easy: lead of bottom	5	1.02	−2.90
125	Mistral: band component (bootstrap)	5	1.99	[−2.34, +0.44]
138	OBQA rank 8: band gap, A frozen against B frozen	5	4.16	+0.52
142	Qwen 7B rank 16: band gap, B trained against frozen	5	4.24	+2.12
10, 110	no mean difference printed	5	$d_z = 2.48$ / 1.68	—

Table S53: Analysis A2, the effect detectable at 80 % power by each band comparison that is not established (post hoc, descriptive). Effects in points unless stated.

With 5 to 8 seeds, the detectable effect is 3 to 5 points, and 2.2 with 15 seeds. Most observed differences, 0.2 to 3.4 points, are smaller, so not established usually means too small for these seeds rather than absent. The two sign reversals read differently. On ARC-Easy, the predicted lead of **bottom** is reversed beyond what the lack of power explains, −2.90 points for a detectable effect of 1.02. On OpenBookQA at rank 2, the reversal of −2.88 stays within the detectable effect of 3.25, and is compatible with noise.

S25.8 Analysis A3: the GPU model behind each reported cell

A3 takes every cell whose score the main paper reports in a table, identifies it in the runs by its published value, and counts the GPU model each run recorded. A3 split some cells across groups of runs, because it grouped runs by every configuration field. Three published values matched no cell, two of them central (0.6443 and 0.5922), and 33 matched two or three cells. A second addendum, A3b, pushed before it ran, groups the runs as the trainer identifies them and leaves out the runs of these additions. A third addendum, A3c, pushed before it ran, settles the values that A3b left ambiguous. They are coincidences of the 4-decimal mean, since on OpenBookQA a five-seed mean moves in steps of 0.0004. For each of their 54 occurrences, the table’s row and column were written down before execution, and A3c reports no problem. A3 and A3b are kept unchanged.

A3c finds 137 distinct reported cells, behind 133 of the 134 published values. 86 of them (63 %) ran entirely on RTX 4000 Ada, 51 (37 %) on several models, and none on a single other model. Of their runs, 577 (82.5 % of the 699 runs with a recorded GPU model) ran on RTX 4000 Ada, 39 on RTX 3090, 33 on RTX 2000 Ada, 27 on RTX A5000 and 23 on RTX A4000. Six runs have no recorded GPU model, all in the two central cells of the format task: three of **top** at rank 2 (0.6443) and three of **bottom** at rank 2 (0.5922). One value is not found: 0.5905, quoted in the captions of Tables 19 and 20 of the main paper. The two central cells were rerun entirely on RTX 4000 Ada in experiment 2, where every established result holds (Table S46).

S25.9 Declared deviations

Six deviations were declared.

1. E1 was launched after the others. Its engine was corrected three times after the commit of its protocol, before its first run.
2. E4 validated its selection on 500 examples instead of 400, for all its arms and rates.

3. A2 reports rows 10 and 110 as a standardised effect only.
4. E3b was added after the reading of E3, and is pre-registered only before its own runs.
5. A3 had a defect found after execution. It was corrected in A3b and then A3c, each committed before execution, and A3 is kept.
6. Environment: the login server was rebuilt, for orchestration only, and PEFT 0.21.0 was loaded separately, for the PEFT arms only.

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